Informatica®
10.5.11

Enterprise Data Catalog Scanner Configuration Guide

Informatica Enterprise Data Catalog Scanner Configuration Guide
10.5.11
August 2026

© Copyright Informatica LLC 2015, 2026

This software and documentation are provided only under a separate license agreement containing restrictions on use and disclosure. No part of this document may be reproduced or transmitted in any form, by any means (electronic, photocopying, recording or otherwise) without prior consent of Informatica LLC.

Informatica and the Informatica logo are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html. Other company and product names may be trade names or trademarks of their respective owners.

U.S. GOVERNMENT RIGHTS Programs, software, databases, and related documentation and technical data delivered to U.S. Government customers are "commercial computer software" or "commercial technical data" pursuant to the applicable Federal Acquisition Regulation and agency-specific supplemental regulations. As such, the use, duplication, disclosure, modification, and adaptation is subject to the restrictions and license terms set forth in the applicable Government contract, and, to the extent applicable by the terms of the Government contract, the additional rights set forth in FAR 52.227-19, Commercial Computer Software License.

Portions of this software and/or documentation are subject to copyright held by third parties, including without limitation: Copyright DataDirect Technologies. All rights reserved. Copyright © Sun Microsystems. All rights reserved. Copyright © RSA Security Inc. All Rights Reserved. Copyright © Ordinal Technology Corp. All rights reserved. Copyright © Aandacht c.v. All rights reserved. Copyright Genivia, Inc. All rights reserved. Copyright Isomorphic Software. All rights reserved. Copyright © Meta Integration Technology, Inc. All rights reserved. Copyright © Intalio. All rights reserved. Copyright © Oracle. All rights reserved. Copyright © Adobe Systems Incorporated. All rights reserved. Copyright © DataArt, Inc. All rights reserved. Copyright © ComponentSource. All rights reserved. Copyright © Microsoft Corporation. All rights reserved. Copyright © Rogue Wave Software, Inc. All rights reserved. Copyright © Teradata Corporation. All rights reserved. Copyright © Yahoo! Inc. All rights reserved. Copyright © Glyph & Cog, LLC. All rights reserved. Copyright © Thinkmap, Inc. All rights reserved. Copyright © Clearpace Software Limited. All rights reserved. Copyright © Information Builders, Inc. All rights reserved. Copyright © OSS Nokalva, Inc. All rights reserved. Copyright Edifecs, Inc. All rights reserved. Copyright Cleo Communications, Inc. All rights reserved. Copyright © International Organization for Standardization 1986. All rights reserved. Copyright © ej-technologies GmbH. All rights reserved. Copyright © Jaspersoft Corporation. All rights reserved. Copyright © International Business Machines Corporation. All rights reserved. Copyright © yWorks GmbH. All rights reserved. Copyright © Lucent Technologies. All rights reserved. Copyright © University of Toronto. All rights reserved. Copyright © Daniel Veillard. All rights reserved. Copyright © Unicode, Inc. Copyright IBM Corp. All rights reserved. Copyright © MicroQuill Software Publishing, Inc. All rights reserved. Copyright © PassMark Software Pty Ltd. All rights reserved. Copyright © LogiXML, Inc. All rights reserved. Copyright © 2003-2010 Lorenzi Davide, All rights reserved. Copyright © Red Hat, Inc. All rights reserved. Copyright © The Board of Trustees of the Leland Stanford Junior University. All rights reserved. Copyright © EMC Corporation. All rights reserved. Copyright © Flexera Software. All rights reserved. Copyright © Jinfonet Software. All rights reserved. Copyright © Apple Inc. All rights reserved. Copyright © Telerik Inc. All rights reserved. Copyright © BEA Systems. All rights reserved. Copyright © PDFlib GmbH. All rights reserved. Copyright © Orientation in Objects GmbH. All rights reserved. Copyright © Tanuki Software, Ltd. All rights reserved. Copyright © Ricebridge. All rights reserved. Copyright © Sencha, Inc. All rights reserved. Copyright © Scalable Systems, Inc. All rights reserved. Copyright © jQWidgets. All rights reserved. Copyright © Tableau Software, Inc. All rights reserved. Copyright© MaxMind, Inc. All Rights Reserved. Copyright © TMate Software s.r.o. All rights reserved. Copyright © MapR Technologies Inc. All rights reserved. Copyright © Amazon Corporate LLC. All rights reserved. Copyright © Highsoft. All rights reserved. Copyright © Python Software Foundation. All rights reserved. Copyright © BeOpen.com. All rights reserved. Copyright © CNRI. All rights reserved.

This product includes software developed by the Apache Software Foundation (http://www.apache.org/), and/or other software which is licensed under various versions of the Apache License (the "License"). You may obtain a copy of these Licenses at http://www.apache.org/licenses/. Unless required by applicable law or agreed to in writing, software distributed under these Licenses is distributed on an "AS IS" BASIS, WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied. See the Licenses for the specific language governing permissions and limitations under the Licenses.

This product includes software which was developed by Mozilla (http://www.mozilla.org/), software copyright The JBoss Group, LLC, all rights reserved; software copyright © 1999-2006 by Bruno Lowagie and Paulo Soares and other software which is licensed under various versions of the GNU Lesser General Public License Agreement, which may be found at http:// www.gnu.org/licenses/lgpl.html. The materials are provided free of charge by Informatica, "as-is", without warranty of any kind, either express or implied, including but not limited to the implied warranties of merchantability and fitness for a particular purpose.

The product includes ACE(TM) and TAO(TM) software copyrighted by Douglas C. Schmidt and his research group at Washington University, University of California, Irvine, and Vanderbilt University, Copyright (©) 1993-2006, all rights reserved.

This product includes software developed by the OpenSSL Project for use in the OpenSSL Toolkit (copyright The OpenSSL Project. All Rights Reserved) and redistribution of this software is subject to terms available at http://www.openssl.org and http://www.openssl.org/source/license.html.

This product includes Curl software which is Copyright 1996-2013, Daniel Stenberg, <daniel@haxx.se>. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http://curl.haxx.se/docs/copyright.html. Permission to use, copy, modify, and distribute this software for any purpose with or without fee is hereby granted, provided that the above copyright notice and this permission notice appear in all copies.

The product includes software copyright 2001-2005 (©) MetaStuff, Ltd. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http://www.dom4j.org/ license.html.

The product includes software copyright © 2004-2007, The Dojo Foundation. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http://dojotoolkit.org/license.

This product includes ICU software which is copyright International Business Machines Corporation and others. All rights reserved. Permissions and limitations regarding this software are subject to terms available at http://source.icu-project.org/repos/icu/icu/trunk/license.html.

This product includes software copyright © 1996-2006 Per Bothner. All rights reserved. Your right to use such materials is set forth in the license which may be found at http:// www.gnu.org/software/ kawa/Software-License.html.

This product includes OSSP UUID software which is Copyright © 2002 Ralf S. Engelschall, Copyright © 2002 The OSSP Project Copyright © 2002 Cable & Wireless Deutschland. Permissions and limitations regarding this software are subject to terms available at http://www.opensource.org/licenses/mit-license.php.

This product includes software developed by Boost (http://www.boost.org/) or under the Boost software license. Permissions and limitations regarding this software are subject to terms available at http:/ /www.boost.org/LICENSE_1_0.txt.

This product includes software copyright © 1997-2007 University of Cambridge. Permissions and limitations regarding this software are subject to terms available at http:// www.pcre.org/license.txt.

This product includes software copyright © 2007 The Eclipse Foundation. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http:// www.eclipse.org/org/documents/epl-v10.php and at http://www.eclipse.org/org/documents/edl-v10.php.

This product includes software licensed under the terms at http://www.tcl.tk/software/tcltk/license.html, http://www.bosrup.com/web/overlib/?License, http:// www.stlport.org/doc/ license.html, http://asm.ow2.org/license.html, http://www.cryptix.org/LICENSE.TXT, http://hsqldb.org/web/hsqlLicense.html, http:// httpunit.sourceforge.net/doc/ license.html, http://jung.sourceforge.net/license.txt , http://www.gzip.org/zlib/zlib_license.html, http://www.openldap.org/software/ release/license.html, http://www.libssh2.org, http://slf4j.org/license.html, http://www.sente.ch/software/OpenSourceLicense.html, http://fusesource.com/downloads/ license-agreements/fuse-message-broker-v-5-3- license-agreement; http://antlr.org/license.html; http://aopalliance.sourceforge.net/; http://www.bouncycastle.org/ licence.html; http://www.jgraph.com/jgraphdownload.html; http://www.jcraft.com/jsch/LICENSE.txt; http://jotm.objectweb.org/bsd_license.html; . http://www.w3.org/

Consortium/Legal/2002/copyright-software-20021231; http://www.slf4j.org/license.html; http://nanoxml.sourceforge.net/orig/copyright.html; http://www.json.org/license.html; http://forge.ow2.org/projects/javaservice/, http://www.postgresql.org/about/licence.html, http://www.sqlite.org/copyright.html, http://www.tcl.tk/software/tcltk/license.html, http://www.jaxen.org/faq.html, http://www.jdom.org/docs/faq.html, http://www.slf4j.org/license.html; http://www.iodbc.org/dataspace/iodbc/wiki/iODBC/License; http://www.keplerproject.org/md5/license.html; http://www.toedter.com/en/jcalendar/license.html; http://www.edankert.com/bounce/index.html; http://www.net-snmp.org/about/license.html; http://www.openmdx.org/#FAQ; http://www.php.net/license/3_01.txt; http://srp.stanford.edu/license.txt; http://www.schneier.com/blowfish.html; http://www.jmock.org/license.html; http://xsom.java.net; http://benalman.com/about/license/; https://github.com/CreateJS/EaselJS/blob/master/src/easeljs/display/Bitmap.js; http://www.h2database.com/html/license.html#summary; http://jsoncpp.sourceforge.net/LICENSE; http://jdbc.postgresql.org/license.html; http://protobuf.googlecode.com/svn/trunk/src/google/protobuf/descriptor.proto; https://github.com/rantav/hector/blob/master/LICENSE; http://web.mit.edu/Kerberos/krb5-current/doc/mitK5license.html; http://jibx.sourceforge.net/jibx-license.html; https://github.com/lyokato/libgeohash/blob/master/LICENSE; https://github.com/hjiang/jsonxx/blob/master/LICENSE; https://code.google.com/p/lz4/; https://github.com/jedisct1/libsodium/blob/master/LICENSE; http://one-jar.sourceforge.net/index.php?page=documents&file=license; https://github.com/EsotericSoftware/kryo/blob/master/license.txt; http://www.scala-lang.org/license.html; https://github.com/tinkerpop/blueprints/blob/master/LICENSE.txt; http://gee.cs.oswego.edu/dl/classes/EDU/oswego/cs/dl/util/concurrent/intro.html; https://aws.amazon.com/asl/; https://github.com/twbs/bootstrap/blob/master/LICENSE; https://sourceforge.net/p/xmlunit/code/HEAD/tree/trunk/LICENSE.txt; https://github.com/documentcloud/underscore-contrib/blob/master/LICENSE, and https://github.com/apache/hbase/blob/master/LICENSE.txt.

This product includes software licensed under the Academic Free License (http://www.opensource.org/licenses/afl-3.0.php), the Common Development and Distribution License (http://www.opensource.org/licenses/cddl1.php) the Common Public License (http://www.opensource.org/licenses/cpl1.0.php), the Sun Binary Code License Agreement Supplemental License Terms, the BSD License (http:// www.opensource.org/licenses/bsd-license.php), the new BSD License (http://opensource.org/licenses/BSD-3-Clause), the MIT License (http://www.opensource.org/licenses/mit-license.php), the Artistic License (http://www.opensource.org/licenses/artistic-license-1.0) and the Initial Developer's Public License Version 1.0 (http://www.firebirdsql.org/en/initial-developer-s-public-license-version-1-0/).

This product includes software copyright © 2003-2006 Joe Walnes, 2006-2007 XStream Committers. All rights reserved. Permissions and limitations regarding this software are subject to terms available at http://xstream.codehaus.org/license.html. This product includes software developed by the Indiana University Extreme! Lab. For further information please visit http://www.extreme.indiana.edu/.

This product includes software Copyright (c) 2013 Frank Balluffi and Markus Moeller. All rights reserved. Permissions and limitations regarding this software are subject to terms of the MIT license.

See patents at https://www.informatica.com/legal/patents.html.

DISCLAIMER: Informatica LLC provides this documentation "as is" without warranty of any kind, either express or implied, including, but not limited to, the implied warranties of noninfringement, merchantability, or use for a particular purpose. Informatica LLC does not warrant that this software or documentation is error free. The information provided in this software or documentation may include technical inaccuracies or typographical errors. The information in this software and documentation is subject to change at any time without notice.

NOTICES

This Informatica product (the "Software") includes certain drivers (the "DataDirect Drivers") from DataDirect Technologies, an operating company of Progress Software Corporation ("DataDirect") which are subject to the following terms and conditions:

1. THE DATADIRECT DRIVERS ARE PROVIDED "AS IS" WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NON-INFRINGEMENT.
2. IN NO EVENT WILL DATADIRECT OR ITS THIRD PARTY SUPPLIERS BE LIABLE TO THE END-USER CUSTOMER FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, CONSEQUENTIAL OR OTHER DAMAGES ARISING OUT OF THE USE OF THE ODBC DRIVERS, WHETHER OR NOT INFORMED OF THE POSSIBILITIES OF DAMAGES IN ADVANCE. THESE LIMITATIONS APPLY TO ALL CAUSES OF ACTION, INCLUDING, WITHOUT LIMITATION, BREACH OF CONTRACT, BREACH OF WARRANTY, NEGLIGENCE, STRICT LIABILITY, MISREPRESENTATION AND OTHER TORTS.

The information in this documentation is subject to change without notice. If you find any problems in this documentation, report them to us at infa_documentation@informatica.com.

Informatica products are warranted according to the terms and conditions of the agreements under which they are provided. INFORMATICA PROVIDES THE INFORMATION IN THIS DOCUMENT "AS IS" WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING WITHOUT ANY WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND ANY WARRANTY OR CONDITION OF NON-INFRINGEMENT.

Publication Date: 2026-09-03

Table of Contents

Preface 8

Informatica Resources 8

- Informatica Network 8
- Informatica Knowledge Base 8
- Informatica Documentation 8
- Informatica Product Availability Matrices 9
- Informatica Velocity 9
- Informatica Marketplace 9
- Informatica Global Customer Support 9

Chapter 1: Overview of Resources 10

Introduction to Scanners 10

Where to configure the scanners 11

Resources and Scanners 14

Resource Types 14

Resource 15

Resources and Schedules 15

Resources and Attributes 15

Creating a Resource 15

Source Metadata and Data Profile Filter 16

Chapter 2: Configuring Cloud Resources 18

Overview of Cloud Resources 18

Amazon Redshift 18

Amazon S3 22

Microsoft Azure Data Lake Store 26

Azure Microsoft SQL Data Warehouse 31

Azure Microsoft SQL Server 34

Google BigQuery 38

Microsoft Azure Blob Storage 42

Salesforce 45

Workday 48

AWS Glue 50

Snowflake 53

Google Cloud Storage 53

Chapter 3: Configuring Data Engineering Resources 56

Overview of Data Engineering Resources 56

Apache Atlas 56

HDFS 59

Hive. 65

Chapter 4: Configuring Application Resources. 69

Overview of Application Resources. 69

SAP R/3. 69

SAP S/4HANA. 72

Chapter 5: Configuring Business Glossary Resources. 78

Overview of Business Glossary Resources. 78

Informatica Axon. 78

Business Glossary. 81

Chapter 6: Configuring Business Intelligence Resources. 83

Overview of Business Intelligence Resources. 83

IBM Cognos. 84

MicroStrategy. 88

Oracle Business Intelligence. 93

QlikView. 97

SAP BusinessObjects. 99

Tableau Server. 105

SAP Business Warehouse (SAP BW). 108

SAP BW/4HANA. 114

Microsoft Power BI. 118

Qlik Sense. 119

TIBCO Spotfire. 121

Chapter 7: Configuring Data Integration Resources. 125

Overview of Data Integration Resources. 125

Custom Lineage. 126

Data Integration Hub. 127

Database Scripts. 128

Informatica Intelligent Cloud Services - Cloud Data Integration. 129

Informatica Master Data Management. 131

Informatica Platform. 133

PowerCenter. 138

SQL Server Integration Service. 144

IBM InfoSphere DataStage. 144

Chapter 8: Configuring Data Modeling Resources. 146

Overview of Data Modeling Resources. 146

Erwin. 146

SAP PowerDesigner. 151

Chapter 9: Configuring Database Management Resources. 154

Overview of Database Management Resources. 154
IBM DB2. 155
IBM DB2 for z/OS. 157
IBM Netezza. 160
JDBC. 163
Microsoft SQL Server. 167
Oracle. 171
SAP HANA. 174
Sybase. 177
Teradata. 180
Configure Case Sensitivity for Database Management Resources. 183

Chapter 10: Configuring Hadoop Multi-Part File Handler Resources. 184

Overview of Hadoop Multi-Part File Handler Resources 184
Hadoop Multi-Part File Handler. 184

Chapter 11: Configuring File Management Resources. 186

Overview of File Management Resources. 186
File System. 186
OneDrive. 190
Microsoft Sharepoint. 193

Chapter 12: Configuring Streaming Source Resources. 197

Overview of Streaming Source Resources. 197
Apache Kafka. 197

Chapter 13: Configuring No SQL Resources. 201

Overview of No SQL Resources. 201
Apache Cassandra. 201
MongoDB. 204

Chapter 14: Configuring Informatica Platform Resources. 207

Informatica Platform Resources Overview. 207
Informatica Data Quality. 207
    Informatica Data Quality Resource Tasks. 211
Informatica Similarity Discovery Resource. 212
    Use Schema Filter and Table Filter in Similarity Discovery. 215
Troubleshooting. 216

Appendix A: JDBC Driver Classes. 219

Examples of JDBC Driver Classes. 219

Index. 220

Preface

Follow the instructions in the *Enterprise Data Catalog Scanner Configuration Guide* to create, configure and manage various resources in Enterprise Data Catalog. In addition to the configuration steps, the guide also includes the required permissions and prerequisites for the various resources. This guide is written for catalog administrators.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Network

The Informatica Network is the gateway to many resources, including the Informatica Knowledge Base and Informatica Global Customer Support. To enter the Informatica Network, visit https://network.informatica.com.

As an Informatica Network member, you have the following options:

- Search the Knowledge Base for product resources.
- View product availability information.
- Create and review your support cases.
- Find your local Informatica User Group Network and collaborate with your peers.

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit https://search.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

Informatica Documentation

Use the Informatica Documentation Portal to explore an extensive library of documentation for current and recent product releases. To explore the Documentation Portal, visit https://docs.informatica.com.

If you have questions, comments, or ideas about the product documentation, contact the Informatica Documentation team at infa_documentation@informatica.com.

Informatica Product Availability Matrices

Product Availability Matrices (PAMs) indicate the versions of the operating systems, databases, and types of data sources and targets that a product release supports. You can browse the Informatica PAMs at https://network.informatica.com/community/informatica-network/product-availability-matrices.

Informatica Velocity

Informatica Velocity is a collection of tips and best practices developed by Informatica Professional Services and based on real-world experiences from hundreds of data management projects. Informatica Velocity represents the collective knowledge of Informatica consultants who work with organizations around the world to plan, develop, deploy, and maintain successful data management solutions.

You can find Informatica Velocity resources at http://velocity.informatica.com. If you have questions, comments, or ideas about Informatica Velocity, contact Informatica Professional Services at ips@informatica.com.

Informatica Marketplace

The Informatica Marketplace is a forum where you can find solutions that extend and enhance your Informatica implementations. Leverage any of the hundreds of solutions from Informatica developers and partners on the Marketplace to improve your productivity and speed up time to implementation on your projects. You can find the Informatica Marketplace at https://marketplace.informatica.com.

Informatica Global Customer Support

You can contact a Global Support Center by telephone or through the Informatica Network.

To find your local Informatica Global Customer Support telephone number, visit the Informatica website at the following link:
https://www.informatica.com/services-and-training/customer-success-services/contact-us.html.

To find online support resources on the Informatica Network, visit https://network.informatica.com and select the eSupport option.

CHAPTER 1

Overview of Resources

This chapter includes the following topics:

- Introduction to Scanners, 10
- Where to configure the scanners, 11
- Resources and Scanners, 14
- Resource Types, 14
- Resource, 15
- Resources and Schedules, 15
- Resources and Attributes, 15
- Creating a Resource, 15
- Source Metadata and Data Profile Filter, 16

Introduction to Scanners

A scanner is a pluggable component of Enterprise Data Catalog that extracts specific metadata, such as source metadata or profile metadata, from external data sources and stores the metadata in the catalog. A scanner typically maps to a single resource type. However, there can be more than one scanner for a resource type.

A scanner performs a scan job on the metadata sources to fetch metadata into the catalog. When Informatica releases newer resource types for specific data sources, you can plug in those scanners to the current installation of Enterprise Data Catalog without having to upgrade the installation.

Where to configure the scanners

You can configure scanners in different tools. For example, you can configure the Salesforce scanner in the Catalog Administrator tool and the Salesforce V2 scanner in the MetaDex tool.

The following table classifies Informatica scanners by the tool in which you configure them:

Scanner	Configuration tool
Amazon Redshift database	Catalog Administrator
Amazon Redshift stored procedures and scripts	MetaDex
Amazon S3	Catalog Administrator
Amazon Web Services (AWS) Glue	Catalog Administrator
Apache Atlas	Catalog Administrator
Apache Cassandra	Catalog Administrator
Apache Hive	Catalog Administrator
Apache HiveQL	MetaDex
Apache Kafka	Catalog Administrator
Azure Microsoft SQL Data Warehouse	Catalog Administrator
Azure Microsoft SQL Server	Catalog Administrator
Business Glossary	Catalog Administrator
COBOL	MetaDex
Data Integration Hub	Catalog Administrator
Databricks	MetaDex
Erwin	Catalog Administrator
Google BigQuery	Catalog Administrator
Google Cloud Storage	Catalog Administrator
Greenplum stored procedures and scripts	MetaDex
HDFS	Catalog Administrator
IBM Cognos	Catalog Administrator
IBM DB2 database	Catalog Administrator
IBM DB2 for z/OS	Catalog Administrator
IBM DB2 i5/OS3 through the JDBC connector	Catalog Administrator

Scanner	Configuration tool
IBM DB2 stored procedures and scripts	MetaDex
IBM InfoSphere DataStage	MetaDex
IBM Netezza database	Catalog Administrator
IBM Netezza stored procedures and scripts	MetaDex
Informatica Axon	Catalog Administrator
Informatica Data Quality	Catalog Administrator
Informatica Intelligent Cloud Services - Cloud Data Integration (IICS-CDI)	Catalog Administrator
Informatica Intelligent Cloud Services (IICS) V2	MetaDex
Informatica Master Data Management	Catalog Administrator
Informatica Platform	Catalog Administrator
JCL	MetaDex
JDBC	Catalog Administrator
MicroStrategy	Catalog Administrator
Microsoft Azure Blob Storage	Catalog Administrator
Microsoft Azure Data Factory	MetaDex
Microsoft Azure Data Lake Store	Catalog Administrator
Microsoft Azure Synapse Analytics stored procedures	MetaDex
Microsoft OneDrive, OneDrive for Business	Catalog Administrator
Microsoft Power BI V2	MetaDex
Microsoft SQL Server Analysis Services (SSAS)	MetaDex
Microsoft SQL Server Integration Services	MetaDex
Microsoft SQL Server Reporting Services (SSRS)	MetaDex
Microsoft SQL Server database	Catalog Administrator
Microsoft SQL Server stored procedures and scripts	MetaDex
Microsoft Sharepoint	Catalog Administrator
MongoDB	Catalog Administrator
MySQL through the JDBC connector	Catalog Administrator

Scanner	Configuration tool
Oracle Business Intelligence	Catalog Administrator
Oracle Data Integrator	MetaDex
Oracle database	Catalog Administrator
Oracle stored procedures and scripts	MetaDex
PowerCenter	Catalog Administrator
Qlik Sense	Catalog Administrator
QlikView	Catalog Administrator
SAP BW/4HANA	Catalog Administrator
SAP Business Object Data Services (BODS)	MetaDex
SAP Business Warehouse (BW)	Catalog Administrator
SAP BusinessObjects	Catalog Administrator
SAP PowerDesigner	Catalog Administrator
SAP R/3	Catalog Administrator
SAP S/4HANA	Catalog Administrator
SAS	MetaDex
Salesforce	Catalog Administrator
Salesforce V2	MetaDex
Snowflake database V2	MetaDex
Snowflake stored procedures and scripts V2	MetaDex
Sybase database	Catalog Administrator
Sybase stored procedures and scripts	MetaDex
TIBCO Spotfire	Catalog Administrator
Tableau	Catalog Administrator
Talend	MetaDex
Teradata database	Catalog Administrator
Teradata stored procedures and scripts	MetaDex
Workday	Catalog Administrator

For more information, see the *Enterprise Data Catalog Scanner Configuration Guide* or the *MetaDex Scanner Configuration Guide*, based on where you configure the scanner.

Resources and Scanners

Scanners attached to resources identify the resource type including the name of the resource type, display name, description, and supported versions. You can configure multiple scanners for different types of metadata. You can configure the scanner properties for a resource when you create the resource. Each scanner has a unique ID that maps to the resource type.

A resource type can have multiple scanners. For example, one scanner might extract the metadata for an Oracle data source directly from the source. Another scanner might derive the metadata about the Oracle data source from other metadata repositories, such as a PowerCenter repository or Model repository.

Resource Types

Resource type is a type of external data source or metadata repository from which the resources extract metadata.

Enterprise Data Catalog supports a list of resources, which are broadly segregated into the following groups:

- Business Intelligence Resources
- Data Integration Resources
- File Management Resources
- Application Resources
- Hadoop Data Resources
- Cloud Data Resources
- Data Engineering Resources
- Business Glossary Resources
- Data Modeling Resources
- Data Management Resources
- Custom Resources
- No SQL Resources
- Data Streaming Resources
- Informatica Platform

Resource

A resource is a catalog object that represents an external data source or metadata repository from where scanners extract metadata. A resource represents an instance of a specific resource type. The basic metadata operations, such as extraction and storage of metadata, are performed at the resource level.

A resource can be of resource types, such as relational databases, Business Glossary classification, and business intelligence sources. A resource might have an associated schedule. Each resource can extract both source metadata and profile metadata from the external data sources.

Resources and Schedules

Resource schedules determine the frequency with which scanners extract metadata from the metadata sources. You can create schedules for specific resources or create reusable schedules that you want to assign to multiple resources.

You can create schedules that have an end date or schedules that run indefinitely. In addition to a start date, you can also set up the start time based on when you want scanners to start extracting metadata.

Resources and Attributes

Resources have attributes that represent specific properties of object types, such Alias Column, Category, Data Set, and Table, based on the data types of the attributes. You can use the resource attributes in Enterprise Data Catalog to search and find relevant information among a large number of data assets and data relationships.

Resource can have associated custom attributes. System attributes include predefined attributes that represent specific properties of object types, such as stored procedure, table, trigger, XML query, schema, and resource. For example, Comment is a system attribute assigned to a Hive table or Hive view. You can create custom attributes based on your business requirements that you want to assign to different resources. You can choose to create custom attributes based on the scanner types that are in use.

Creating a Resource

When you create a resource, you can specify the resource type, type of metadata that Enterprise Data Catalog extracts, and an optional schedule for the resource. You can also assign custom attributes to the resource when you create it.

Note: It is recommended that you validate the connection after you configure a resource. Click **Test Connection** to validate the connection parameters.

1. Launch the Catalog Administrator.
2. Click **New** > **Resource**.

   The **New Resource** wizard appears.
3. Enter a name and an optional description for the resource.

4. Optionally, if you want to assign custom attribute values to the resource, enter values for attributes, such as the department, data owner, data steward, and subject matter experts in the **Additional Properties** section.
    - Select **Propagate** under each custom attribute type if you want to apply custom attribute values to all the assets loaded through the resource.
5. Click **Choose** to open the **Select Resource Type** dialog box.
6. Select a resource type, and click **Select**.

   More fields appear in the wizard based on the resource type you selected.
7. Based on the resource type, configure the connection properties.
8. Click **Next** to move to the **Metadata Load Settings** page.

   You can choose the type of metadata that you want Enterprise Data Catalog to extract from the source systems.
9. Configure the required source metadata and profiling metadata parameters.
10. Click **Next** to move to the **Custom Attributes** page, and configure the attribute settings. You can select the custom attributes that you want to associate with the resource.
11. Click **Next** to move to the **Data Provisioning** page.
12. Optionally, enable data provisioning if you want to create provisioning tasks in Enterprise Data Catalog.
13. Click **Next** to move to the **Permissions** page, and restrict the permissions for the resource. You can also assign or modify the resource permissions from the security settings.
14. Click **Next** to go to the **Schedule** page.
15. Optionally, select schedules for source metadata load and profiling metadata. You can create a global schedule if required.
16. Click **Save**, or Click **Save and Run**.

Source Metadata and Data Profile Filter

In a resource that contains multiple tables and views, you can choose to extract source metadata from specific tables and views, and extract profile metadata and data from specific tables and views.

You can configure the filter field if one of the following conditions is true:

- You want to extract metadata only for a few tables and views from a list of all of the tables and views in a resource.
- You want to include a few tables and views from the next resource run.
- You want to exclude a few tables and views from the resource run.
- You want to exclude a few tables and views from the next resource run.

In a resource, you can use the following filters to include and exclude tables and views:

- Source Metadata Filter. In the **Metadata and Load Settings** tab in the Resource wizard, enter the tables and views that you want to include or exclude in a resource run. The scanner extracts the source metadata based on the tables and views in the Source Metadata Filter field.
- Data Profile Filter. In the **Metadata and Load Settings** tab in the Resource wizard, enter the tables and views that you want to include or exclude in a profile run. The profile scanner extracts profile metadata based on the tables and views in the Data Profile Filter field.

To include tables and views, you can enter full names of tables and views, or use the % wildcard character with the name or pattern. Use semicolons (;) to separate the table names and view names. If the pattern contains a special character, such as a whitespace, enclose the pattern within single quotes. For example, HR_id; 'HR Emp%'; HR_job%.

Note: The Data Profile Filter supports regular expressions such as % , ^ , $, . , | , and (a | b), and the Source Metadata Filter supports the % expression. The filter constraint applies to Azure Microsoft SQL Server, Azure Microsoft SQL Data Warehouse, Hive, and all relational database sources.

You can choose one or more of the following patterns to identify the tables and views:

- <name>%. Includes tables and views that start with <name>.
  For example, A% includes all the tables and views that start with the character A.
- %<name>. Includes tables and views that end with <name>.
  For example, %id includes all the tables and views that end with the text 'id'.
- %<name>%. Includes tables and views that contain <name>.
  For example, %DB% includes all tables and views that contain the keyword "DB".
- <name>. Includes tables and views that match <name>.
  For example, HR_id filter includes any table or view in the resource that matches HR_id.

To exclude tables and views from the scan, prefix the term NOT before the name or pattern. For example, NOT %ID; NOT 'ID%'. Use semicolons (;) to separate the table names and view names.

Troubleshooting

What happens if I include a table in the Source Metadata Filter that depends on an external object?

When a filtered table or view depends on an external object such as a database, the resource scanner extracts the metadata of the external object even when the external object does not meet the filter conditions because the filter does not apply to the dependent external object.

How do I know which tables and views were excluded in the profile scan?

By default, the log files do not list all the tables and views that were excluded in a scan. To view the list of excluded tables and views, enable the *Debug* log level and run the scan. The log files list all the excluded tables and views in the scan.

CHAPTER 2

Configuring Cloud Resources

This chapter includes the following topics:

- Overview of Cloud Resources, 18
- Amazon Redshift, 18
- Amazon S3, 22
- Microsoft Azure Data Lake Store, 26
- Azure Microsoft SQL Data Warehouse, 31
- Azure Microsoft SQL Server, 34
- Google BigQuery, 38
- Microsoft Azure Blob Storage, 42
- Salesforce, 45
- Workday, 48
- AWS Glue, 50
- Snowflake, 53
- Google Cloud Storage, 53

Overview of Cloud Resources

The Cloud resource group includes resources related to cloud. When you create and run a cloud resource, the cloud scanner extracts and migrates the metadata from the associated resource to the catalog. You can view the migrated results in Enterprise Data Catalog.

Amazon Redshift

Amazon Redshift is an Internet hosting service and data warehouse product. Amazon Redshift is part of Amazon Web Services, the cloud-computing platform offered by Amazon.

Objects Extracted

- Tables
- Views

Permissions to Configure the Resource

If you create a new user, ensure that you configure `read` permission on the Amazon Redshift data source for the user account.

Prerequisites

Obtain JDBC driver file

Verify that you are using the correct version of the Amazon Redshift JDBC driver:

- For Enterprise Data Catalog versions 10.4.1.3 and later, use version 2.0 of the Amazon Redshift JDBC driver.
- For Enterprise Data Catalog versions prior to 10.4.1.3, use version 1.0 of the Amazon Redshift JDBC driver.

Copy the files to the `RedShiftScanner` directory in the `<INFA_HOME>/services/CatalogService/ScannerJars/externalDependencies` directory. You need not recycle the Catalog Service.

Note: Verify that ANTLR version 4.8.1 is included in the driver JAR file.

Alternatively, you can perform the following steps to complete the prerequisites:

1. Verify that you are using the correct version of the Amazon Redshift JDBC driver:
   - For Enterprise Data Catalog versions 10.4.1.3 and later, use version 2.0 of the Amazon Redshift JDBC driver.
   - For Enterprise Data Catalog versions prior to 10.4.1.3, use version 1.0 of the Amazon Redshift JDBC driver.

   Note: Verify that ANTLR version 4.8.1 is included in the driver JAR file.
2. Include the `<Informatica Installation directory>/externaljdbcjars/<redshift-jdbc version>.jar` file in the `redshiftJars.zip` file.
3. Copy the `redshiftJars.zip` file to the `<INFA_HOME>/services/CatalogService/ScannerBinaries` directory.
4. Open the `<INFA_HOME>/services/CatalogService/ScannerBinaries/CustomDeployer/scannerDeployer.xml` file and add the following lines in the file:

   ```
   <ExecutionContextProperty isLocationProperty="true"
   dependencyToUnpack="redshiftJars.zip">
   <PropertyName>RedShiftScanner_DriverLocation</PropertyName>
   <PropertyValue>scanner_miti/RedShift/Drivers</PropertyValue> </
   ExecutionContextProperty
   ```
5. Save the `scannerDeployer.xml` file.
6. If you want to enable profiling for the Amazon Redshift resource, copy the downloaded Amazon Redshift JDBC driver file to the `<INFA_HOME>/connectors/thirdparty/informatica.amazonredshift/common/` directory.
7. Recycle the Catalog Service.

Update the JDBC driver file

To replace the obsolete JDBC driver jar with latest driver jar, perform the following steps:

1. Open the `<INFA_HOME>/services/CatalogService/ScannerBinaries/` directory.
2. Replace the `redshift-jdbc41` driver jar with `redshift-jdbc42` in the `redshiftJars.zip` file.
3. Recycle the Catalog Service.

Basic Information

The **General** tab includes the following basic information about the resource:

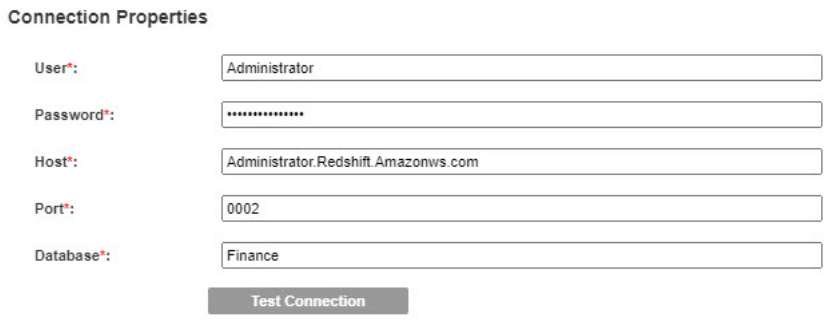
Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
User	The user name used to access the database.
Password	The password associated with the user name.
Host	Host name or IP address of Amazon Redshift service.
Port	Amazon Redshift server port number. Default is 5439.
Database	The name of the database instance.

The following image shows sample connection properties on the **General** tab:



The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Source Metadata	Extracts metadata from the data source.
Import System Objects	Select this option to specify that the system objects must be imported.
Schema	Click **Select...** to specify the Amazon Redshift schemas that you want to import. You can use one of the following options from the **Select Schema** dialog box to import the schemas: - Select from List: Use this option to select the required schemas from a list of available schemas. - Select using regex: Provide an SQL regular expression to select schemas that match the expression.
S3 Bucket Name	Provide a valid Amazon S3 bucket name for the Amazon Redshift data source. You must provide this value if you want to enable profiling for Amazon Redshift. If you do not want to enable profiling, retain the default value. Bucket name should use the access key or private key specified in DIS connection.
Case Sensitive	Specifies that the resource is configured for case insensitivity. Select one of the following values: - True. Select this check box to specify that the resource is configured as case sensitive. - False. Clear this check box to specify that the resource is configured as case insensitive. The default value is False.
Memory	The memory required to run the scanner job. Select one of the following values based on the data set size imported: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure the scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of a scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. The default value is 1.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.
Agent Options	Specify the Enterprise Data Catalog Agent options to run the scanner job.

You can enable data discovery for an Amazon Redshift resource. For more information, see the Enable Data Discovery topic.

You can enable composite data domain discovery for an Amazon Redshift resource. For more information, see the Composite Data Domain Discovery topic.

Amazon S3

Amazon S3 is a simple storage service offered by Amazon Web Services (AWS) that provides object storage through a web service interface.

Objects Extracted

Enterprise Data Catalog extracts files from Amazon S3 data source. You can also extract metadata from an Amazon S3 compatible storage such as Scality RING.

Note: In Enterprise Data Catalog version 10.4.1.2 or later, you can use the Amazon S3 resource to connect to an Amazon S3 data source by using a temporary session token.

Permissions to Configure the Resource

Configure the `read` permission on the Amazon S3 data source for the user account that you use to access the data source. Configure the access permission for the user account if you use a user account that is different from the user account used to create the Amazon S3 data source.

Prerequisites

Before you use the Amazon S3 resource, perform the following steps to run the certificates-importer.jar utility to import the required root certificates:

1. Disable Catalog Service.
2. Navigate to the directory where the utility is located: `<Informatica installation directory>/services/CatalogService/ScannerBinaries/certificates-importer`
3. From the `certificates-importer` directory, download the following JAR file: `certificates-importer.jar`
4. In the command prompt, run the following command: `java -jar certificates-importer.jar`
5. Provide the following arguments:
   a. Path to the directory that contains certificates.
   b. Path to the Informatica truststore.
   c. Your keystore password.
   d. Optionally: `-f`
   Use this argument to if you want the utility to import all certificates without your interaction.
6. Enable Catalog Service.

Supported File Types

The Amazon S3 resource enables you to extract metadata from structured, unstructured, and extended unstructured files.

The resource supports the following structured file types:

- AVRO files
- Delimited files
- Text files
- JSON files
- Parquet files
- XML files

The resource supports the following unstructured file types:

- Apple files
- Compressed files
- Email
- Microsoft Excel
- Microsoft PowerPoint
- Microsoft Word
- Apache OpenOffice files
- PDF files
- Web pages.

Assign `read` permission to the files to extract metadata.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog service or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
S3 Supported Type	Indicates if the data source is an Amazon S3 compatible storage such as Scality RING.
REST Endpoint URL	The REST end point URL of the Amazon S3 compatible storage to extract metadata.
Amazon Web Services Bucket URL	Amazon Web Services URL to access a bucket.
Are Credentials Temporary	Enables the resource to connect and extract metadata from an Amazon S3 data source by using temporary credentials.
AWS Access Key ID	Amazon Web Services access key ID to sign requests that you send to Amazon Web Services. **Note:** Specify the Amazon S3 compatible storage access key ID to sign requests that you send to the Amazon S3 compatible storage.
AWS Secret Access Key	Amazon Web Services secret access key to sign requests that you send to Amazon Web Services. **Note:** Specify the Amazon S3 compatible storage secret access key to sign requests that you send to the Amazon S3 compatible storage.

Property	Description
AWS Session Token	The temporary session token to connect to the Amazon S3 data source.
Amazon Web Services Bucket Name	Amazon Web Services bucket name that Enterprise Data Catalog needs to scan.
Source Directory	The source directory from where metadata must be extracted.

The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Source Metadata	Extracts metadata from the data source.
File Types	Select any or all of the following file types from which you want to extract metadata: - All. Use this option to specify if you want to extract metadata from all file types. - Select. Use this option to specify that you want to extract metadata from specific file types. Perform the following steps to specify the file types: 1. Click **Select**. The **Select Specific File Types** dialog box appears. 2. Select the required files from the following options: - Extended unstructured formats. Use this option to extract metadata from file types such as audio files, video files, image files, and ebooks. - Structured file types. Use this option to extract metadata from file types, such as Avro, Parquet, JSON, XML, text, and delimited files. - Unstructured file types. Use this option to extract metadata from file types such as Microsoft Excel, Microsoft PowerPoint, Microsoft Word, web pages, compressed files, emails, and PDF. 3. Click Select. **Note:** You can select **Specific File Types** option in the dialog box to select files under all the categories.
Other File Types	Extracts basic file metadata such as, file size, path, and time stamp, from file types not present in the **File Types** property.
Treat Files Without Extension As	Select one of the following options to identify files without an extension: - None - Avro - Parquet
Enter File Delimiter	Specify the file delimiter if the file from which you extract metadata uses a delimiter other than the following list of delimiters: - Comma (,) - Horizontal tab (\t) - Semicolon (;) - Colon (:) - Pipe symbol (|) Verify that you enclose the delimiter in single quotes. For example, `'$'`. Use a comma to separate multiple delimiters. For example, `'$','%','&'`

Property	Description
First Level Directory	Specify a directory or a list of directories under the source directory. If you leave this option blank, Enterprise Data Catalog imports all the files from the specified source directory. To specify a directory or a list of directories, you can perform the following steps: 1. Click **Select...**. The **Select First Level Directory** dialog box appears. 2. Use one of the following options to select the required directories: - **Select from list**: select the required directories from a list of directories. - **Select using regex**: provide an SQL regular expression to select schemas that match the expression. **Note:** If you want to select multiple directories, you must separate the directories with a semicolon (;).
Incremental Scan	Scans only the files that you added or modified after the last run. Discovery incremental profiling picks the file received from metadata extraction. **Note:** You can use incremental profiling only when metadata is enabled. For more information about incremental scanning, see the FAQ: What is an incremental scan? Knowledge Base article.
Recursive Scan	Recursively scans the subdirectories under the selected first-level directories. Recursive scan is required for partitioned file discovery.
Enable Partitioned File Discovery	Identifies and publishes horizontally partitioned files under the same directory and files organized in hierarchical Hive-style directory structures as a single partitioned file.
Non Strict Mode	Detects partitions in parquet files when compatible schemas are identified in the files.
Case Sensitive	Specifies that the resource is configured for case sensitivity. Select one of the following values: - True. Select this check box to specify that the resource is configured as case sensitive. - False. Clear this check box to specify that the resource is configured as case insensitive. The default value is True.
Memory	The memory required to run the scanner job. Select one of the following values based on the data set size imported: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal

Property	Description
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - -Dscannerloglevel=<DEBUG/INFO/ERROR>. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - -Dscanner.container.core=<No. of core>. Increases the core for the scanner container. The value should be a number. - -Dscanner.yarn.app.environment=<key=value>. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - -Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>. Increases the scanner container memory when pmem is enabled. Default value is 1. - -DmaxPartFilesToValidatePerTable=<number>. Validates the specified number of part files in the partitioned table. Default value is 10. - -DmaxPartFilesToValidatePerPartition=<number>. Validates the specified number of part files for each partition in the partition table. Default value is 5. - -DexcludePatterns=<comma separated regex patterns>. Excludes the files while parsing partition tables based on the regex pattern. By default, file names that start with a period and an underscore are excluded.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

You can enable data discovery for an Amazon S3 resource. For more information, see the Enable Data Discovery topic.

You can enable composite data domain discovery for an Amazon S3 resource. For more information, see the Composite Data Domain Discovery topic.

Microsoft Azure Data Lake Store

Microsoft Azure Data Lake is a scalable data storage and analytics service. The service is hosted on Azure. Configure an Azure Data Lake Store (ADLS) resource type to extract metadata from the data lake store.

When you create a Azure Data Lake Store resource, you can access the files and folders in the following Azure storage products:

Azure Data Lake Store or Data Lake Storage Gen1

To access this repository, Enterprise Data Catalog uses service-to-service or OAuth 2.0 authentication. To use the OAuth 2.0 authentication, you must create an Azure Active Directory (AD) application, and use the client ID and client key from the application for authentication. Enterprise Data Catalog uses SDK to access the repository contents.

Azure Data Lake Storage Gen2

Azure Blob storage supports Azure Data Lake Storage Gen2. This is a hierarchical file system. When you create a Azure Data Lake Store resource and choose the **Azure Data Lake Storage Gen2** option, you need to enter the user account ID and one of the keys provided in the **Access keys** section. In the Azure portal, you can view the two keys that are generated for each Azure Data Lake Storage Gen2 storage account in the **Settings** > **Access keys** section. To access the files and folders in this hierarchal file system, Enterprise Data Catalog uses REST APIs. Azure uses Shared Key authorization to authenticate the requests. In Enterprise Data Catalog, access and runtime is two times faster for *Azure Data Lake Storage Gen2* as compared to *Data Lake Storage Gen1* storage.

Objects Extracted

- File
- Folder
- Container

Permissions to Configure the Resource

If you create a new user, ensure that you configure `read` permission on the data source for the new user account.

Supported File Types

The Microsoft Azure Data Lake Store resource enables you to extract metadata from structured, unstructured, and extended unstructured files.

The structured files supported are:

- AVRO files
- Delimited files
- Text files
- JSON files
- Parquet files
- XML files

The unstructured files supported are:

- Apple files
- Compressed files
- Email

The extended unstructured files are:

- VB files
- ASP files
- TIF files
- LOG files
- CSS files
- ASPX files
- DLL files
- GIF files
- SQL files

Assign `read` and `write` permissions to the files to extract metadata.

Prerequisites

Before you create the resource, ensure that you have met the following prerequisites:

1. Merge the certificates in `<INFA_HOME>/java/jre/lib/security/cacerts` to `<INFA_HOME>/services/shared/security/ infa_truststore.jks` file.
2. Move the `infa_truststore.jks` file to a common location accessible to all the nodes in the cluster.

3. In the HDFS configuration properties of the Ambari interface, update the `infa_truststore.jks` file path in the **ssl.client.truststore.location** property and update the `infa_truststore.jks` password in the **ssl.client.truststore.password** property.
4. Restart the Informatica Cluster Service.

   Note: Ensure that you configure the required permissions for the ADLS storage in Azure Active Directory.

Basic Information

The **General** tab includes the following basic information about the resource:

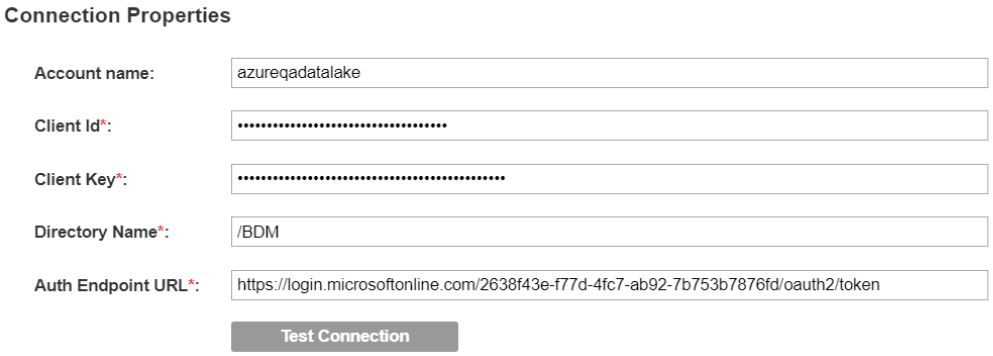
Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
Account Name	Enter the storage account name that you created in the Azure portal.
ADLS Source Type	Choose **Data Lake Store Gen 1** or **Data Lake Store Gen 2** option.
Client Id	Enter the client ID to connect to the Microsoft Azure Data Lake Store. Use the value listed for the application ID in the Azure portal. This option appears when you choose the **Data Lake Store Gen 1** option as the ADLS Source Type.
Client Key	Enter the client key to connect to the Microsoft Azure Data Lake Store. Use the Azure Active Directory application key value in the Azure portal as the client key. This option appears when you choose the **Data Lake Store Gen 1** option as the ADLS Source Type.
Directory Name	Directory name of the Azure Data Lake Store.
Auth EndPoint URL	The OAuth 2.0 token endpoint URL in the Azure portal. This option appears when you choose the **Data Lake Store Gen 1** option as the ADLS Source Type.
Storage Account Key	Enter key1 or key2 as the storage account key. Navigate to the **Settings** > **Access keys** section in Azure portal to view the storage account keys. This option appears when you choose the **Data Lake Store Gen 2** option as the ADLS Source Type.

The following image shows sample connection properties on the **General** tab:



The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Source Metadata	Extracts metadata from the data source.
File Types	Select any or all of the following file types from which you want to extract metadata: - All. Use this option to specify if you want to extract metadata from all file types. - Select. Use this option to specify that you want to extract metadata from specific file types. Perform the following steps to specify the file types: 1. Click **Select**. The **Select Specific File Types** dialog box appears. 2. Select the required files from the following options: - Extended unstructured formats. Use this option to extract metadata from file types such as audio files, video files, image files, and ebooks. - Structured file types. Use this option to extract metadata from file types, such as Avro, Parquet, JSON, XML, text, and delimited files. - Unstructured file types. Use this option to extract metadata from file types such as Microsoft Excel, Microsoft PowerPoint, Microsoft Word, web pages, compressed files, emails, and PDF. 3. Click Select. **Note:** You can select **Specific File Types** option in the dialog box to select files under all the categories.
Other File Types	Extract basic file metadata such as, file size, path, and time stamp, from file types not present in the **File Types** property.
Treat Files Without Extension As	Select one of the following options to identify files without an extension: - None - Avro - Parquet
Enter File Delimiter	Specify the file delimiter if the file from which you extract metadata uses a delimiter other than the following list of delimiters: - Comma (,) - Horizontal tab (\t) - Semicolon (;) - Colon (:) - Pipe symbol (|) Verify that you enclose the delimiter in single quotes. For example, `'$'`. Use a comma to separate multiple delimiters. For example, `'$','%','&'`

Property	Description
First Level Directory	Specify a directory or a list of directories under the source directory. If you leave this option blank, Enterprise Data Catalog imports all the files from the specified source directory. To specify a directory or a list of directories, you can perform the following steps: 1. Click **Select...**. The **Select First Level Directory** dialog box appears. 2. Use one of the following options to select the required directories: - **Select from list**: select the required directories from a list of directories. - **Select using regex**: provide an SQL regular expression to select schemas that match the expression. **Note:** If you want to select multiple directories, you must separate the directories with a semicolon (;).
Recursive Scan	Recursively scans the subdirectories under the selected first-level directories. Recursive scan is required for partitioned file discovery.
Enable Partitioned File Discovery	Identifies and publishes horizontally partitioned files under the same directory and files organized in hierarchical Hive-style directory structures as a single partitioned file.
Non Strict Mode	Detects partitions in parquet files when compatible schemas are identified in the files.
Case Sensitive	Specifies that the resource is configured for case sensitivity. Select one of the following values: - True. Select this check box to specify that the resource is configured as case sensitive. - False. Clear this check box to specify that the resource is configured as case insensitive. The default value is True.
Memory	The memory required to run the scanner job. Select one of the following values based on the data set size imported: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1. - `-DmaxPartFilesToValidatePerTable=<number>`. Validates the specified number of part files in the partitioned table. Default value is 10. - `-DmaxPartFilesToValidatePerPartition=<number>`. Validates the specified number of part files for each partition in the partition table. Default value is 5. - `-DexcludePatterns=<comma separated regex patterns>`. Excludes the files while parsing partition tables based on the regex pattern. By default, file names that start with a period and an underscore are excluded.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

You can enable data discovery for an Azure Data Lake Store. For more information, see the Enable Data Discovery topic. You can enable composite data domain discovery for an Azure Data Lake Store. For more information, see the Composite Data Domain Discovery topic.

Azure Microsoft SQL Data Warehouse

Azure Microsoft SQL Data Warehouse is a fully managed cloud data warehouse for enterprises.

Objects Extracted

- Tables
- Views
- Extended stored procedure
- Table (user-defined)
- Synonym
- Functions
- UNIQUE constraint
- Triggers
- Procedures
- Index
- SQL table-valued-function
- Users
- Partition Scheme
- Partition Function
- XML Schema Collections
- User defined Data type
- Rules
- Defaults
- Constraints
- SQL_INLINE_TABLE_VALUED_FUNCTION
- SQL_SCALAR_FUNCTION
- SEQUENCE
- Database
- Schema
- Stored procedures

Permissions to Configure the Resource

Before you create an Azure Microsoft SQL resource, configure the permissions for the Azure Microsoft SQL Server database user account that you use to connect to Azure Microsoft SQL Server. Enterprise Data Catalog uses SQL Server authentication to connect to the Azure Microsoft SQL Server database. The user account that you use to connect to Azure Microsoft SQL Server must be an SQL Server login account. Configure the VIEW DEFINITION permission for the user account. Configuring this permission lists all the schemas from where the user can load metadata.

Prerequisites

Verify that you configure the VIEW DEFINITION and CONNECT permissions for the Microsoft SQL Server database user. You must verify that you configure `sqljdbc_auth.dll` in the PATH environment variable. The version of the .dll file must match the version of sqljdbc4.jar that you use.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

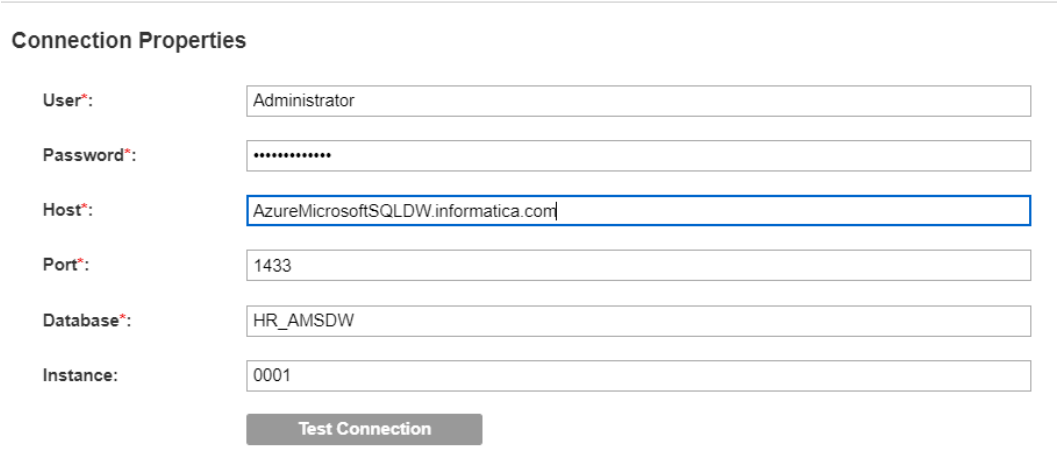
Resource Connection Properties

The following tables list the properties that you need to configure to add an Azure Microsoft SQL Data Warehouse resource:

The **General** tab includes the following properties:

Property	Description
User	The user name that you need to specify to connect to the Microsoft SQL Server database. If you use a Microsoft SQL bridge to connect to the database, and leave the user name empty, by default, the integrated security signature is used to connect to the database. An integrated security uses the following signature for connection: jdbc:sqlserver://;integratedSecurity=true instead of the jdbc:sqlserver://;user=userid;password=userpassword signature.
Password	The password for the database user.
Host	Host name or IP address of the machine where Microsoft SQL Server is running.
Port	The port number of the Microsoft SQL Server database engine service. Default is 1433. It is recommended that you specify the port number when you connect using the **Instance** property. If you specify both the **Port** and the **Instance** properties, Enterprise Data Catalog uses the **Port** property.
Database	The name of the database from which you want to import metadata. Enterprise Data Catalog imports the tables and schemas from the database.
Instance	Optional. The instance name of the Microsoft SQL Server. You can alternatively specify the port number of the instance. It is recommended that you specify the port number of the instance.

The following image shows sample connection properties on the **General** tab:



The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Source Metadata	Extracts metadata from the data source.
Enable Source Metadata	Select the option to enable and configure the resource to extract metadata.
Schema	Optional. Specify the schema that you want to import. Use semicolons (;) to separate multiple schemas. If you do not specify schemas, Enterprise Data Catalog imports all schemas.
Source Metadata Filter	You can include or exclude tables and views from the resource run. Use semicolons (;) to separate the table names and view names. For more information about the filter field, see "Source Metadata and Data Profile Filter" on page 16.
Azure Blob Container Name	Name of the Azure storage blob container.
Case Sensitive	Specifies that the resource is configured for case insensitivity. Select one of the following values: - True. Select this check box to specify that the resource is configured as case sensitive. - False. Clear this check box to specify that the resource is configured as case insensitive. Default value is False.
Memory	The memory required to run the scanner job. Select one of the following values based on the imported data set size: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal

Property	Description
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - -Dscannerloglevel=<DEBUG/INFO/ERROR>. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - -Dscanner.container.core=<No. of core>. Increases the core for the scanner container. The value should be a number. - -Dscanner.yarn.app.environment=<key=value>. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - -Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>. Increases the scanner container memory when pmem is enabled. Default value is 1.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

You can enable data discovery for an Azure Microsoft SQL Data Warehouse resource. For more information, see the Enable Data Discovery topic.

You can enable composite data domain discovery for an Azure Microsoft SQL Data Warehouse resource. For more information, see the Composite Data Domain Discovery topic.

Azure Microsoft SQL Server

Azure Microsoft Azure SQL Database is a managed cloud database.

Objects Extracted

- Tables
- Views
- Extended stored procedure
- Table (user-defined)
- Synonym
- UNIQUE constraint
- Triggers
- Procedures
- Index
- SQL table-valued-function
- Users
- Partition Scheme
- Partition Function
- XML Schema Collections
- User defined Data type
- Rules
- Defaults

- Constraints
- SQL_INLINE_TABLE_VALUED_FUNCTION
- SQL_SCALAR_FUNCTION
- SEQUENCE
- Database
- Schema
- Stored procedures
- Functions

Permissions to Configure the Resource

Before you create an Azure Microsoft SQL resource, configure the permissions for the Azure Microsoft SQL Server database user account that you use to connect to Azure Microsoft SQL Server. Enterprise Data Catalog uses SQL Server authentication to connect to the Azure Microsoft SQL Server database. The user account that you use to connect to Azure Microsoft SQL Server must be an SQL Server login account. Configure the VIEW DEFINITION permission for the user account. Configuring this permission lists all the schemas from where the user can load metadata.

Prerequisites

Make sure that you configure the VIEW DEFINITION and CONNECT permissions for the Microsoft SQL Server database user. You must also make sure that you configure `sqljdbc_auth.dll` in the PATH environment variable. The version of the .dll file must match the version of sqljdbc4.jar that you use.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

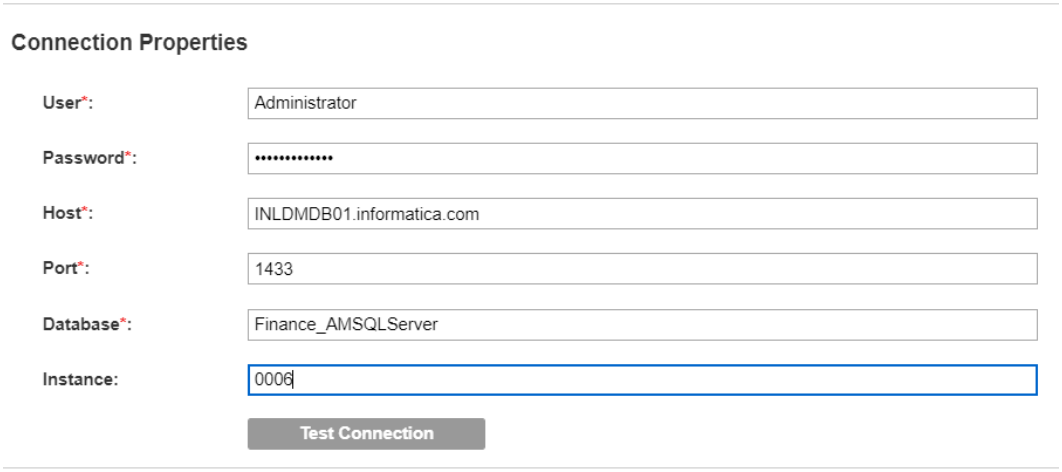
Resource Connection Properties

The following tables list the properties that you need to configure to add an Azure Microsoft SQL Server resource.

The **General** tab includes the following properties:

Property	Description
User	The username that you need to specify to connect to the Microsoft SQL Server database. If you use a Microsoft SQL bridge to connect to the database and leave the username empty, Enterprise Data Catalog uses the integrated security signature to connect to the database. An integrated security uses the following signature for connection: jdbc:sqlserver://;integratedSecurity=true instead of the jdbc:sqlserver://;user=userid;password=userpassword signature.
Password	The password that you use for the database user.
Host	Hostname or IP address of the machine where Microsoft SQL Server is running.
Port	The port number of the Microsoft SQL server database engine service. Default is 1433. It is recommended that you specify the port number when you connect using the **Instance** property. If you specify both the **Port** and the **Instance** properties, Enterprise Data Catalog uses the **Port** property.
Database	The name of the database from which you want to import metadata. Enterprise Data Catalog imports the tables and schemas from the database.
Instance	Optional. The instance name of the Microsoft SQL Server. You can alternatively specify the port number of the instance. It is recommended that you specify the port number of the instance.

The following image shows sample connection properties on the **General** tab:



The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Source Metadata	Select to extract metadata from the data source.
Import system objects	Optional. Select this option to import system objects. By default, Enterprise Data Catalog does not import system objects.

Property	Description
Schema	(Optional) Click **Select** to specify the schemas that you want to import. You can use one of the following options from the Select Schema dialog box to import the schemas: - Select from List. Select the required schemas from a list of available schemas. - Select using regex. Provide an SQL regular expression to select schemas that match the expression. If you do not specify the schemas, Enterprise Data Catalog imports all schemas.
Source Metadata Filter	You can include or exclude tables and views from the resource run. Use semicolons (;) to separate the table names and view names.
Import stored procedures	Select this property to import stored procedures. Enterprise Data Catalog does not import stored procedures by default.
Case Sensitive	Specifies that the resource is configured for case insensitivity. Select one of the following values: - True. Select this check box to specify that the resource is configured as case sensitive. - False. Clear this check box to specify that the resource is configured as case insensitive. The default value is False.
Memory	The memory required to run the scanner job. Select one of the following values based on the imported data set size: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

You can enable data discovery for an Azure Microsoft SQL Server resource. For more information, see the Enable Data Discovery topic.

You can enable composite data domain discovery for an Azure Microsoft SQL Server resource. For more information, see the Composite Data Domain Discovery topic.

Google BigQuery

You can use the Google BigQuery resource to collect metadata from the assets in Google BigQuery.

Objects Extracted

The Google BigQuery resource extracts metadata from the following assets in a Google BigQuery data source:

- Project
- Dataset
- Table
- View

Permissions to Configure the Resource

Make sure that you perform the following steps before you configure the Google BigQuery resource:

- Assign the `dataViewer` role to the service account that you use to access the Google Cloud Platform project.
- Configure `bigquery.tables.list` and the `bigquery.jobs.get` permissions for the service account that you use to access the Google Cloud Platform project.

Connect to a Google BigQuery Data Source Enabled for SSL

By default, all cloud resources are SSL-enabled. To connect to a Google BigQuery data source enabled for SSL, perform the following steps:

1. Download the Google BigQuery SSL certificates using a web browser.

   Note: Make sure that you import the Google Trust Services certificate in the certification path.
2. Copy the certificates to the `<INFA_HOME>/services/shared/security/` directory.
3. Go to the `<INFA_HOME>/source/java/jre/bin` directory and run the following `keytool` command to import each copied certificate as a trusted certificate in the Informatica domain keystore:

   ```
   keytool -import -file <INFA_HOME>/services/shared/security/<certificate>.cer -alias
   <alias name> -keystore <INFA_HOME>/services/shared/security/infa_truststore.jks -
   storepass <Informatica domain keystore password>
   ```

You can also import certificates from the Informatica domain truststore. If you are unable to connect to the truststore, run the below command:

```
$INFAINSTALL/java/jre/bin/keytool -importkeystore -srckeystore $INFAINSTALL/java/jre/lib/
security/cacerts -destkeystore $INFAINSTALL/services/shared/security/infa_truststore.jks
-srcstorepass changeit -deststorepass pass2038@infaSSL
```

For custom certificates, you can use the com.infa.ldm.certificates.importer.jar utility packaged here:
`<INFA_HOME>/services/CatalogService/ScannerBinaries/certificates-importer/certificates-importer.jar`

Input the following arguments to the utility jar file:

1. Directory containing certificates
2. Path to the truststore
3. Store pass value
4. -f

   Note: -f is an optional argument. You can provide this argument if you want to overwrite certificates that already exist in the truststore.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
Project ID	Name of the Google Cloud Platform project that you want to access.
Private Key	The private key associated with the service account.
Client Email	The client email address associated with the service account.

The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Source Metadata	Extracts metadata from the data source.
Scan Hidden Datasets	Extracts metadata from hidden and anonymous datasets.
Dataset	Select the datasets that you want to use to import metadata from Google BigQuery tables in the project. Default is all datasets.
Source Metadata Filter	You can include or exclude tables and views from the resource run. Use semicolons (;) to separate the table names and view names. For more information about the filter field, see "Source Metadata and Data Profile Filter" on page 16.
Case Sensitive	Specifies that the resource is configured for case sensitivity. Select one of the following values: - True. Select this check box to specify that the resource is configured as case sensitive. - False. Clear this check box to specify that the resource is configured as case insensitive. The default value is True.

Property	Description
Memory	Specifies the memory required to run the scanner job. Select one of the following values based on the data set size that you plan to import into the catalog: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. The key-value pair that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. The default value is 1.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

Identity and Access Management (IAM) Authentication for Google BigQuery

Assign users the following roles to run the profile in hybrid connection mode:

- BigQuery Data Editor
- BigQuery Job User

As Google Cloud Storage is used for staging Google BigQuery tables in the hybrid connection mode, you must also assign users with the following storage roles to run the profile:

- storage.objects.create
- storage.objects.delete
- storage.objects.get

Sampling Options

You can perform sampling on the following types of tables:

- Native tables
- Clustered tables
- Partitioned tables with require_partition_filter option set to false
- External tables

  Note: You can perform sampling only on external tables created on top of Google Cloud Storage.

You should use the hybrid connection mode for profiling and extraction of metadata from the Google BigQuery data source. The hybrid connection mode uses standard SQL language to query Google BigQuery. Google BigQuery suggests to use standard SQL language over legacy SQL language. The simple connection mode uses legacy SQL language and must be avoided. Views that are standard SQL compliant are also supported in the hybrid connection mode.

The following table describes the sampling options to determine the number of rows to run the profile job on the Google BigQuery resource:

Sampling Option	Description	Pushdown to Google BigQuery
First N rows	Runs the profile on N number of rows in the resource. In the **Number of First N Sampling Rows** field that appears, enter the number of rows to run the profile on. **Note:** For optimal performance, consider using this sampling option.	Yes Sample pushdown query: `LIMIT <N>`
Random Percentage	Runs the profile on a percentage of data blocks in the data object. In the **Random Percentage** field that appears, enter the number of data blocks to run the profile on. Google BigQuery tables are organized into data blocks. The `TABLESAMPLE` clause works by randomly selecting a percentage of data blocks from the table and reading all of the rows in the selected blocks. The sampling granularity is limited by the number of data blocks. Google BigQuery charges you for reading the data that is sampled. Google BigQuery does not cache the results of a query that includes a `TABLESAMPLE` clause, so each execution incurs the cost of reading the data from storage. For more information on table sampling in Google BigQuery, see Table Sampling in Google BigQuery.	Yes Sample pushdown query: `TABLESAMPLE SYSTEM (<N> PERCENT)`
Limit N rows	Runs the profile based on the number of rows in the data object. In the **Number of Rows to Limit** field that appears, enter the number of rows to run the profile on. The Data Integration Service reads all rows and then runs a sampling algorithm.	No
Auto Random rows	Runs the profile on a random sample of rows. Enterprise Data Catalog computes the number of random rows based on the number of source rows. The Data Integration Service reads all rows and then runs a sampling algorithm.	No
Random N rows	Runs the profile on the configured number of random rows. In the **Random Sampling Rows** field that appears, enter the number of rows that you want to run the profile on. The Data Integration Service reads all rows and then runs a sampling algorithm.	No
All rows	Runs the profile on all the rows in the data source.	No

Note: The Google BigQuery resource does not support the following features:

- During profiling, the Geography, and Byte data type columns are skipped.
- The Google BigQuery resource does not read multiple records when you use the legacy SQL query.

Microsoft Azure Blob Storage

Configure a Microsoft Azure Blob Storage resource type to extract metadata from the Microsoft Azure Blob Storage.

Objects Extracted

Enterprise Data Catalog extracts only files from Microsoft Azure Blob Storage source.

Permissions to Configure the Resource

Configure `read` permission on the Microsoft Azure Blob Storage data source for the user account that you use to access the data source.

Supported File Types

The Microsoft Azure Blob Storage resource enables you to extract metadata from structured, unstructured, and extended unstructured files.

The structured files supported are:

- AVRO files
- Delimited files
- Text files
- JSON files
- Parquet files
- XML files

The unstructured files supported are:

- Apple files
- Compressed files
- Email

The extended unstructured files are:

- VB files
- ASP files
- TIF files
- LOG files
- CSS files
- ASPX files
- DLL files
- GIF files
- SQL files

Assign `read` and `write` permissions to the files to extract metadata.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

Provide the **Account Key** for Microsoft Azure Blob Storage in Informatica Administrator. Use the value listed for **key 1** under **Access Keys** in the Azure portal.

The **General** tab includes the following properties:

Property	Description
Blob Endpoint URL	Microsoft Azure Blob Storage URL to access a container. Use the value listed for the Primary BLOB Service endpoint URL in the Azure portal.
Account Name	Name of the Microsoft Azure Blob Storage.
Container Name	Name of the container that contains the blobs.
Source Directory	The source directory from where metadata needs to be extracted.
Shared Access Signature Token	A token that provides access to the Microsoft Azure Blob Storage resource. Use the value for the Shared Access Signature in the Azure portal.

The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Source Metadata	Extracts metadata from the data source.
Blob Prefix	Use this option to sort the blobs based on the prefix of the Microsoft Azure Blob name. This value is case sensitive.

Property	Description
File Types	Select any or all of the following file types from which you want to extract metadata: - All. Use this option to specify if you want to extract metadata from all file types. - Select. Use this option to specify that you want to extract metadata from specific file types. Perform the following steps to specify the file types: 1. Click **Select**. The **Select Specific File Types** dialog box appears. 2. Select the required files from the following options: - Extended unstructured formats. Use this option to extract metadata from file types such as audio files, video files, image files, and ebooks. - Structured file types. Use this option to extract metadata from file types, such as Avro, Parquet, JSON, XML, text, and delimited files. - Unstructured file types. Use this option to extract metadata from file types such as Microsoft Excel, Microsoft PowerPoint, Microsoft Word, web pages, compressed files, emails, and PDF. 3. Click Select. **Note:** You can select **Specific File Types** option in the dialog box to select files under all the categories.
Other File Types	Extracts basic file metadata such as, file size, path, and time stamp, from file types not present in the **File Types** property.
Treat Files Without Extension As	Select one of the following options to identify files without an extension: - None - Avro - Parquet
Enter File Delimiter	Specify the file delimiter if the file from which you extract metadata uses a delimiter other than the following list of delimiters: - Comma (,) - Horizontal tab (\t) - Semicolon (;) - Colon (:) - Pipe symbol (|) Verify that you enclose the delimiter in single quotes. For example, `'$'`. Use a comma to separate multiple delimiters. For example, `'$','%','&'`
First Level Directory	Specify a directory or a list of directories under the source directory. If you leave this option blank, Enterprise Data Catalog imports all the files from the specified source directory. To specify a directory or a list of directories, you can perform the following steps: 1. Click **Select...**. The **Select First Level Directory** dialog box appears. 2. Use one of the following options to select the required directories: - **Select from list**: select the required directories from a list of directories. - **Select using regex**: provide an SQL regular expression to select schemas that match the expression. **Note:** If you want to select multiple directories, you must separate the directories with a semicolon (;).
Include Subdirectory	Select this option to import all the files in the subdirectories under the source directory.
Case Sensitive	Specifies that the resource is configured for case sensitivity. Select one of the following values: - True. Select this check box to specify that the resource is configured as case sensitive. - False. Clear this check box to specify that the resource is configured as case insensitive. The default value is True.

Property	Description
Memory	The memory required to run the scanner job. Select one of the following values based on the data set size imported: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value must be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

You can enable data discovery for a Microsoft Azure Blob Storage resource. For more information, see the Enable Data Discovery topic. You can enable composite data domain discovery for a Microsoft Azure Blob Storage resource. For more information, see the Composite Data Domain Discovery topic.

Salesforce

Use a Salesforce connection to connect to a Salesforce object. The Salesforce connection is an application connection type.

Note: The Salesforce V2 resource has the following new features:

- Metadata extraction for Salesforce Sales Cloud and Service Cloud from the following types of input:
  - Objects
  - Views
  - Triggers
  - Asset Descriptions
- Metadata filtering
- Connection through proxy
- Salesforce API version 52
- Additional OAuth 2.0 Authentication Flow

For more information, see "Salesforce" in the *MetaDex Scanner Configuration Guide*.

Objects Extracted

Enterprise Data Catalog extracts only tables from Salesforce source.

Prerequisites

Perform the following steps to complete the prerequisites:

1. Export the Salesforce certificates that are in the DER format from the following web pages:
   - https://login.salesforce.com
   - Salesforce user page
2. Import the certificates to the `<INFA_HOME>/services/shared/security/infa_truststore.jks` file.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The following table describes the connection properties for the Salesforce resource type:

Property	Description
Authentication Type	The authentication type to connect to Salesforce.
Refresh Token	Enter the Refresh Token of Salesforce.
Consumer Key	The Consumer Key obtained from Salesforce, required to generate the Refresh Token. For more information about how to generate the Consumer Key, see the Salesforce documentation.
Consumer Secret	The Consumer Secret obtained from Salesforce, required to generate the Refresh Token. For more information about how to generate the Consumer Secret, see the Salesforce documentation.
Username	Salesforce username.
Password	Password and the Salesforce security token for the Salesforce user name. **Note:** Make sure that you provide the password and the Salesforce token as a string.
Service URL	URL of the Salesforce service that you want to access.

The following table describes the **Advanced** property for source metadata settings on the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Extracts metadata from the data source.
Case Sensitive	Specifies that the resource is configured for case insensitivity. - Select the check box to specify that the resource is configured as case sensitive. - Clear the check box to specify that the resource is configured as case insensitive. By default the resource is configured as case insensitive.
Memory	The memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1. - `-DenableSalesforceRelationships=true`. Enables you to view all the relationships for an object.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

You can enable data discovery for a Salesforce resource. For more information, see the Enable Data Discovery topic.

You can enable composite data domain discovery for a Salesforce resource. For more information, see the Composite Data Domain Discovery topic.

Configuring a Proxy Connection to Access the Salesforce Resource

You can configure a proxy connection for the resource to access the Salesforce data source. A proxy connection secures access to the data source.

For more information about configuring a proxy connection to access the Salesforce data source, see the knowledge base article *Configuring a Proxy Connection to Access the Salesforce Data Source*.

Workday

You can use the Workday resource to collect metadata from assets in Workday.

Objects Extracted

The Workday resource extracts metadata from the following assets in a Workday data source:

- Service
- Entity*
- Report
- Operation
- Data source
- Property*
- Business objects

Note: *Enterprise Data Catalog extracts the Workday Service metadata as an XSD schema. In the schema, an Entity represents the complex XML elements and the Property represents the simple XML elements.

Permissions to Configure the Resource

Make sure that you use a user account with Workday administrator privileges when you configure a Workday resource.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The following table shows the connection properties that you must configure in the **General** tab for the resource:

Property	Description
Username	Username to access the Workday tenant. A Workday tenant represents a secure instance of the Workday software.
Password	Password to access the Workday tenant.

Property	Description
Tenant	URL of the of the Workday instance that you want to access.
Workday Domain	Name of the Workday domain that includes the Workday assets that you want to access. A domain in Workday represents a collection of related assets secured using a domain security policy. For example, a domain can include assets such as reports, data sources for the report, and properties of the data sources.

The following table shows metadata extraction settings that you can configure in the **Metadata Load Settings** tab for the resource:

Property	Description
Enable Source Metadata	Extracts metadata from the data source.
Version	Version of the Workday Web Services API. Based on the version you select, the list of available Workday services changes.
Services	Workday services from which you want to extract metadata. A service in Workday represents a module that caters to a feature. For example, the feature to manage payroll related services might be provided by the Payroll module.
Import Business Objects, Data Sources, and Reports	Imports metadata related to business objects, data sources, and reports from the Workday data source.
Case Sensitive	Specifies that the resource is configured for case sensitivity. Select one of the following values: - True. Select this check box to specify that the resource is configured as case sensitive. - False. Clear this check box to specify that the resource is configured as case insensitive. The default value is True.
Memory	Specifies the memory required to run the scanner job. Select one of the following values based on the data set size that you plan to import into the catalog: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal

Property	Description
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. The key-value pair that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. The default value is 1.
Track Data Source Changes	Select the option to view metadata source change notifications in Enterprise Data Catalog.

AWS Glue

Amazon Web Services Glue is an ETL service of the Amazon Web Services ecosystem that uses data moved across different stores. Glue captures the metadata of multiple data stores that are part of the Amazon Web Services ecosystem.

Objects Extracted

The Glue resource extracts metadata from the following assets in a Glue data source:

- Databases
- Tables
- Columns
- Derived metadata from Glue Catalog. You can also view the lineage between Glue assets and derived assets.

  Note: If the derived asset is a struct data type, the Catalog displays the asset as a field along with the existing struct hierarchy. For example, a `person.json` file includes first name, last name, phone, and address information. The address includes nested fields such as city, state, and pin code. Enterprise Data Catalog extracts the person asset as a field along with the nested fields.
- Jobs
  You can extract Glue job metadata, but Enterprise Data Catalog does not parse that metadata for lineage.

Note: Athena, a query based service provided by Amazon S3 is created as a reference JDBC resource that can interact with data sources such as Tableau and others.

Permissions to Configure the Resource

To access Glue, make sure that you perform one of the following steps before you configure the resource:

- Assign the `AWSGlueServiceRole` role to the Amazon Web Services IAM user.
- Create a custom policy with the following permissions to the Glue service, and then assign the custom policy to an Amazon Web Services IAM user:

  - GetConnection

- GetConnections
- GetDatabase
- GetDatabases
- GetJob
- GetJobs
- GetPartition
- GetPartitions
- GetTable
- GetTables

Connect to a Glue Data Source Enabled for SSL

To connect to a Glue data source enabled for SSL, perform the following steps:

1. Download the Glue SSL certificates using a web browser.

   Note: Make sure that you import the Glue Trust Services certificate in the Certificates directory.
2. Copy the certificates to the `<INFA_HOME>/services/shared/security/` directory.
3. Go to the `<INFA_HOME>/source/java/jre/bin` directory and then run the following `keytool` command to import each copied certificate as a trusted certificate in to the Informatica domain keystore:

```
keytool -import -file <INFA_HOME>/services/shared/security/<certificate>.cer -alias
<alias name> -keystore <INFA_HOME>/services/shared/security/infa_truststore.jks -
storepass <Informatica domain keystore password>
```

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
Role-based Authentication	Option to use the Amazon Elastic Compute Cloud (Amazon EC2) instance profile credentials when Enterprise Data Catalog is installed on an Amazon EC2 instance.
AWS Access Key	Access Key of the Amazon Web Services account.
AWS Secret Key	Secret key of the Amazon Web Services account.
AWS Region	Amazon Web Services region from where you want to scan the Glue Catalog.

The following table describes the properties that you can configure in the **Source Metadata** section of the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Enables metadata extraction
Database Filter	Filter that enables you to include or exclude databases in the resource run. You can also specify a regular expression that represents databases you want to include or exclude.
Table Filter	Filter that enables you to enter a suitable combination of regular expression and wildcard characters to include or exclude specific assets in the resource run that match the regular expression format. You can also enter table names to include them in the resource run. Use a semicolon to separate the wildcard patterns and table names.
Enable Reference Resources	Option to extract metadata about assets that are not included in this resource, but referred to in the resource. Examples include source and target tables in PowerCenter mappings, and source tables and files from Tableau reports.
Create Athena Resources	Indicates whether or not to create an Athena data source.
Retain Unresolved Reference Assets	Option to retain unresolved reference assets in the catalog after you assign connections. Retaining unresolved reference assets help you view the complete lineage. The unresolved assets include deleted files, temporary tables, and other assets that are not present in the primary resource.
Auto Assign Connections	Indicates whether the connections must be assigned automatically.
Memory	The memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, INFO, or ERROR. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value must be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key value pair that you need to set in the Yarn environment. Use a comma to separate the multiple key value pairs. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. The default value is 1.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

Snowflake

Snowflake is an analytic data warehouse provided as Software-as-a-Service (SaaS). Snowflake provides a data warehouse that is faster, easier to use, and far more flexible than the traditional data warehouse offerings. The Snowflake data warehouse uses a SQL database engine with a unique architecture designed for the Cloud.

Deprecation Notice: Effective in 10.5.0.1, the Snowflake scanner is deprecated and Informatica intends to drop support in a future release.

Informatica recommends that you update all Snowflake scanners to leverage the Advanced Scanners framework. You do not need an additional license.

However, with an additional license for Snowflake stored procedures, you can create an advanced scanner configuration that derives lineage from the following additional functionalities:

- Stored procedures and functions, both based on SQL and JavaScript
- Pipes and stages with external storage systems

For information about the Snowflake Advanced Scanner and its functionality, see *Advanced Scanners Configuration Guide*.

Google Cloud Storage

Google Cloud Storage is an online file storage web service that stores and accesses data from the Google Cloud platform.

Objects Extracted

The Google Cloud Storage resource extracts metadata from the following assets in a Google Cloud Storage data source:

- File
- Folder

Connect to a Google Cloud Storage Data Source Enabled for SSL

To connect to a Google Cloud Storage data source enabled for SSL, perform the following steps:

1. Download the Google Cloud Storage SSL certificates using a web browser.

   Note: Make sure that you import the Google Cloud Storage Trust Services certificate in the Certificates directory.
2. Copy the certificates to the `<INFA_HOME>/services/shared/security/` directory.
3. Go to the `<INFA_HOME>/source/java/jre/bin` directory and then run the following `keytool` command to import each copied certificate as a trusted certificate in to the Informatica domain keystore:

   ```
   keytool -import -file <INFA_HOME>/services/shared/security/<certificate>.cer -alias
   <alias name> -keystore <INFA_HOME>/services/shared/security/infa_truststore.jks -
   storepass <Informatica domain keystore password>
   ```

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
Project ID	ID of the Google Cloud platform project
Private Key	Private key of the Google Cloud platform service account
Client Email	Client email ID of the Google Cloud platform service account
Bucket Name	Bucket name of the stored objects that must be extracted.
Source Directory	Path of the source directory for metadata extraction. Use slash as a suffix to the source directory path.

The following table describes the properties that you can configure in the **Source Metadata** section of the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Enables metadata extraction
File Types	Enables you to extract metadata from all the files or specific files.
Specific File Types	Select the file type for metadata extraction.
Enter File Delimiter	Specify the file delimiter. Enterprise Data Catalog supports the following delimiters: Comma (,), Colon (:), Semicolon (;), Tab (\t) and Pipe (|). Use single quotes to specify another file delimiter.
Other File Types	Extracts file metadata, such as file size, file path, and time stamp from other file types.
First Level Directory	Option to add a constraint while importing a first-level directory from the source directory. An empty string indicates that files from the source directory are imported. A file name indicates that all files from the source directory along with the files from the specified directory are imported.

Property	Description
Include Subdirectory	Option to include all the directories within the selected first-level directory while extracting metadata. If the first-level directory is empty, all the directories within the source directory are included in the metadata extraction.
Case Sensitive	Specifies that the resource is configured for case insensitivity. Select the check box to configure the resource as case sensitive. Clear the check box to configure the resource as case insensitive. By default the resource is configured as case sensitive.
Memory	The memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, INFO, or ERROR. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value must be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key value pair that you need to set in the Yarn environment. Use a comma to separate the multiple key value pairs. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. The default value is 1.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

CHAPTER 3

Configuring Data Engineering Resources

This chapter includes the following topics:

- Overview of Data Engineering Resources, 56
- Apache Atlas, 56
- HDFS, 59
- Hive, 65

Overview of Data Engineering Resources

The Data Engineering resource group includes resources related to data engineering. When you create and run a data engineering resource, the data engineering scanner extracts and migrates the metadata results from the associated resource to the catalog. You can view the migrated results in Enterprise Data Catalog.

Apache Atlas

Apache Atlas is the governance and metadata framework for Hadoop. Apache Atlas has a scalable and extensible architecture that can be plugged into many Hadoop components to manage their metadata in a central repository.

Objects Extracted

The Apache Atlas resource extracts data object links from the following data sources:

- Relational databases
- Hive
- HDFS

Note:

- Ensure that you configure resources in Enterprise Data Catalog for the data sources from which you want to extract data object links.

- The cluster must have access to the internet as the Apache Atlas resource metadata extraction requires a few binaries.

Prerequisites

If the cluster is Kerberos-enabled, perform the following steps:

1. Copy the `krb5.conf` and keytab files from the cluster to the following locations on the Informatica domain:
   - `<INFA_HOME>/services/shared/security`
   - `<INFA_HOME>/java/jre/lib/security`
2. Specify the `krb5.conf` file path in the Informatica domain and the Informatica Cluster Service nodes.
3. Add the Service Principal Name (SPN) and the keytab properties to the Data Integration Service properties.
4. Run the kinit command on the Informatica domain with the required SPN and keytab file.
5. Run the kinit command on the cluster with the required SPN and keytab file.
6. Restart the Data Integration Service and recycle the Catalog Service.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The following table describes the connection properties for the Atlas resource type:

Property	Description
URL	URL to access Apache Atlas.
Authentication	Select one of the following options to specify the authentication type configured for Apache Atlas: - Simple. Specify the following parameters: - Login. Specify the username configured to access Apache Atlas. - Password. Specify the password configured to access Apache Atlas. - Kerberos. Specify the following parameters: - Kerberos configuration file. Click **Choose** to select and upload the Kerberos configuration file used for authentication. - Kerberos Keytab file. Click **Choose** to select and upload the Kerberos keytab file used for authentication. - Principal. Specify the Kerberos principal used for authentication.

Property	Description
Entities filter	Filters the entities set that the Atlas Bridge uses to import metadata and build lineage. The filter uses Atlas basic or Advanced search syntax.
Lineage direction	Allows you to set lineage extract direction.

The following image shows sample connection properties on the **General** tab:



The following table describes the **Additional** and **Advanced** properties for source metadata settings on the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Select to extract metadata from the data source.
Auto Assign Connections	Select this option to specify that the connection must be assigned automatically.
Enable Reference Resources	Option to extract metadata about assets that are not included in this resource, but referred to in the resource. Examples include source and target tables in PowerCenter mappings, and source tables and files from Tableau reports.
Retain Unresolved Reference Assets	Option to retain unresolved reference assets in the catalog after you assign connections. Retaining unresolved reference assets help you view the complete lineage. The unresolved assets include deleted files, temporary tables, and other assets that are not present in the primary resource.
Memory	Specify the memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal

Property	Description
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Agent Options	Specify the Enterprise Data Catalog Agent options to run the scanner job. Use the following arguments to configure the parameters: - `-entities.limit <max number of entities>`. Specifies the maximum number of entities that the entities filter returns during filtering For example: `-entities.limit 500` - `-request.limit <number of entities per one request>` Specifies the number of entities that the entities filter obtains for one offset request. For example: `-request.limit 500`

HDFS

Use a HDFS resource to import metadata from CSV, XML, and JSON files.

Objects Extracted

The HDFS resource extracts metadata from files in a HDFS data source.

Permissions to Configure the Resource

Configure `read` permission on the HDFS data source for the user account that you use to access the data source.

Supported File Types

The HDFS resource enables you to extract metadata from structured, unstructured, and extended unstructured files.

The structured files supported are:

- AVRO files
- Delimited files
- Text files
- JSON files
- Parquet files
- XML files

The unstructured files supported are:

- Apple files
- Compressed files

- Email

The extended unstructured files are:

- VB files
- ASP files
- TIF files
- LOG files
- CSS files
- ASPX files
- DLL files
- GIF files
- SQL files

Assign `read` and `write` permissions to the files to extract metadata.

Prerequisites

If the domain is SSL-enabled and the cluster is Kerberos-enabled, perform the following steps:

1. Copy the `krb5.conf`, `infa_truststore.jks` and keytab files from the cluster to the following locations on the Informatica domain:
   - `<INFA_HOME>/services/shared/security`
   - `<INFA_HOME>/java/jre/lib/security`
2. Specify the `krb5.conf` file path in the Informatica domain and the Informatica Cluster Service nodes.
3. Add the Service Principal Name (SPN) and the keytab properties to the Data Integration Service properties.
4. Run the kinit command on the Informatica domain with the required SPN and keytab file.
5. Run the kinit command on the cluster with the required SPN and keytab file.
6. Restart the Data Integration Service and the Catalog Service.

Choose Hadoop Connection for Existing HDFS Resources

If you have a HDFS resource that uses the Hive engine as the run-time environment, edit the resource, choose Blaze, Spark, or Databricks as the **Run on** option, select a Hadoop connection, and run the resource to view the profile results.

Import the HDFS Resource

Perform the following steps if you use a HDFS resource:

1. If you want to enable profiling for the HDFS resource, create a cluster configuration based on the type of cluster that you use.
2. Edit the HDFS resource to assign the new HDFS connection in the Source Connection field.
3. Select **Blaze**, **Spark**, or **Databricks** as the **Run on** option.
4. Select the Hadoop connection name.
5. If the Hive data source is located in a Kerberos-enabled cluster, make sure that you perform the following steps:
   a. Use Informatica Administrator to configure the Hadoop Kerberos Service Principal Name and the Hadoop Kerberos Keytab in the properties for the Data Integration Service.

b. Set the value of Data Integration Service Hadoop Distribution Directory to /data/

c. Create the cluster configuration if the cluster configuration was not created prior to upgrade. If the cluster configuration was created prior to upgrade, refresh the cluster configuration using the Informatica Administrator or the CLI.

d. Use Informatica Administrator to append the following line to the JVM command line options for the Data Integration Service: -Djava.security.krb5.conf=/data. The JVM options are located in the Processes section of the Data Integration Service.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
Cluster Configuration Details	Select one of the following options to specify details to configure the resource: - Provide Configuration Details - Load from Configuration Archive File
Storage Type	Select one of the following options to specify the type of storage from which where you want to extract metadata: - DFS. Distributed File System - WASB. Windows Azure Blob Storage. Configure the following options if you select WASB: - Azure Storage Account URI. The fully qualified URI to access data stored in WASB. - Azure Storage Account Name. Name of the storage account. - Azure Storage Account Key. Key to access the storage account. - ABFS. Azure Blob File system. Configure the following options if you select ABFS: - Azure Storage Account URI. The fully qualified URI to access data stored in ABFS. - Azure Storage Account Name. Name of the storage account. - Azure Storage Account Key. Key to access the storage account.
Name Node URI 1	URI to the active HDFS NameNode. The active HDFS NameNode manages all the client operations in the cluster.
HA Cluster	Select **Yes** if the cluster is configured for high availability, and configure the following properties: - Name Node URI 2. URI to the secondary HDFS NameNode. The secondary HDFS NameNode stores modifications to HDFS as a log file appended to a native file system file. - HDFS Service Name. The service name configured for HDFS.

Property	Description
Distribution Type	Select one of the following Hadoop distribution types for the HDFS resource: - Cloudera - Hortonworks - IBM BigInsights - Azure HDInsight - Amazon EMR - MapR FS. Complete the prerequisites before you select MapR as a distribution type. See the *Prerequisites for MapR FS Distribution Type* section for more details. **Note:** This property is used if you select the **Load from Configuration Archive File** option to configure the resource.
MAPR Home	Specify the MapR client installation path. **Note:** This property is used if you selected **MapR FS** as the Hadoop distribution type.
User Name/User Principal	User name to connect to HDFS. Specify the Kerberos Principal if the cluster is enabled for Kerberos.
Source Directory	The source location from where metadata must be extracted.
Configuration Archive file	A zip file that contains the resource configuration properties in the following xml files: - core-site.xml - hdfs-site.xml - mapred-site.xml - yarn-site.xml **Note:** This property is used if you select the **Load from Configuration Archive File** option to configure the resource.
HDFS Transparent Encryption	Select **Yes** if transparent encryption is enabled for HDFS. Provide the fully qualified URI to the Key Management Server key provider in the **Key Management Server Provider URI** box. If you choose **Yes**, you need to import the SSL certificate into the Informatica domain `infa_truststore.jks` truststore file located in the `<INFA_HOME>/services/shared/security` location.
Kerberos Cluster	Select **Yes** if the cluster is enabled for Kerberos. If the cluster is enabled for Kerberos, provide the following details: - Hadoop RPC Protection. Select any of the following options based on the Remote Procedure Call (RPC) protection value configured for the cluster: - authentication - integrity - privacy Default is **authentication**. - HDFS Service Principal. The service principal name of HDFS service. - Keytab File. The path to the Kerberos Principal keytab file. Make sure that the keytab file is present at the specified location on Informatica domain host and cluster hosts of the Catalog Service.

For more information about configuring Azure storage access for Azure HDInsight distribution type, see the *Data Engineering Integration Guide*.

The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Source Metadata	Extracts metadata from the data source.
File Types	Select any or all of the following file types from which you want to extract metadata: - All. Use this option to specify if you want to extract metadata from all file types. - Select. Use this option to specify that you want to extract metadata from specific file types. Perform the following steps to specify the file types: 1. Click **Select**. The **Select Specific File Types** dialog box appears. 2. Select the required files from the following options: - Extended unstructured formats. Use this option to extract metadata from file types such as audio files, video files, image files, and ebooks. - Structured file types. Use this option to extract metadata from file types such as JSON, Avro, Parquet, XML, text, and delimited files. - Unstructured file types. Use this option to extract metadata from file types such as Microsoft Excel, Microsoft PowerPoint, Microsoft Word, web pages, compressed files, emails, and PDF. 3. Click Select. **Note:** You can select **Specific File Types** option in the dialog box to select files under all the categories.
Other File Types	Extracts basic file metadata such as, file size, path, and timestamp, from file types that are not listed in the **File Types** property.
Treat Files Without Extension As	Select one of the following options to identify files without an extension: - None - Avro - Parquet
Enter File Delimiter	Specify the file delimiter if the file from which you extract metadata uses a delimiter other than the following list of delimiters: - Comma (,) - Horizontal tab (\t) - Semicolon (;) - Colon (:) - Pipe symbol (|) Make sure that you enclose the delimiter in single quotes. For example, `'$'`. Use a comma to separate multiple delimiters. For example, `'$','%','&'`
First Level Directory	Specifies that all the directories must be selected. If you want specific directories to be selected, use the **Select Directory** option. This option is disabled if you had selected the **Include Subdirectories** option on the **General** tab.
Include Subdirectory	Type the required directories in the text box or click **Select...** to choose the required directories. This option is disabled if you had selected the **Include Subdirectories** option on the **General** tab or the **Select all Directories** option listed above.
Non Strict Mode	Detects partitions in parquet files when compatible schemas are identified in the files.
Case Sensitive	Identifies whether the resource is configured for case sensitivity. Select one of the following values: - True. Select this check box to specify that the resource is configured as case sensitive. - False. Clear this check box to specify that the resource is configured as case insensitive. The default value is True.

Property	Description
Memory	The memory required to run the scanner job. Select one of the following values based on the data set size imported: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1. - `Djava.security.auth.login.config={MAPR_HOME}/conf/mapr.login.conf`. Extracts metadata from a MapR Hadoop distribution that accesses data from a HDFS data source. The MapR client installed on the Informatica domain and the cluster must be of version 6.1.0 and the MapR Hadoop distributed version must be MEP 6.x.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

You can enable data discovery for an HDFS resource. For more information about enabling data discovery, see the Enable Data Discovery topic.
You can enable composite data domain discovery for an HDFS resource. For more information about enabling composite data domain discovery, see the Composite Data Domain Discovery topic.

Prerequisites for MapR FS Distribution Type

Complete the following prerequisites before you select MapR FS as a distribution type:

- Verify that the MapR client is installed on the Informatica Cluster Service nodes.
- Verify that the MapR client is installed in the same location across the Informatica Cluster Service nodes.
- Configure the MapR client to generate the mapr-clusters.conf and other files that are required to connect to the MapR cluster. Run the `/opt/mapr/server/configure.sh -N my.cluster.com -c -C mynode01:7222 -HS mynode02` command to configure the MapR client.

Hive

Apache Hive is a data warehouse software built on Apache Hadoop to provide data query and analysis. Apache Hive provides a SQL-like interface to query data stored in various databases and file systems that integrate with Hadoop.

Objects Extracted

The Hive resource extracts metadata from the following assets in a Hive data source:

- Tables
- Views
- Database
- Schema
- Connection details for views of different schemas

Permissions to Configure the Resource

Configure `read` permission on the Hive data source for the user account that you use to access the data source.

Prerequisites

If the domain is SSL-enabled and the cluster is Kerberos-enabled, perform the following steps:

1. Copy the `krb5.conf` and `infa_truststore.jks` files from the cluster, and the `hive.service.keytab` file from `/etc/security/keytabs` to the following locations on the Informatica domain:
   - `<INFA_HOME>/services/shared/security`
   - `<INFA_HOME>/java/jre/lib/security`
2. Specify the `krb5.conf` file path in the Informatica domain and the Informatica Cluster Service nodes.
3. Add the Service Principal Name (SPN) and the keytab properties to the Data Integration Service properties.
4. Run the kinit command on the Informatica domain with the required SPN and keytab file.
5. Run the kinit command on the cluster with the required SPN and keytab file.
6. Restart the Data Integration Service and the Catalog Service.

Choose Hadoop Connection for Existing Hive Resources

If you have a Hive resource that uses the Hive engine as the run-time environment, edit the resource, choose Blaze, Spark, or Databricks as the **Run on** option, select a Hadoop connection, and run the resource to view the profile results.

Import the Hive Resource

Perform the following steps if you use a Hive resource:

1. If you want to enable profiling for the Hive resource, create a cluster configuration based on the type of cluster that you use.
2. Edit the Hive resource to assign the new Hive connection in the Source Connection field.
3. Select **Blaze**, **Spark**, or **Databricks** as the **Run on** option.
4. Select the Hadoop connection name.

5. If the Hive data source is located in a Kerberos-enabled cluster, make sure that you perform the following steps:
    a. Use Informatica Administrator to configure the Hadoop Kerberos Service Principal Name and the Hadoop Kerberos Keytab in the properties for the Data Integration Service.
    b. Set the value of Data Integration Service Hadoop Distribution Directory to /data/
    c. Create the cluster configuration if the cluster configuration was not created prior to upgrade. If the cluster configuration was created prior to upgrade, refresh the cluster configuration using the Informatica Administrator or the CLI.
    d. Use Informatica Administrator to append the following line to the JVM command line options for the Data Integration Service: -Djava.security.krb5.conf=/data. The JVM options are located in the Processes section of the Data Integration Service.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The following table describes the connection properties:

Property	Description
Hadoop Distribution	Select one of the following Hadoop distribution types for the Hive resource: - Cloudera - Hortonworks - MapR - Amazon EMR - Azure HDInsight - IBM BigInsights
URL	JDBC connection URL used to access the Hive server.
User	The Hive user name.
Password	The password for the Hive user name.
Keytab file	Path to the keytab file if Hive uses Kerberos for authentication.
User proxy	The proxy user name to be used if Hive uses Kerberos for authentication.

Property	Description
Kerberos Configuration File	Specify the path to the Kerberos configuration file if you use Kerberos-based authentication for Hive.
Enable Debug for Kerberos	Select this option to enable debugging options for Kerberos-based authentication.

For information about configuring Azure storage access for Azure HDInsight distribution type, see *Data Engineering Integration Guide*.

The following table describes the **Additional** and **Advanced** properties for source metadata settings on the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Select to extract metadata from the data source.
Schema	Click **Select...** to specify the Hive schemas that you want to import. You can use one of the following options from the **Select Schema** dialog box to import the schemas: - Select from List: Use this option to select the required schemas from a list of available schemas. - Select using regex: Provide an SQL regular expression to select schemas that match the expression.
Source Metadata Filter	You can include or exclude tables and views from the resource run. Use semicolons (;) to separate the table names and view names. For more information about the filter field, see "Source Metadata and Data Profile Filter" on page 16.
Table	Specify the name of the Hive table that you want to import. If you leave this property blank, Enterprise Data Catalog imports all the Hive tables.
SerDe jars list	Specify the path to the Serializer/DeSerializer (SerDe) jar file list. You can specify multiple jar files by separating the jar file paths using a semicolon (;).
Worker Threads	Specify the number of worker threads to process metadata asynchronously. You can leave the value empty if you want Enterprise Data Catalog to calculate the value. Enterprise Data Catalog assigns a value between one and six based on the JVM architecture and number of available CPU cores. You can use the following points to decide the value to use: - You can provide a value that is greater than or equal to one and lesser than six to specify the number of worker threads required. - If you specify an invalid value, Enterprise Data Catalog shows a warning and uses the value one. - If your machine has more memory, you can specify a higher value to process more metadata asynchronously. **Note:** Specifying a higher value might impact performance of the system.
Case Sensitive	Specifies that the resource is configured for case insensitivity. Select one of the following values: - True. Select this check box to specify that the resource is configured as case sensitive. - False. Clear this check box to specify that the resource is configured as case insensitive. The default value is False.

Property	Description
Memory	Specify the memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.
Auto Assign Connections	Indicates whether the connections must be assigned automatically.
Enable Reference Resources	Extracts metadata about assets that are not included in this resource, but referred to in the resource. Examples include source and target tables in PowerCenter mappings, and source tables and files from Tableau reports.
Retain Unresolved Reference Assets	Retains unresolved reference assets in the catalog after you assign connections. Retaining unresolved reference assets help you view the complete lineage. The unresolved assets include deleted files, temporary tables, and other assets that are not present in the primary resource.

You can enable data discovery for a Hive resource. For more information about enabling data discovery, see the Enable Data Discovery topic.

You can enable composite data domain discovery for a Hive resource. For more information about enabling composite data discovery, see the Composite Data Domain Discovery topic.

Configure Hive Resource with Apache Knox Gateway

Enterprise Data Catalog supports Knox if you configure Hive for Knox. Verify that you install Informatica and Hive hosting service on the same cluster.

Note: You cannot deploy Enterprise Data Catalog on a cluster if you configure all the services on the nodes for Knox. Verify that you configure Knox for Hive service and not for other services running on the nodes.

CHAPTER 4

Configuring Application Resources

This chapter includes the following topics:

- Overview of Application Resources, 69
- SAP R/3, 69
- SAP S/4HANA, 72

Overview of Application Resources

The Informatica Application resource group includes the SAP resource. When you create and run the SAP resource, the SAP scanner extracts and migrates the metadata results in the associated SAP to the catalog. You can view the migrated metadata results in Enterprise Data Catalog.

SAP R/3

You must complete the prerequisites listed before configuring the properties for an SAP R/3 resource.

Objects Extracted

The SAP R/3 resource extracts metadata from the following assets in a SAP R/3 data source:

- Application components
- Packages
- Tables
- Columns
- Primary and foreign keys

Permissions to Configure the Resource

To configure SAP R/3, complete the following tasks:

1. Install Scanner transports.

For information about installing the transports that Enterprise Data Catalog uses, see "Informatica Developer Transports" in the PowerExchange for SAP NetWeaver Transport Versions Installation Notice.

2. Configure a user-authorization profile using the following values:

Authorization Object	Description	Class	Field Values
S_RFC	Authorization check for RFC access.	Cross Application Authorization objects	Activity: 16 (Execute) Name of RFC to be protected: *. Type of RFC object to be protected: FUGR

The SAP R/3 administrator needs to create the product and development user authorization profile. For more information about user authorization profile, see Create Profiles in the *PowerExchange for SAP NetWeaver User Guide*.

3. Install SAP SDK libraries.

Prerequisites

1. Create an Event Details Record (EDR) connection for SAP R/3 using Informatica Administrator. For more information about creating connections, see the *Informatica Administrator Guide*.
2. Disable the Catalog Service.
3. Download the `sapjco3.jar` file and copy the file to the following location: `<Informatica installation directory>/services/CatalogService/access/WEB-INF/lib`.
4. If you want to enable profiling, copy the `sapjco3.jar` file to the following location: `<Informatica installation directory>/services/shared/jars/thirdparty`.
5. Download the `libsapjco3.so` file and copy the file to the following location: `<Informatica installation directory>/services/shared/bin`.
6. Include the `libsapjco3.so` and `sapjco3.jar` files in the `SAPJCO.zip` file.
7. Copy the `SAPJCO.zip` file to the following location: `<Informatica installation directory>/services/CatalogService/ScannerBinaries`.
8. Enable the Catalog Service.
9. Restart the Informatica domain.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
Username	The user name to access SAP R/3 system.
Password	The password associated with the user name.
Application server host	Host name or IP address of the system that hosts SAP R/3.
System number	System number of the SAP R/3 system to which you want to connect.
Client	The SAP R/3 client to access data from the SAP R/3 system.
Language	Specify the language to be used while importing metadata using the resource.
Enable Streaming for Data Access	Choose this option to extract data by using the HTTP protocol. If you do not choose this option, the scanner extracts the SAP data to a staging file.
Encoding	Default is UTF-8 character encoding for metadata imported from the resource. You cannot change the default setting for this property.

The following table describes the properties that you can configure in the **Source Metadata** section of the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Extracts metadata from the data source.
Repository Objects	Imports the repository objects such as resources, information, and activities from the SAP R/3 system.
Memory	Specifies the memory required to run the scanner job. Select one of the following values based on the data set size imported: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on the **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.

You can enable data discovery for an SAP R/3 resource. For more information about enabling data discovery, see the Enable Data Discovery topic.

You can enable composite data domain discovery for an SAP R/3 resource. For more information about enabling composite data discovery, see the Composite Data Domain Discovery topic.

SAP S/4HANA

SAP Business Suite 4 SAP HANA (SAP S/4HANA) is an ERP business suite that can be used to perform transactions and analyze business data in real time.

Objects Extracted

You can use the SAP S/4HANA resource to extract metadata from the following assets in an SAP S/4HANA data source:

- Application Components
- Packages
- Classes
- Data Elements
- Data Sources
- Domains
- Function modules
- Programs
- RFC Connections
- Tables
- Fields
- Transaction Codes
- Views such as database views, CDS views, append views, and external views

Note: If the asset type does not include lineage, hierarchy, categories, and relationship information, you cannot view the following tabs in the Catalog:

- Lineage and Impact
- Relationship
- Contains

You can also view the detailed lineage between the function module and data source assets in the Catalog.

Permissions to Configure the Resource

To configure SAP S/4HANA, perform the following tasks:

1. Install Scanner transports.
2. Configure a user-authorization profile.

Installing Scanner Transports

To extract metadata from SAP S/4HANA, you must import the Scanner transports that Enterprise Data Catalog uses to extract metadata from SAP S/4HANA. The transports are located in the `SAP_Scanner_Binaries.zip` file that is downloaded from the Informatica installer location.

Note: To extract profiling data from SAP S/4HANA, you must import the Scanner transports located in the following folder where you downloaded PowerCenter: `<download location>\saptrans\mySAP`.

Configuring User-Authorization Profiles

To access metadata from SAP S/4HANA, the SAP S/4HANA administrator needs to create the following user authorization profiles:

Authorization Object	Description	Field Values
S_RFC	Authorization check for RFC access	ACTVT: 16 (Execute) RFC_NAME: DDIF_FIELDINFO_GET RFCPING RFC_GET_FUNCTION_INTERFACE RFC_METADATA_GET SEU_COMPONENT ZINFA_ERP RFC_TYPE: FUGR, FUNC

Prerequisites

1. Use Informatica Administrator to create an Event Details Record (EDR) connection for SAP.
   For more information about creating connections, see "Connections" in the *Informatica Administrator for Enterprise Data Catalog* guide.

   Note: Ensure that in the Enterprise Application section, you choose the "SAP" connection type and provide the SAP S/4HANA details with Staging and FTP connection details.
2. Disable Catalog Service.
3. Download the `sapjco3.jar` file and copy the file to the following locations:
   - *Install_directory*`/services/CatalogService/access/WEB-INF/lib`
   - *Install_directory*`/services/shared/jars/thirdpatry`

   Note: The SAP Java Connector version supported is 3.1.4.
4. Download the `libsapjco3.so` file and copy the file to the following locations:
   - *Install_directory*`/services/shared/bin`
   - *Install_directory*`/server/bin`
5. Create a `SAPJCO.zip` file, and then add the following files to the first level directory of the .zip file:
   - `libsapjco3.so`
   - `sapjco3.jar`
6. Copy the .zip file to the following location: *Install_directory*`/services/CatalogService/ScannerBinaries`
7. Enable the Catalog Service.

   Note: If you use the SAP S/4HANA resource to extract profiling data, you must restart the Informatica domain.

Prerequisites to use Secure Network Communications (SNC) connection

1. Disable the Catalog Service.
2. Download the `sapjco3.jar` file and copy the file to the following locations:
   - `<Install_directory>/services/CatalogService/access/WEB-INF/lib`
   - `<Install_directory>/services/shared/jars/thirdpatry`

   Note: The SAP Java Connector version supported is 3.1.4.
3. Download the `libsapjco3.so` file and copy the file to the following locations:
   - `<Install_directory>/services/shared/bin`
   - `<Install_directory>/server/bin`
4. Configure the SNC connection type with X.509 certificate. For more information about the steps to configure SNC connection type, see the SAP documentation.
5. Create and set the *SECUDIR* environment variable for the Catalog Service as the location of the `libsapcrypto.so` file.
6. Include the following files in the `SAPJCO.zip` file:
   - `cred_v2`
   - `libsapcrypto.so`
   - `sapjco3.jar`
   - `libsapjco3.so`
   - X.509 certificate
7. Copy the `SAPJCO.zip` file to the following location: `<Install_directory>/services/CatalogService/ScannerBinaries`
8. Enable the Catalog Service.
9. In a multi-node cluster, verify that you perform the following tasks:
   - Copy the Personal Security Environment (PSE) file into the same path on all the cluster nodes.
   - Configure `read` permission on the SAP S/4HANA data source for the user account that you use to access the data source.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
SAP Server Type	The type of SAP server installation: - SAP Central Instance - SAP Load Balancer (Message Server)
Server Host	Host name or IP address of the system that hosts the SAP S/4HANA.
Login Group	Login group is a group of servers that belong to an SAP system. Login groups distribute user logins to individual instances or groups of SAP instances or application servers. Login groups are maintained in the transaction SAP Logon Groups (SMLG). The option appears when you select SAP Load Balancer (Message Server) as the SAP Server Type.
Message Server Port	Default port of the Message server. The option appears when you select SAP Load Balancer (Message Server) as the SAP Server Type.
System Number	System number of the SAP S/4HANA. The option appears when you select SAP Central Instance as the SAP Server Type.
SAP SID	The SAP System Identifier of the SAP instance.
Client	The SAP S/4HANA client to access data from the SAP S/4HANA.
Language	The language for metadata import from the resource. The selected login language is used to extract the SAP descriptions.
Connection Type	The type of SAP server connection: - Basic - Secure Network Communications (SNC). You can use the connection type to extract metadata from the SAP S/4HANA data source.
Username	SAP S/4HANA system user name.
Password	SAP S/4HANA system password.
SNC Partner Name	The SNC name of the communication partner.
X.509 Certificate File Name	The file name of the X.509 certificate as the log in ticket.
SNC My Name	The SNC name of the user sending the Remote Function Call (RFC) from Enterprise Data Catalog.

Property	Description
SNC QoP	Specifies the level of protection to use for the connection. Following are the available protection levels: - 1. Authentication only - 2. Integrity protection - 3. Privacy protection - 8. Use the value from the `snc/data_protection/use` location on the application server - 9. Use the value from the `snc/data_protection/max` location on the application server The default protection level is 3.
Auto Assign Connections	Indicates whether the connections must be assigned automatically.

The following table describes the properties that you can configure in the **Source Metadata** section of the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Enables metadata extraction
Repository Objects	Filter to extract metadata based on the SAP S/4HANA application components, and packages. The resource performs a full metadata scan if you do not select an SAP S/4HANA application component or package.
Asset Types	Select the asset types that you want to load into the catalog. Use the options to refine the Repository Objects.
Enable Reference Resource	Extracts metadata about assets that are not included in the selected component or package, but referred to in the selected component or package.
Retain Unresolved Reference Assets	Retains unresolved reference assets in the catalog after you assign connections. Retaining unresolved reference assets help you to view the complete lineage. The unresolved assets include assets that are not present in the primary resource.
Memory	The memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal

Property	Description
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, INFO, or ERROR. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value must be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the multiple key value pairs. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1. **Note:** Contact the Informatica Global Customer Support before you change the default JVM parameters.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

You can use the SAP S/4HANA resource to extract the following profiling information for a Table asset during staging:

- Domain Discovery
- Column Profiling
- Relationship Discovery

Note: Effective in version 10.5.0.1, the SAP S/4 HANA resource does not support extraction of profiling information from columns of type string and string with variable or fixed length.

For more information about enabling data discovery, see the Enable Data Discovery topic.

CHAPTER 5

Configuring Business Glossary Resources

This chapter includes the following topics:

- Overview of Business Glossary Resources, 78
- Informatica Axon, 78
- Business Glossary, 81

Overview of Business Glossary Resources

The Business Glossary resource group includes resources related to business glossary. When you create and run a business glossary resource, the business glossary scanner extracts and migrates the metadata results from the associated resource to the catalog. You can view the migrated results in Enterprise Data Catalog.

Informatica Axon

Informatica Axon is a knowledge repository and governance tool that stores the core data items and business context of your organization. Owners, stewards, subject matter experts and other responsible stakeholders collaborate to progressively chart the business reality of data, its lineage, and usage across processes, policies, projects, and regulation.

Objects Extracted

The Informatica Axon resource extracts metadata from the following assets:

- Glossary
- Policy
- Data Set
- System
- Attribute. Enterprise Data Catalog extracts the manually created as well as the curated attributes.

You can read the details of the native users, LDAP users, and SSO users associated with the assets in the Catalog.

You can view the following relationships in the Catalog:

- Parent-child relationship between Axon artifacts and extracted objects
- Semantic relationship between the glossary assets

Permissions

Enterprise Data Catalog recommends that you configure the `SuperAdmin` permission for the user account that you use to access the Axon data source.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The following table describes the connection properties for the Informatica Axon resource:

Property	Description
Username	Axon user name or LDAP user name to access the Axon application.
Password	Axon password or LDAP password to access the Axon application.
Host	The fully qualified host name to access Axon application. **Note:** Ensure that you are using Informatica Axon host details instead of the Load Balancer host.
Port	Port number to access Axon application. **Note:** Ensure that you are using Informatica Axon port details instead of the Load Balancer port.
Enable Secure Communication	Select the property if Axon application is enabled for SSL.

The following table describes the **Additional** and **Advanced** properties for source metadata settings on the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Select to extract metadata from the data source.
Axon Status	Select the objects with the Axon Status values that you want to load into the catalog.

Property	Description
Glossary Lifecycle	Select the glossary objects with the Lifecycle values that you want to load into the catalog.
System Lifecycle	Select the system objects with the Lifecycle values that you want to load into the catalog.
Data Set Lifecycle	Select the data set objects with the Lifecycle values that you want to load into the catalog.
Policy Lifecycle	Select the policy objects with the Lifecycle values that you want to load into the catalog.
Memory	Specify the memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.

Unsupported Features

- You cannot view the logical to physical relationship between the following Axon artifacts and the objects extracted in the Catalog:
  - System and Resource
  - Attribute and Field
- You cannot view the logical data flow relationship between the following assets in the Catalog:
  - Attribute and Attribute
  - System and System
- You cannot view the semantic relationship between the following assets in the Catalog:
  - Policy and Glossary
  - Policy and System
  - System and Glossary
- You cannot view the Glossary to Axon artifact relationship between the following assets in the Catalog:
  - Glossary and Data Set
  - Glossary and Attribute

Business Glossary

Business Glossary contains online glossaries of business terms and policies that define important concepts within an organization. Configure a Business Glossary Classification resource type to extract metadata from Business Glossary.

Objects Extracted

The Business Glossary resource extracts metadata from the following assets:

- Category
- Term
- Glossary
- Policy

Basic Information

The **General** tab includes the following basic information about the resource:

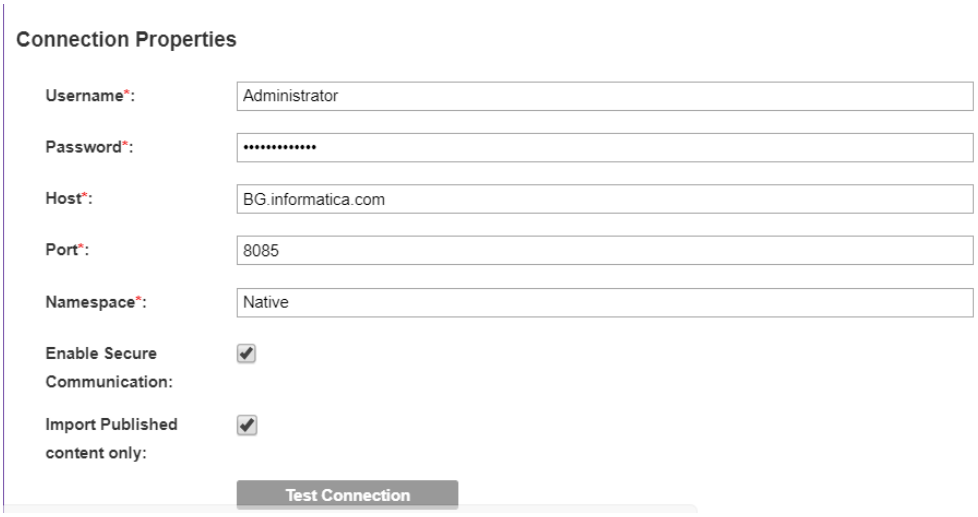
Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The following table describes the connection properties for the Business Glossary Classification resource:

Property	Description
Username	Name of the user account used that connects to the Analyst tool.
Password	Password for the user account that connects to the Analyst tool.
Host	Name of the Analyst tool business glossary from which you want to extract metadata. Each resource can extract metadata from one business glossary.
Port	Port number on which the Analyst tool runs.
Namespace	Name of the security domain to which the Analyst tool user belongs. If the domain uses LDAP authentication or Kerberos authentication, enter the security domain name. Otherwise, enter `Native`.
Enable Secure Communication	Enable secure communication from the Analyst tool to the Analyst Service.
Import Published Content Only	Select this option to specify that you want to import only the published content. If you do not select this option, Enterprise Data Catalog imports all content.

The following image shows sample connection properties on the **General** tab:



The following table describes the **Additional** and **Advanced** properties for source metadata settings on the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Select to extract metadata from the data source.
Glossary	Name of the business glossary resource that you want to import.
Memory	Specify the memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.

Note:

- Make sure that you log into the Analyst Tool once before you run the Business Glossary resource.
- If you modify the properties of a Business Glossary resource after creating the resource, make sure that you run the resource again.

CHAPTER 6

Configuring Business Intelligence Resources

This chapter includes the following topics:

- Overview of Business Intelligence Resources, 83
- IBM Cognos, 84
- MicroStrategy, 88
- Oracle Business Intelligence, 93
- QlikView, 97
- SAP BusinessObjects, 99
- Tableau Server, 105
- SAP Business Warehouse (SAP BW), 108
- SAP BW/4HANA, 114
- Microsoft Power BI, 118
- Qlik Sense, 119
- TIBCO Spotfire, 121

Overview of Business Intelligence Resources

The Business Intelligence resource group includes resources related to business intelligence. When you create and run a business intelligence resource, the business intelligence scanner extracts and migrates the metadata results from the associated resource to the catalog. You can view the migrated results in Enterprise Data Catalog.

IBM Cognos

IBM Cognos is a web-based integrated business intelligence suite by IBM. It provides a toolset for reporting, analytics, scorecarding, and monitoring of events and metrics.

Objects Extracted

Enterprise Data Catalog extracts metadata of the following schema objects from an IBM Cognos Business Intelligence Suite resource:

- Axis
- Body
- Crosstab
- Crosstab Columns
- Crosstab Node
- Crosstab Node Member
- Data Sources
- DataAttribute
- Database Query
- Dimension
- Field
- Filter
- Folder
- Footer
- Header
- Hierarchy
- Namespace
- Package
- Page
- PromptAnswer
- Queries
- Query
- Query Item
- Query Package
- Rectangle
- Report
- Table
- Text
- Variables
- Cognos Column
- OlapSchema Namespace
- Connection Model

- crosstab corner
- FrameworkManager Model
- QueryStudio Model
- ReportStudioModel
- Namespaces
- ConnectionPackage Data Source
- Repository
- StoreConnection Data Sources
- Alias column
- Alias Table
- Business layer view
- Cell
- Class
- Businessobjectscolumn
- Connection folder
- Contexts
- Crystal report
- Crystal report model
- Custom hierarchies
- Data provider
- Database connection
- Default hierarchies
- Target
- Dimension result
- Dimension variable
- Enterprise folder
- Formulas
- Group Name Fields
- Join
- Key
- Logical column
- Measure result
- Query filter
- Report chart
- Report field
- Report folder
- Report page
- Report table
- Repository

- Schema
- Section
- Special fields
- Summary fields
- Group Name Fields
- Universe folder
- Universe model
- Universe query
- View
- Web intelligence document
- Web intelligence model
- Connection
- Expand mapping
- Expression
- Group
- Joiner
- Lookup
- Lookup port
- Mapping
- Source qualifier
- Derived table

Prerequisites

You must install the Enterprise Data Catalog Agent to extract metadata from IBM Cognos running on Microsoft Windows.

Permissions to Configure the Resource

Before you create an IBM Cognos resource, you must configure the permissions for the user account that you use to connect to IBM Cognos.

You must configure the `read`, `execute`, and `traverse` permissions for all objects for which Enterprise Data Catalog extracts metadata. For example, reports, queries, analysis, packages, and connections.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.

Information	Description
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
Agent URL	URL to the Enterprise Data Catalog agent that runs on a Microsoft Windows Server. **Note:** Make sure that you specify the URL in the following format: `http://<hostname>:<connector_port>/MIMBWebServices`
Version	Indicates the Cognos server version.
Dispatcher URL	URL used by the framework manager to send requests to Cognos.
Namespace	Defines a collection of user accounts from an authentication provider.
User	User name used to connect to the Cognos server.
Password	Password for the user account to connect to the Cognos server.
Detailed Lineage	Select to extract and ingest metadata related to transformation logic for assets that include transformations. A transformation indicates generation, modification, or passage of data between source and target connections. A transformation logic displays the mappings or data flow relation types between source assets and target assets related to the asset you select in Enterprise Data Catalog.
Add Dependent Objects	Use to import dependent objects to the selection. Selecting this option requires a complete scan of report dependencies on the Cognos server. You can select any of the following options for this property: - None: only imports the selected Cognos objects. - Packages referenced by selected reports: imports the reports and associated source packages. - All: imports the source packages when a report is selected and imports the dependent reports when a source package is selected.
Folder Representation	Specifies how the folders from Cognos framework manager must be represented. You can select from the following options: - Ignore: ignores the folders. - Flat: represents the folders as diagrams, but does not retain the hierarchy. - Hierarchical: represents folders as diagrams and retains the hierarchy.
Transformer Import Configuration	The XML file that describes mappings between Cognos Content Manager data sources and PowerPlay Transformer models.
Worker Threads	Number of worker threads required to retrieve metadata asynchronously.
Auto Assign Connections	Specifies to automatically assign the connection.

The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Source Metadata	Select to extract metadata from the data source.
Content Browsing Mode	Specifies the content to be retrieved while searching the Cognos repository. You can select any of the following options: - Packages Only: retrieves the packages and folders and does not retrieve the reports. - Connections Only: retrieves the list of connections. - All: retrieves the packages, folders, queries, and reports. - Content: allows you to reduce the scope of import to a smaller set of objects than the whole set of objects on the server.
Content	Specifies the hierarchy for the content objects.
Memory	Specifies the memory required to run the scanner job. Select one of the following values based on the data set size imported: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. The default value is 1.

MicroStrategy

MicroStrategy is an enterprise business intelligence application software. The MicroStrategy platform supports interactive dashboards, scorecards, highly formatted reports, ad hoc query, thresholds and alerts, and automated report distribution.

Objects Extracted

Enterprise Data Catalog extracts metadata from reports in a MicroStrategy Business Intelligence resource.

You can view the following assets in the catalog for a MicroStrategy Business Intelligence resource:

- (Project) Metric
- (Project) Subtotal
- (Report) Metric
- (Report) Subtotal

- Attribute
- Attribute Form
- Calculated Report Field
- Consolidation
- Consolidation Element
- Custom Group
- Database Instance
- Document
- Document Report
- Fact
- Fact Measure
- Filter
- Folder
- Hidden Folder
- Hierarchy
- HTML Document
- HTML Document Report
- Logical Table
- Lookup
- Metric Measure
- MicroStrategy Column
- Project
- Project Folder
- Prompt
- Prompt Answer
- Report
- Report Field
- Report Graph
- Report Grid
- Report Page
- Report Page Body
- Report Text
- Resource
- Schema
- Security Filter
- Subtotals
- Subtotal Measure
- Project Source
- Warehouse Table

Prerequisites

You must install the Enterprise Data Catalog Agent to extract metadata from MicroStrategy running on Microsoft Windows. Also, you must enable the Enterprise Data Catalog Agent to interact with the desktop. For information on how to enable the Enterprise Data Catalog Agent to interact with the desktop, see the How to enable the EDC Agent to Interact with the Desktop KB article.

Permissions to Configure the Resource

Configure the following administration privileges for the MicroStrategy project user account for MicroStrategy version 7.x, and the `SecurityRole` for MicroStartegy version 9.x or later:

- Use server cache
- Use Developer
- View SQL
- Administer Caches
- Load and unload project
- Bypass All Object Security Access Checks

  Note: Configure the privilege to assign administration privileges for all the objects in the MicroStrategy project that you want to import.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
Agent URL	URL to the Enterprise Data Catalog agent that runs on a Microsoft Windows Server. **Note:** Make sure that you specify the URL in the following format: `http://<hostname>:<connector_port>/MIMBWebServices`
Version	Select the version of MicroStrategy from the drop-down list. You can select the **Auto detect** option if you want Enterprise Data Catalog to automatically detect the version of the MicroStrategy resource.
Project Source	Name of the MicroStrategy project source to which you want to connect.
Login User	The user name used to connect to the project source.

Property	Description
Login Password	The password associated with the user name.
Default Language	Specify the language to be used while importing metadata from the resource.
Import Schema Only	Select this option to import the project schema without the reports and documents.
Incremental import	Load changes since the previous resource load or load complete metadata. - Select the option to load recent changes only. - Clear the option to perform a complete load of the metadata.
Data Model Tables Design Level	Select one of the following options to specify the design for the imported tables: - Physical: the imported tables appear in the physical view of the model. - Logical and Physical: the imported tables appear in the logical and physical view of the model.
Detailed Lineage	Select to extract and ingest metadata related to transformation logic for assets that include transformations. A transformation indicates generation, modification, or passage of data between source and target connections. A transformation logic displays the mappings or data flow relation types between source assets and target assets related to the asset you select in Enterprise Data Catalog.
Project(s)	Select the names of the projects to which you want to connect from the project source.
Auto Assign Connections	Specifies to automatically assign the connection.

The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Source Metadata	Select to extract metadata from the data source.
Memory	Specifies the memory required to run the scanner job. Select one of the following values based on the data set size imported: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal

Property	Description
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

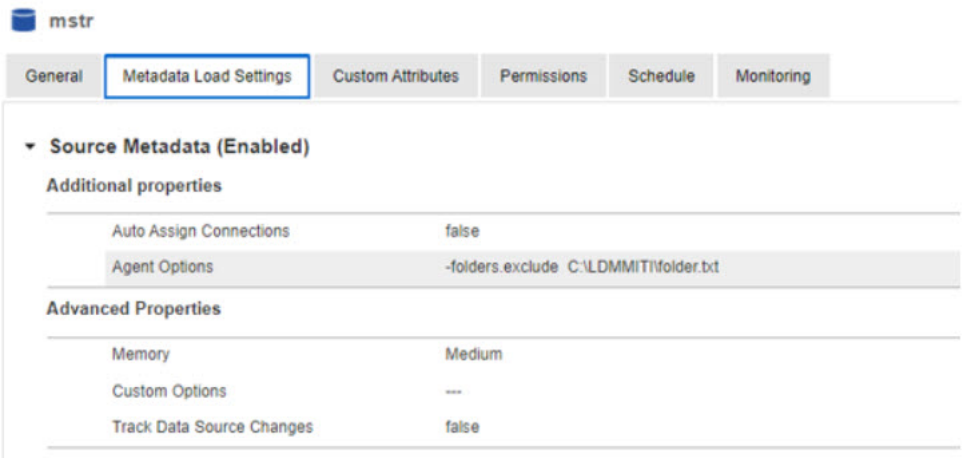
Excluding Folders That Store Metadata Objects Wthin a Project

By default, you cannot specify folders for metadata extraction within a project for a Microstrategy scanner. You must load reports from, for example, personal or migration folders.

The following solution allows you to exclude the chosen folders:

1. In the agent location of Enterprise Data Catalog, create a TXT file.
   Sample file location: <Windows drive>\<FolderName>\<FileName.txt>
2. In the TXT file, enter the location of the folder that you want to exclude for metadata extraction: `\<ProjectName>\<FolderName>\<Folder to exclude>`
3. In Catalog Administrator, open the Microstrategy resource.
4. On the **General** tab, in the **Project(s)** field, select the required project, and click **Save**.
5. On the **Metadata Load Settings** tab, in the **Agent Options** field, enter the following command: `-folders.exclude <Windows drive>\<FolderName>\<FileName.txt>` and click **Save**.

   The following image shows an example of the **Agent Options** field:



Oracle Business Intelligence

Oracle Business Intelligence is a business intelligence tool that includes analysis and reporting capabilities. You can create and configure an the resource to extract metadata from the Oracle Business Intelligence Enterprise Edition. Enterprise Data Catalog extracts report objects that includes presentations, business, and physical models.

Objects Extracted

Enterprise Data Catalog extracts metadata of the following schema objects from an Oracle Business Intelligence resource:

- Alias
- Alias Colum
- Alias Key Column
- AliasPhysicalTable
- Alias Table
- AnswersPage
- AnswersReport
- Axis
- BusinessModel
- Business Model and Mapping
- Catalog
- Connection Model
- Condition
- Compound View
- Criteria
- Criteria Column
- Criteria Hierarchical Column
- Data Attribute Column
- Dashboard
- Dashboard Page
- Database
- Derived Logical Column
- Derived Presentation Column
- Design Package Subject Area
- Dimension Logical Column
- DynamicVariable
- External Column
- External Connection
- External Schema
- External Table
- FactLogicalTable

- Field
- Filters View
- Folder
- Funnel
- Gauge
- Graph Chart
- Hierarchy
- Html View
- Key
- Key Logical Column
- LogicalColumn
- LogicalDimension
- Logical Fact Table
- LogicalJoin
- LogicalKeyColumn
- LogicalLevel
- LogicalTable
- Measure
- Model Content
- Physical
- PhysicalColumn
- Physical Key Column
- PhysicalForeignKey
- PhysicalJoin
- PhysicalSelectTable
- PhysicalTable
- PivotTable
- Presentation
- Presentation Table
- Presentation Fact Table
- Presentation Level
- Presentation Variable
- PresentationColumn
- Presentation Hierarchy
- PresentationTable
- Primary Key Column
- Report
- Report Field
- Repository

- RepositoryInitializationBlock
- RepositoryInitializationBlocks
- RepositoryVariables
- Schema
- SessionInitBlock
- Session Initialization Blocks
- SessionVariable
- SessionVariables
- StaticVariable
- StaticVariables
- Subject Area
- TableView
- Ticker
- TitleView
- Top View
- Variables
- View

Permissions to Configure the Resource

Enterprise Data Catalog connects to the OBIEE Presentation Server to extract report metadata. Ensure that the OBIEE user account for which you provide the user name and password has read permission on the metadata that Enterprise Data Catalog extracts from OBIEE.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
Version	Select one of the following options to specify the version of OBIEE: - Auto Detect. Select this option to let Enterprise Data Catalog detect the version of OBIEE. - OBIEE 11.x. Select this option to specify the OBIEE version as 11.x.
Server URL	The OBIEE Presentation Server URL. If you use SSL, you must make sure that Enterprise Data Catalog trusts the server certificate of the OBIEE Presentation Server.
Login User	The username used to log on to the OBIEE Presentation Server. Make sure that the username you use has the necessary permissions to import metadata.
Login Password	The password associated with the username.
Optimize for large models	Select this option to optimize the import of metadata for large OBIEE repository models. If you select this option, Enterprise Data Catalog does not import metadata for the following assets: - Foreign keys - Joins - Relationships - Logical foreign keys In addition, Enterprise Data Catalog does not store expression tree objects with lineage links. If you do not select this option, Enterprise Data Catalog imports the entire repository model, resulting in a high consumption of memory.
Worker threads	Specify the number of worker threads to process metadata asynchronously. You can leave the value empty if you want Enterprise Data Catalog to calculate the value. Enterprise Data Catalog assigns a value between one and six based on the JVM architecture and number of available CPU cores. You can use the following points to decide the value to use: - You can provide a value that is greater than or equal to one and lesser than six to specify the number of worker threads required. - If you specify an invalid value, Enterprise Data Catalog shows a warning and uses the value one. - If your machine has more memory, you can specify a higher value to process more metadata asynchronously. **Note:** Specifying a higher value might impact performance of the system.
File	The Oracle Business Intelligence Repository RPD file where the metadata is stored. **Note:** - For OBIEE version 11.x, you must convert the RPD file to an XML format. Enterprise Data Catalog does not support the UDML file format for OBIEE version 11.x. - For OBIEE version 10.x, you must convert the RPD file to a UDML file. Click **Choose** to select the RPD file. Alternatively, you can place the RPD file in the **Variable values file** text box from your file browser using a drag-and-drop operation.
Variable values file	(Optional) The file that defines the list of RPD variable values. Click **Choose** to select the file that includes the variable values. Alternatively, you can place the file in the **Variable values file** text box from your file browser using a drag-and-drop operation.
Auto assign connections	Select this option to specify that the connection must be assigned automatically.

The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Source Metadata	Select to extract metadata from the data source.
Repository Subset	Click **Select**. The **Select Repository** subset dialog box appears. Select the folders from where you want to import metadata for reports from the Oracle Business Intelligence Presentation Server.
Memory	Specifies the memory required to run the scanner job. Select one of the following values based on the data set size imported: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. The default value is 1.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog

QlikView

QlikView is a flexible business intelligence platform that transforms data into knowledge. Configure the connection properties to create a QlikView resource.

Objects Extracted

The QlikView resource extracts the following objects:

- Connection
- Expressions
- Fields
- Groups
- System Fields
- System Variables
- Tables
- Variables
- Text Fields

Prerequisites

You must install the Enterprise Data Catalog Agent to extract metadata from QlikView running on Microsoft Windows.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
AgentURL	URL to the Enterprise Data Catalog agent that runs on a Microsoft Windows Server.
Directory	Path to the directory from where Enterprise Data Catalog needs to import the QlikView files. The directory should be present in the same machine where the Microsoft Windows agent is running.
Miscellaneous Options	Specify one of the following miscellaneous options: - ODBC connection database type - Replace DIRECTORY statement - Require QVW execution log Use a comma to separate the string values. For example, `DatabaseType=Oracle, RequireLog=no, DIRECTORY []=[E:\\Qlik_View].`

The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Source Metadata	Select to extract metadata from the data source.
Auto Assign Connections	Select this option to specify that the connection must be assigned automatically.
Memory	Specifies the memory required to run the scanner job. Select one of the following values based on the data set size imported: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal

Property	Description
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. The default value is 1.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

SAP BusinessObjects

You can configure an SAP BusinessObjects resource type to extract metadata from SAP BusinessObjects. SAP BusinessObjects is a business intelligence tool that includes components for performance management, planning, reporting, analysis, and enterprise data management.

Objects Extracted

Enterprise Data Catalog extracts metadata of the following schema objects from an SAP Business Objects data source:

- Body
- Cell
- Class
- Context
- Crystal Report
- Crystal Report Model
- Custom Hierarchies Folder
- Data Providers
- Derived Table
- EnterpriseFolder
- Field
- Filter
- Folder
- Footer
- Header
- Hierarchy
- Join
- Key

- Measure
- Page
- Query
- Rectangle
- Report Chart
- Schema
- Universe Query
- UniverseFolder
- Web Intelligence Document
- Web Intelligence Model
- Dimension
- Dimension Result
- Rectangle
- Dimension variable
- alias column
- section
- report field
- Text
- Measure result
- Business Object Table
- Report page
- Axis
- Report Table
- Formulas
- Special field
- Group Name Fields
- Alias table
- Report folder
- Summary fields
- Logical column
- Query filter
- Connectionfolder
- Connection model
- Database connection
- Business layer view
- Context
- Contexts
- Default hierarchies
- Detail

- Repository
- View
- Queries
- Connection Model
- ConnectionPackage Data Source
- Crosstab
- Package
- Page
- Crosstab Columns
- Crosstab Corner
- Crosstab Node
- Crosstab Node Member
- DataAttribute
- Data Sources
- Database query
- FrameworkManager Model
- Namespace
- Query item
- Query package
- Report
- ReportStudio Model
- Repository
- StoreConnection Data Sources
- Text
- Connection
- Expanded Mapping
- Expression
- Group
- Joiner
- Lookup
- Lookup Port
- Mapping
- Source Qualifier
- Target
- OlapSchema Namespace
- UniverseModel

Prerequisites

You must install the Enterprise Data Catalog Agent to extract metadata from SAP BusinessObjects running on Microsoft Windows. Also, you must enable the Enterprise Data Catalog Agent to interact with the desktop.

For information on how to enable the Enterprise Data Catalog Agent to interact with the desktop, see the How to enable the EDC Agent to Interact with the Desktop KB article.

Permissions to Configure the Resource

You must remove `write-protection` permissions from any SAP Business Object universe that you want to export. For more information about removing Business Object permissions, see the SAP Business Objects documentation.

Make sure that the user who logs in to the SAP Business Objects repository belongs to the Universe Designer Users group and has read access to all the Business Objects metadata. If the Business Objects repository contains web intelligence reports in the **Favorites** or **Personal** folders, make sure that you include the user in the **Administrators** group. You must create a custom security group and provide permission for the user to view Web intelligence reports.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The following table describes the properties for the BusinessObjects resource type:

Property	Description
Agent URL	Host name and port number of the Enterprise Data Catalog agent that runs on a Microsoft Windows Server. **Note:** Make sure that you specify the URL in the following format: `http://<hostname>:<connector_port>/MIMBWebServices`
Version	Version of the SAP Business Objects repository.

System	Name of the BusinessObjects repository. For BusinessObjects 11.x and 12.x, specify the name of the BusinessObjects Central Management Server.
	Specify the server name in the following format:
	`<server name>:<port number>`
	If the Central Management Server is configured on a cluster, specify the cluster name in the following format:
	`<host name>:<port>@<cluster name>`
	Default port is 6400.
	Note: If the version of the BusinessObjects repository is 14.0.6, do not specify a port number in the repository name. If you specify the port number, Enterprise Data Catalog cannot extract the Web Intelligence reports.

Property	Description
Authentication mode	The authentication mode for the user account that logs in to the BusinessObjects repository. Specify one of the following values: - Enterprise. Log in using the BusinessObjects Enterprise authentication mode. - LDAP. Log in using LDAP authentication configured to BusinessObjects. Default is Enterprise.
User Name	User name to log in to the BusinessObjects repository.
Password	Password of the user account for the BusinessObjects repository.
Add dependent objects	Choose the documents that depend on the universe you selected. Specify one of the following values: - True. Imports the documents that depend on the specified universe. - False. Ignores the documents that depend on the specified universe. **Note:** Dependency information is retrieved from the Business Objects repository metadata cache. If the Enterprise Data Catalog load does not reflect modified or moved reports, refresh the cache by loading these reports and refreshing the queries.
Add specific objects	Specifies additional objects to the universe. Specify one of the following values: - None. Ignores all objects. - Universe independent Documents. Imports documents that do not depend on any universe. Default is none.
Crystal CORBA port	Specifies the client port number on which the Crystal SDK communicates with the Report Application Server (RAS). The RAS server uses the port to send metadata to the local client computer. If you do not specify a port, the server randomly selects a port for each execution.
Class representation	Controls how the import of the tree structure of classes and sub classes occur. The Enterprise Data Catalog agent imports each class containing objects as a dimension or as a tree of packages. Specify one of the following values: - As a flat structure. Creates no packages. - As a simplified flat structure. Creates a package for each class with a sub class. - As a full tree structure. Creates a package for each class. Default is As a flat structure.
Detailed Lineage	Select to extract and ingest metadata related to transformation logic for assets that include transformations. A transformation indicates generation, modification, or passage of data between source and target connections. A transformation logic displays the mappings or data flow relation types between source assets and target assets related to the asset you select in Enterprise Data Catalog.

Property	Description
Worker Threads	Number of worker threads that the Enterprise Data Catalog agent uses to extract metadata asynchronously. Leave blank or enter a positive integer value. If left blank, the Enterprise Data Catalog agent calculates the number of worker threads. The Enterprise Data Catalog agent uses the JVM architecture and number of available CPU cores on the Enterprise Data Catalog agent machine to calculate the number of threads. If you specify a value that is not valid, the Enterprise Data Catalog agent uses one worker thread. Reduce the number of worker threads if the Enterprise Data Catalog agent generates out-of-memory errors during metadata extraction. Increase the number of worker threads if the Enterprise Data Catalog agent machine has a large amount of available memory, for example, 10 GB or more. If you specify too many worker threads, performance can decrease. Default is blank.
Auto Assign Connections	Choose to automatically assign the database schemas to the resource that you create for SAP BusinessObjects source.

The following table describes the **Additional** and **Advanced** properties for source metadata settings on the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Select to extract metadata from the data source.
Repository browsing mode	Specifies the available objects in the SAP BusinessObjects repository. select one of the following options: - Universes only. Imports the metadata for the SAP BusinessObjects universes. - Connection only. Imports the metadata for the connections to the databases that are published to the SAP BusinessObjects repository. - All. Imports both the universes and connections from the SAP BusinessObjects repository.
Repository subset	Specifies the objects stored in a remote SAP BusinessObjects repository.
Memory	Specify the memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal

Property	Description
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. The default value is 1.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

Tableau Server

Tableau is a business intelligence tool that allows you to connect to data, and create dashboards that can be shared. Tableau has a desktop component and a server component.

Objects Extracted

Enterprise Data Catalog extracts metadata from the server component of Tableau:

- Bin
- Calculation Dimension
- Calculation Measure
- Tableau Column
- Columns
- Condition
- Connection
- Dashboard
- Data Source
- Data Source Filter
- Data Source Joins
- Data Sources
- Dimension
- Dimension Table
- Directory
- Fact Table
- Field
- File
- Filter

- Group
- Hierarchy
- Hierarchy field
- Join
- Mark
- Marks
- Measure
- Namespace
- Page
- Parameter Dimension
- Parameter Measure
- Parameters
- Project
- Proxy Field
- Proxy Table
- Published Data Source
- Query
- Relationship
- Row
- Rows
- Session
- Set
- Shared Filter
- Story
- Tableau Table
- Tableau Data Source
- Tableau Server
- Tableau Workbook
- Temporary Field
- Temporary Table
- Unpublished Worksheet
- Workbook
- Workbook Data Source
- Worksheet

Permissions to Configure the Resource

To connect to Tableau Server, Enterprise Data Catalog uses the credentials of a user created on the Tableau Server. Make sure that you configure the user account with the following license levels and permissions:

- Interactor license level.

- View and Download permissions for all projects, workbooks, and data sources for which you want to extract metadata.

  Note: With a Personal Access Token and Personal Access Token Secret, you can use an SSO-enabled Tableau resource. For more information, see the HOW TO Access an SSO enabled Tableau resource in Enterprise Data Catalog KB article.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
Server	The host name or the IP address where the Tableau server runs.
Site	Specify the site if the Tableau server has multiple sites installed. The value is case sensitive.
Username	The user name to connect to the Tableau server.
Password	The password associated with the user name.
Worker Threads	Number of worker threads required to retrieve metadata asynchronously.
Auto Assign Connections	Specifies to automatically assign the connection.

The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Source Metadata	Select to extract metadata from the data source.
Enable Reference Resources	Option to extract metadata about assets that are not included in this resource, but referred to in the resource. Examples include source and target tables in PowerCenter mappings, and source tables and files from Tableau reports.
Retain Unresolved Reference Assets	Option to retain unresolved reference assets in the catalog after you assign connections. Retaining unresolved reference assets help you view the complete lineage. The unresolved assets include deleted files, temporary tables, and other assets that are not present in the primary resource.

Property	Description
Group By	Specify to group workbooks in the following categories: - Project - My Workbooks - All Workbooks
Repository Objects	Imports the repository objects such as workbooks and data sources. For any workbooks, the dependent data sources are also imported.
Memory	Specifies the memory required to run the scanner job. Select one of the following values based on the data set size imported: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1. - `-DhandleIdCollisionForTableau=true`. Handles the duplicate object conflicts with the extracted Tableau objects.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

SAP Business Warehouse (SAP BW)

SAP Business Warehouse is a model driven data warehouse that is based on the SAP NetWeaver ABAP environment. The SAP Business Warehouse collects, transforms and stores data generated by SAP and non-SAP applications. This data is accessible through built-in reporting, business intelligence and analytic tools as well as third-party software. The SAP Business Warehouse resource is also called as the Classic SAP Business Warehouse.

Objects Extracted

The SAP Business Warehouse resource extracts metadata from the following assets in a SAP Business Warehouse data source:

- Info Area
- Data Source
- Info Objects
- Info Object Catalog

- Source System
- Info Source
- Characteristics
- Key Figures
- Dimension
- Navigational Attributes
- InfoSet
- DSO
- Info Cube
- Multi Provider
- Transformation
- Transformation Rule
- Query
- Query Element or Field
- Workbook
- Advanced DSO
- Composite Provider
- DTP
- InfoObject as InfoProvider

Note: If the asset type does not include lineage, hierarchy, categories, and relationship information, you cannot view the following tabs in the Catalog:

- Lineage and Impact
- Relationship
- Contains

Permissions to Configure the Resource

To configure SAP Business Warehouse, complete the following tasks:

1. Install Scanner transports.
2. Configure a user-authorization profile.

Installing Scanner Transports

To extract metadata from SAP Business Warehouse, you must import the Scanner transports that Enterprise Data Catalog uses to extract metadata from SAP Business Warehouse. The transports are located in the `SAP_Scanner_Binaries.zip` file that is downloaded from the Informatica installer location.

Configuring User-Authorization Profiles

The SAP Business Warehouse administrator needs to create the product and development user authorization profiles. The following table describes the user authorization profiles:

Authorization Object	Description	Class	Field Values
S_RFC	Authorization check for RFC access	Cross Application Authorization Objects	ACTVT: 16 (Execute) RFC_NAME: DDIF_FIELDINFO_GET RFCPING RFC_GET_FUNCTION_INTERFACE RFC_METADATA_GET ZINFA_BW_FG RFC_TYPE : FUGR, FUNC
S_RS_COMP	Business explorer components	Business Warehouse	ACTVT : 03,16 RSINFOAREA: * (Full Authorization) RSINFOCUBE: * (Full Authorization) RSZCOMPID: * (Full Authorization) RSZCOMPTP: Full Authorization
S_RS_COMP1	Business explorer components: Enhancements to the owner	Business Warehouse	ACTVT : 03,16 RSZCOMPID: * (Full Authorization) RSZOWNER: * (Full Authorization) RSZCOMPTP: * (Full Authorization)
S_RS_ISET	Data warehouse workbench InfoSet	Business Warehouse	ACTVT : 03 RSINFOAREA - * (Full Authorization) RSINFOSET - * (Full Authorization) RSISETOBJ - * (Full Authorization)
S_RS_ADMWB	Data warehouse workbench objects	Business Warehouse	ACTVT : 03 RSADMWBOBJ - * (Full Authorization)

Prerequisites

1. Disable the Catalog Service.
2. Download the `sapjco3.jar` file and copy the file to the following location: `<Install_directory>/services/CatalogService/access/WEB-INF/lib`.

   Note: The SAP Java Connector version supported is 3.1.4.

3. Download the `libsapjco3.so` file and copy the file to the following location: `<Install_directory>/services/shared/bin`.
4. Create a `SAPJCO.zip` file and add the following file to the first level directory of the .zip file: `libsapjco3.so`
5. Copy the .zip file to the following location: `<Install_directory>/services/CatalogService/ScannerBinaries`
6. Enable Catalog Service.

Prerequisites to use Secure Network Communications (SNC) connection

1. Disable the Catalog Service.
2. Download the `sapjco3.jar` file and copy the file to the following location: `<Install_directory>/services/CatalogService/access/WEB-INF/lib`

   Note: The SAP Java Connector version supported is 3.1.4.
3. Download the `libsapjco3.so` file and copy the file to the following location: `<Install_directory>/services/shared/bin`
4. Configure the SNC connection type with X.509 certificate. For more information about the steps to configure SNC connection type, see the SAP documentation.
5. Create and set the *SECUDIR* environment variable for the Catalog Service as the location of the `libsapcrypto.so` file.
6. Include the following files in the `SAPJCO.zip` file:
   - `cred_v2`
   - `libsapcrypto.so`
   - `sapjco3.jar`
   - `libsapjco3.so`
   - X.509 certificate
7. Copy the `SAPJCO.zip` file to the following location: `<Install_directory>/services/CatalogService/ScannerBinaries`
8. Enable the Catalog Service.
9. In a multi-node cluster, verify that you perform the following tasks:
   - Copy the Personal Security Environment (PSE) file into the same path on all the cluster nodes.
   - Configure `read` permission on the SAP BW data source for the user account that you use to access the data source.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.

Information	Description
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
SAP Server Type	The type of SAP server installation: - SAP Central Instance - SAP Load Balancer (Message Server)
Server Host	Host name or IP address of the system that hosts the SAP Business Warehouse.
Login Group	Login group is a group of servers that belong to an SAP system. Login groups distribute user logins to individual instances or groups of SAP instances or application servers. Login groups are maintained in the transaction SAP Logon Groups (SMLG). The option appears when you select SAP Load Balancer (Message Server) as the SAP Server Type.
Message Server Port	Default port of the Message server. The option appears when you select SAP Load Balancer (Message Server) as the SAP Server Type.
System Number	System number of the SAP Business Warehouse. The option appears when you select SAP Central Instance as the SAP Server Type.
SAP SID	The SAP System Identifier of the SAP instance.
Client	The SAP Business Warehouse client to access data from the SAP Business Warehouse.
Language	The language for metadata import from the resource. The selected login language is used to extract the SAP descriptions.
Connection Type	The type of SAP server connection: - Basic - Secure Network Communications (SNC). You can use the connection type to extract metadata from the SAP BW data source.
Username	SAP Business Warehouse system user name.
Password	SAP Business Warehouse system password.
SNC Partner Name	The SNC name of the communication partner.
X.509 Certificate File Name	The file name of the X.509 certificate as the log in ticket.
SNC My Name	The SNC name of the user sending the Remote Function Call (RFC) from Enterprise Data Catalog.

Property	Description
SNC QoP	Specifies the level of protection to use for the connection. Following are the available protection levels: - 1. Authentication only - 2. Integrity protection - 3. Privacy protection - 8. Use the value from the `snc/data_protection/use` location on the application server - 9. Use the value from the `snc/data_protection/max` location on the application server The default protection level is 3.
Auto Assign Connections	Indicates whether the connections must be assigned automatically.

The following table describes the properties that you can configure in the **Source Metadata** section of the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Enables metadata extraction
Source Extraction	The type of source metadata extraction: - Full. Extracts the full metadata from the source. - Based on Object Modified Date. Extracts metadata that is modified or created after the specified date. You must perform at least one full extraction before extracting metadata based on object modified date. - Based on Query Execution Date. Extracts metadata based on the query execution date. The corresponding dependant metadata objects are also extracted.
Memory	The memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, INFO, or ERROR. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value must be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the multiple key value pairs. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. The default value is 1.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

SAP BW/4HANA

SAP BW/4HANA is an enterprise data warehouse platform that captures, stores, and consolidates enterprise data.

Objects Extracted

The SAP BW/4HANA resource extracts metadata from the following assets in an SAP BW/4HANA data source:

- Info Area
- Data Source
- Info Objects
- Source System
- Info Source
- Characteristics
- Key Figures
- Navigational Attributes
- Transformation
- Transformation Rule
- Query
- Query Element or Field
- Advanced DSO
- Composite Provider
- DTP
- Open ODS view
- Aggregation Level
- Open Hub Destination
- InfoObject as InfoProvider

Note: If the asset type does not include lineage, hierarchy, categories, and relationship information, you cannot view the following tabs in the Catalog:

- Lineage and Impact
- Relationship
- Contains

Permissions to Configure the Resource

To configure SAP BW/4HANA, perform the following tasks:

1. Install Scanner transports.
2. Configure a user-authorization profile.

Installing Scanner Transports

To extract metadata from SAP BW/4HANA, you must import the Scanner transports that Enterprise Data Catalog uses to extract metadata from SAP BW/4HANA. The transports are located in the `SAP_Scanner_Binaries.zip` file that is downloaded from the Informatica installer location.

Configuring User-Authorization Profiles

The SAP Business Warehouse administrator needs to create the product and development user authorization profiles. The following table describes the user authorization profiles:

Authorization Object	Description	Class	Field Values
S_RFC	Authorization check for RFC access	Cross Application Authorization Objects	ACTVT: 16 (Execute) RFC_NAME: DDIF_FIELDINFO_GET RFCPING RFC_GET_FUNCTION_INTERFACE RFC_METADATA_GET ZINFA_BW4H_FG RFC_TYPE : FUGR, FUNC
S_RS_COMP	Business explorer components	Business Warehouse	ACTVT : 03,16 RSINFOAREA: * (Full Authorization) RSINFOCUBE: * (Full Authorization) RSZCOMPID: * (Full Authorization) RSZCOMPTP: Full Authorization
S_RS_COMP1	Business explorer components: Enhancements to the owner	Business Warehouse	ACTVT : 03,16 RSZCOMPID: * (Full Authorization) RSZOWNER: * (Full Authorization) RSZCOMPTP: * (Full Authorization)

Prerequisites

1. Disable Catalog Service.
2. Download the `sapjco3.jar` file, and then copy the file to the following location: `<Install_directory>/services/CatalogService/access/WEB-INF/lib`.

   Note: The SAP Java Connector version supported is 3.1.4.
3. Download the `libsapjco3.so` file and copy the file to the following location: `<Install_directory>/services/shared/bin`.
4. Create a `SAPJCO.zip` file and add the following file to the first level directory of the .zip file: `libsapjco3.so`
5. Copy the .zip file to the following location: `<Install_directory>/services/CatalogService/ScannerBinaries`
6. Enable Catalog Service.

Prerequisites to use Secure Network Communications (SNC) connection

1. Disable the Catalog Service.

2. Download the `sapjco3.jar` file and copy the file to the following location: `<Install_directory>/services/CatalogService/access/WEB-INF/lib`

   Note: The SAP Java Connector version supported is 3.1.4.
3. Download the `libsapjco3.so` file, and then copy the file to the following location: `<Install_directory>/services/shared/bin`
4. Configure the SNC connection type with X.509 certificate. For more information about the steps to configure SNC connection type, see the SAP documentation.
5. Create and set the *SECUDIR* environment variable for the Catalog Service as the location of the `libsapcrypto.so` file.
6. Include the following files in the `SAPJCO.zip` file:
   - `cred_v2`
   - `libsapcrypto.so`
   - `sapjco3.jar`
   - `libsapjco3.so`
   - X.509 certificate
7. Copy the `SAPJCO.zip` file to the following location: `<Install_directory>/services/CatalogService/ScannerBinaries`
8. Enable the Catalog Service.
9. In a multi-node cluster, verify that you perform the following tasks:
   - Copy the Personal Security Environment (PSE) file into the same path on all the cluster nodes.
   - Configure `read` permission on the SAP BW/4HANA data source for the user account that you use to access the data source.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
SAP Server Type	The type of SAP server installation: - SAP Central Instance - SAP Load Balancer (Message Server)
Server Host	Host name or IP address of the system that hosts the SAP Business Warehouse.

Property	Description
Login Group	Login group is a group of servers that belong to an SAP system. Login groups distribute user logins to individual instances or groups of SAP instances or application servers. Login groups are maintained in the transaction SAP Logon Groups (SMLG). The option appears when you select SAP Load Balancer (Message Server) as the SAP Server Type.
Message Server Port	Default port of the Message server. The option appears when you select SAP Load Balancer (Message Server) as the SAP Server Type.
System Number	System number of the SAP Business Warehouse. The option appears when you select SAP Central Instance as the SAP Server Type.
SAP SID	The SAP System Identifier of the SAP instance.
Client	The SAP Business Warehouse client to access data from the SAP Business Warehouse.
Language	The language for metadata import from the resource. The selected login language is used to extract the SAP descriptions.
Connection Type	The type of SAP server connection: - Basic - Secure Network Communications (SNC). You can use the connection type to extract metadata from the SAP BW/4HANA data source.
Username	SAP Business Warehouse system user name.
Password	SAP Business Warehouse system password.
SNC Partner Name	The SNC name of the communication partner.
X.509 Certificate File Name	The file name of the X.509 certificate as the log in ticket.
SNC My Name	The SNC name of the user sending the Remote Function Call (RFC) from Enterprise Data Catalog.
SNC QoP	Specifies the level of protection to use for the connection. Following are the available protection levels: - 1. Authentication only - 2. Integrity protection - 3. Privacy protection - 8. Use the value from the `snc/data_protection/use` location on the application server - 9. Use the value from the `snc/data_protection/max` location on the application server The default protection level is 3.
Auto Assign Connections	Indicates whether the connections must be assigned automatically.

The following table describes the properties that you can configure in the **Source Metadata** section of the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Enables metadata extraction
Source Extraction	The type of source metadata extraction: - Full. Extracts the full metadata from the source. - Based on Object Modified Date. Extracts metadata that is modified or created after the specified date. You must perform at least one full extraction before extracting metadata based on object modified date. - Based on Query Execution Date. Extracts metadata based on the query execution date. The corresponding dependant metadata objects are also extracted.
Memory	The memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, INFO, or ERROR. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value must be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the multiple key value pairs. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. The default value is 1.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

Microsoft Power BI

Microsoft Power BI is a business intelligence tool that allows you to connect data and create dashboards that can be shared.

Deprecation Notice: Effective in version 10.5.0.1, the Microsoft Power BI resource is deprecated. Informatica intends to drop support for the Microsoft Power BI resource in a future release.

Informatica suggests that you configure the new Microsoft Power BI scanner that leverages the Advanced Scanners framework. You do not need an additional license.

For more information, see "Microsoft Power BI" in the *Advanced Scanners Configuration Guide*.

Qlik Sense

Qlik Sense is a business intelligence and visual analytics platform that supports a range of use cases including centrally deployed analytic applications, dashboards, custom, and embedded analytics within a scalable framework.

Objects Extracted

The Qlik Sense resource extracts metadata from the following assets in a Qlik Sense data source:

- Applications
- Sheets
- Stories
- Measures
- Dimensions
- Charts
- Fields
- Tables (staging data table)

Prerequisites

Add the machine details of the Qlik Sense host to the `/etc/hosts` file in the domain machine and the cluster machine where the Catalog Service is running.

You must install the Enterprise Data Catalog Agent to extract metadata from Qlik Sense running on Microsoft Windows.

Permissions to Configure the Resource

Assign the `RootAdmin` role to the Qlik Sense user to access all the assets in the Qlik Sense resource.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
Agent URL	URL of the external MITI agent. **Note:** To extract metadata from the Qlik Sense data source, you must install the Enterprise Data Catalog Agent on the Qlik Sense server.
Engine API URL	URL of the Qlik API Engine.
User	User name of the administrator to connect to the Qlik Sense server.
User Directory	User directory of the Qlik Sense server.
Certificate	Location of the Qlik Sense client certificate.
Certificate Password	Password of the Qlik Sense client certificate. Specify the password if the Qlik Sense client certificate is generated with a password.

The following table describes the properties that you can configure in the **Source Metadata** section of the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Enables metadata extraction.
Applications	Select the required applications created in the Qlik Sense source system.
Log Folder for Selected Applications	Path of the Qlik Sense log folder that is common for the selected applications.
Memory	The memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article in **How To-Library Articles** tab in the Informatica Doc Portal

Property	Description
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - -Dscannerloglevel=<DEBUG/INFO/ERROR>. Changes the log level of scanner to values, such as DEBUG, INFO, or ERROR. Default value is INFO. - -DagentLogLevel=<NONE/FATAL/ERROR/WARNING/STATUS/INFO/DEBUG>. Changes the Enterprise Data Catalog Agent log level to values such as NONE, FATAL, ERROR, WARNING, STATUS, INFO, and DEBUG. Default log level is DEBUG. Use the ERROR log level when the Qlik Sense data set is huge. - -Dscanner.container.core=<No. of core>. Increases the core for the scanner container. The value must be a number. - -Dscanner.yarn.app.environment=<key=value>. Key pair value that you need to set in the Yarn environment. Use a comma to separate the multiple key value pairs. - -Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>. Increases the scanner container memory when pmem is enabled. Default value is 1. **Note:** You can enable miscellaneous options such as DatabaseType, FILEPATH, RequireLog, and m for a Qlik Sense resource using the Custom Options.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.
Auto Assign Connections	Indicates whether the connections must be assigned automatically.
Enable Reference Resources	Extracts metadata about assets that are not included in this resource, but referred to in the resource. Examples include source and target tables in PowerCenter mappings, and source tables and files from Tableau reports.

Unsupported Features

You cannot view the external lineage for applications that generate a QVD file for a Qlik Sense resource.

TIBCO Spotfire

TIBCO Spotfire is a business intelligence tool that you can use to explore data through dashboards and advanced analytics.

Objects Extracted

The TIBCO Spotfire resource extracts metadata from the following assets:

- Reports (Analysis documents)
- Pages
- Page objects such as Charts, Tables, Axis, Cross Table, Graphical Table, and Fields.
- Custom Queries
- Data Columns
- Data Connections
- Data Functions
- Data Producers
- Data Sources

- Data Tables
- Data Transformations
- Expressions
- Filters
- Folders
- Scripts

Permissions to Configure the Resource

Configure `read` permission for the user account used to access TIBCO Spotfire data source.

Lineage Support for TIBCO Spotfire Assets

You can view the column level lineage between the Data Sources, Data Tables, and Report objects in the catalog.

You can extract lineage from the following asset types in TIBCO Spotfire:

- Oracle
- Microsoft SQL Server
- Azure Microsoft SQL Data Warehouse
- MongoDB
- PostgreSQL
- Teradata
- Files such as CSV and TXT
- JDBC
- Snowflake

Connect to a TIBCO Spotfire Source Enabled for SSL

To connect to a TIBCO Spotfire data source enabled for SSL, perform the following steps:

1. Run the following command to generate the TIBCO Spotfire SSL certificates:

```
openssl s_client -connect <TIBCO Spotfire hostname>:<portnumber> -showcerts </dev/
null | sed -ne '/-BEGIN CERTIFICATE-/,/-END CERTIFICATE-/p' > <Path on the
Informatica domain machine>/<certificate>.crt
```

2. Go to the `<INFA_HOME>/source/java/jre/bin` directory and run the following `keytool` command to import each copied certificate as a trusted certificate in the Informatica domain keystore:

```
keytool -import -file <Path on the Informatica domain machine>/<certificate>.cer -
alias <alias name> -keystore <INFA_HOME>/services/shared/security/
infa_truststore.jks -storepass <Informatica default truststore password>
```

3. Restart the Catalog Service.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.

Information	Description
Resource type	The type of the resource.
Execute On	You can choose to run the resource on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
Server URL	URL to connect to the TIBCO Spotfire Server.
OAuth Client ID	OAuth client ID to connect to the TIBCO Spotfire Server.
OAuth Client Secret	OAuth client secret key to connect to the TIBCO Spotfire Server.

The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Source Metadata	Extracts metadata from the data source.
Library	Imports library objects such as reports and data sources. For a report object, the resource imports the dependent data sources.
Memory	Configures the memory required to run the scanner job. Select one of the following values based on the data set size imported: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	Configures the scanner container using JVM parameters. You can use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value must be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when you enable `pmem`. Default value is 1. - `-DhandleIdCollisionForTableau=true`. Handles the duplicate object conflicts with the extracted TIBCO Spotfire objects.
Track Data Source Changes	Displays metadata source change notifications in Enterprise Data Catalog.
Auto Assign Connections	Indicates whether the connections are assigned automatically.

Property	Description
Enable Reference Resources	Extracts metadata about assets that are not included in this resource, but referred to in the resource. Examples include source tables and files from TIBCO Spotfire reports.
Retain Unresolved Reference Assets	Retains unresolved reference assets in the catalog after you assign connections. Retaining unresolved reference assets help you view the complete lineage. The unresolved assets include deleted files, temporary tables, and other assets that are not present in the primary resource.

Limitations of the Resource

The TIBCO Spotfire resource includes the following limitations:

- The resource does not parse few complex SQL queries and code written in programming languages such as JavaScript, R, and Python.
- Enterprise Data Catalog supports only OAuth 2.0 authentication to connect to the TIBCO Spotfire data source.
- The resource does not extract information link from libraries as the TIBCO Spotfire API does not provide this information during the metadata scan.

CHAPTER 7

Configuring Data Integration Resources

This chapter includes the following topics:

- Overview of Data Integration Resources, 125
- Custom Lineage, 126
- Data Integration Hub, 127
- Database Scripts, 128
- Informatica Intelligent Cloud Services - Cloud Data Integration, 129
- Informatica Master Data Management, 131
- Informatica Platform, 133
- PowerCenter, 138
- SQL Server Integration Service, 144
- IBM InfoSphere DataStage, 144

Overview of Data Integration Resources

The Data Integration resource group includes the data integration resources. When you create and run a data integration resource, the data integration scanner extracts and migrates the metadata from the associated resource to the catalog. You can view the migrated results in Enterprise Data Catalog.

Custom Lineage

You can create a custom lineage resource to view the data lineage information for the assets in your organization. A custom lineage resource uses CSV files provided by you that include lineage data for your enterprise. You can use this option if you do not have an ETL tool supported by Enterprise Data Catalog.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The following tables list the properties that you must configure to add a custom lineage resource:

The **General** tab includes the following properties:

Property	Description
File	The CSV file or the .zip file that includes the CSV files with the lineage data. Click **Choose** to select the required CSV file or .zip file that you want to upload. Ensure that the CSV files in the .zip file are not stored in a directory within the .zip file. If you want to select multiple CSV files, you must include the required CSV files in a .zip file and then select the .zip file for upload. **Note:** Make sure that the CSV file includes the following parameters in the header: - Association - From Connection - To Connection - From Object - To Object

The following image shows sample connection properties on the **General** tab:



The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Source Metadata	Select to extract metadata from the data source.
Auto Assign Connections	Specifies to automatically assign the connection.
Memory	Specifies the memory required to run the scanner job. Select one of the following values based on the data set size imported: - Low - Medium - High **Note:** Enterprise Data Catalog restricts the size of the `lineage.csv` file based on the following memory size you specify: - Low. Maximum file size is 500 MB - Medium memory. Maximum file size is 1024 MB - High memory. Maximum file size is 2048 MB To override the restriction on the file size, use the `-DcustomScannerMaxFileSize` JVM option. For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1. - `-DcustomScannerMaxFileSize=<file size>`. Overrides the restriction on the `lineage.csv` file size. **Note:** If you specify a very large file size, you might encounter a Java out-of-memory error.

Data Integration Hub

Data Integration Hub is an application integration solution that you can use to share and synchronize data between multiple applications.

Objects Extracted

The Data Integration Hub resource extracts metadata from the following assets in a Data Integration Hub data source:

- Publication
- Topic
- Subscription
- Application

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
DIH Console URL	URL of the Data Integration Hub console.
User Name	User name used to connect to the Data Integration Hub server.
Password	Password for the user account that connects to the Data Integration Hub server.

The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Source Metadata	Extracts metadata from the data source.
Memory	The memory required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

Database Scripts

Create a Database Scripts resource to extract lineage information from database scripts.

Deprecation Notice: Effective in version 10.5.0.1, the Database Scripts resource is deprecated. Informatica intends to drop support for the following Database Scripts resources in a future release:

- IBM DB2 Scripts
- Microsoft SQL Server Scripts
- IBM Netezza Scripts
- Oracle Script
- Snowflake Scripts
- Sybase Scripts
- Teradata Scripts

Informatica suggests that you configure advanced scanners for these sources in the Advanced Scanners tool before we drop support for the resources. When you create an advanced scanner, you can configure additional functionality such as embedded procedural calls and additional lineage detail. For more information, see the *Advanced Scanners Configuration Guide*.

Informatica Intelligent Cloud Services - Cloud Data Integration

Informatica Intelligent Cloud Services - Cloud Data Integration (IICS-CDI) is an on-demand subscription service that provides access to applications, databases, platforms, and flat files hosted on premise or on a cloud. IICS-CDI runs at a hosting facility.

Permissions to Configure the Resource

To access synchronization task, mapping task, and PowerCenter task objects in the IICS-CDI, the cloud user must have Cloud REST API license and `read` permission on the tasks.

Prerequisites

Before you add an IICS-CDI resource, perform the following steps:

1. Create an organization for your company on the IICS-CDI website, define the organization hierarchy, and configure the organization properties.

   Note: To create an organization, you must have a REST API license. If you do not have a REST API license, contact Informatica Global Customer Support.
2. Create a subscription account on the Informatica Cloud.
3. Verify that the machine where you install the Informatica Cloud Secure Agent meets the minimum system requirements. The Informatica Cloud Secure Agent is a lightweight program that runs all tasks and enables secure communication across the firewall between your organization and the Informatica Cloud.
4. Download, install, and register the Informatica Cloud Secure Agent using the Informatica Cloud user name and password.
5. Create the following tasks on Informatica Cloud:
   a. Mapping tasks
      A mapping defines reusable data flow logic that you can use in Mapping Configuration tasks. Use mapping to define data flow logic that is not available in Data Synchronization tasks, such as specific ordering of logic or joining sources from different systems. When you configure a mapping, you describe the flow of data from source and target. You can add transformations to transform data, such as an Expression transformation for row-level calculations or a Filter transformation to remove data from the data flow.
   b. PowerCenter tasks
      The PowerCenter task allows you to import PowerCenter workflows in to Informatica Cloud and run them as Informatica Cloud tasks.
   c. Data synchronization tasks
      The Data Synchronization task allows you to synchronize data between a source and target.

Note: A IICS-CDI resource imports all the tasks to Enterprise Data Catalog the first time metadata is extracted from the resource. During the subsequent extract operations, the resource imports only the updated tasks to Enterprise Data Catalog.

For more information about the prerequisites, see the *Informatica Cloud User Guide* and the *Informatica Cloud Administrator Guide*.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
Cloud URL	The URL to access the IICS-CDI.
Username	The user name to connect to the IICS-CDI.
Password	The password associated with the user name.
Auto Assign Connections	Select this option to specify that the connection must be assigned automatically.

The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Source Metadata	Select to extract metadata from the data source.
Detailed Lineage	Select to extract and ingest metadata related to transformation logic for assets that include transformations. A transformation indicates generation, modification, or passage of data between source and target connections. A transformation logic displays the mappings or data flow relation types between source assets and target assets related to the asset you select in Enterprise Data Catalog.
Memory	Specifies the memory required to run the scanner job. Select one of the following values based on the data set size imported: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal

Property	Description
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - -Dscannerloglevel=<DEBUG/INFO/ERROR>. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - -Dscanner.container.core=<No. of core>. Increases the core for the scanner container. The value should be a number. - -Dscanner.yarn.app.environment=<key=value>. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - -Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>. Increases the scanner container memory when pmem is enabled. Default value is 1.
Auto Assign Connections	Select this option to specify that the connection must be assigned automatically.
Enable Reference Resources	Extracts metadata about assets that are not included in this resource, but referred to in the resource.

Informatica Master Data Management

Multidomain MDM creates and maintains master data, which is a collection of core entities along with their attributes and their values that are critical to a business.

Objects Extracted

You can use the Informatica MDM resource to extract metadata from the following assets in an Informatica MDM data source:

- Business entities
- Business entity views
- Staging tables
- Landing tables
- Relationships
- Base objects
- Cross reference tables
- Attribute information of all business entities

You can view lineage for the following assets in the Catalog:

- Landing tables and staging tables
- Staging tables and cross reference tables
- Cross reference tables and base objects
- Base objects and business entities

Note: Enterprise Data Catalog version 10.4.1 supports Multidomain MDM version 10.4.0 or later.

Prerequisites

- Register the Operational Reference Store (ORS) on the Informatica MDM Hub Console.
- Configure the Informatica Data Director (IDD) application on the ORS using the Informatica Data Director Configuration Manager.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
SSL Enabled	Indicates whether the MDM Provisioning tool uses SSL.
Provisioning Host	Enter the host name to connect to the MDM Provisioning tool.
Provisioning Port	Enter the port number of the MDM Provisioning tool.
Provisioning Login User Name	Enter the user name to connect to the MDM Provisioning tool.
Provisioning Login Password	Enter the password to connect to the MDM Provisioning tool.
Database ID	Enter the database ID of the schema registered on the Informatica MDM Hub Console. **Note:** The database ID of the schema must be in lowercase.

The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Source Metadata	Extracts metadata from the data source.
Memory	Specifies the memory required to run the scanner job. Select one of the following values based on the data set size imported: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal

Property	Description
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - -Dscannerloglevel=<DEBUG/INFO/ERROR>. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - -Dscanner.container.core=<No. of core>. Increases the core for the scanner container. The value should be a number. - -Dscanner.yarn.app.environment=<key=value>. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - -Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>. Increases the scanner container memory when pmem is enabled. The default value is 1.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

Viewing Detailed Lineage between Informatica MDM and Informatica Platform Resources

To view the detailed lineage between Informatica MDM and Informatica Platform resources for data quality, and data integration mappings, perform the following steps:

1. Create and run an Informatica MDM resource.
2. Create an Informatica Platform resource, and enable **Detailed Lineage** for the resource.
3. Run the Informatica Platform resource.
4. Use the **Connection Assignment** option to configure the MDM and Informatica Platform resources as destination or target data sources. For more information about assigning connections, see the Managing Connections topic.

   Note: Verify that the resource scan status of both the resources is complete before performing connection assignment.

Informatica Platform

Create a resource based on the Informatica Platform resource type to extract metadata from the Model repository. You need to specify the Data Integration Service connection details when you configure the resource.

Objects Extracted

Enterprise Data Catalog extracts only mapping metadata from Informatica Platform.

Prerequisites

If the domain is SSL-enabled and cluster is Kerberos enable, perform the following steps:

- Copy the SSL certificate in all the cluster machines. Ensure that the infa_truststore.jks file path is common across all the cluster machines.
- To get the encrypted password, navigate to <INFAHOME>/server/bin file path, and then type pmpasswd <Truststore Password> in the command line.

- Configure the following JVM parameters for the resource:

  - -DINFA_TRUSTSTORE=<trust store path>

  - -DINFA_TRUSTSTORE_PASSWORD=<trust store encrypted key>

Source and Target Objects for Mapping

The Informatica Platform resource can use the following source and target objects:

- Flat file
- Oracle
- Amazon Redshift
- Amazon S3
- Microsoft SQL Server
- Sybase
- Hive
- IBM DB2
- Teradata JDBC
- Teradata ODBC
- HDFS
- Azure Data Lake Store
- Microsoft Azure Blob Storage
- Azure Microsoft SQL Data Warehouse
- SAP HANA

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The following table describes the Data Integration Service connection properties:

Property	Description
Target version	The version number of the Informatica platform. You can choose any of the following Informatica versions: - 9.6.1 HotFix 3 - 9.6.1 HotFix4 - 10.0 - 10.1 - 10.1.1 - 10.1.1 HF1 - 10.2 - 10.2 HF2 - 10.2.1 - 10.4.1 - 10.5
Domain Name	Name of the Informatica domain.
Data Integration Service Name	Name of the Data Integration Service.
Username	Username for the Data Integration Service connection.
Password	Password for the Data Integration Service connection.
Security Domain	Name of the LDAP security domain if the Informatica domain contains an LDAP security domain.
Host	Host name for the informatica domain.
Port	Port number of the Informatica domain.
Application Name	Name of the Data Integration Service application. By default, all the application names are selected. To select a single application name, click **Select...** to select the name of the application from the **Select Application Name** dialog box. **Note:** This property applies to target versions 10.0 or later.
Param Set for Mappings in Application	Parameter set for mappings configured for the Data Integration Service application. Click **Select...** to select the parameter set from the **Select Param Sets for Mappings in Application** dialog box. **Note:** This property applies to target versions 10.0, 10.1, 10.1.1 or 10.2.
Parameter Set or Parameter File	Specify whether you want to use a parameter set or parameter file in the Data Integration Service application. **Note:** This property is applies to target versions 10.2.1 or later.
Parameter Sets	Parameter set for mappings or workflows configured for the Data Integration Service application. The Data Integration Service application can contain a single parameter set or multiple parameter sets based on the parameter sets configured for the Data Integration Service application. You can use this property if you select a single Data Integration Service application. Click **Select...** to select the parameter set from the **Select Param Sets** dialog box. **Note:** This property is applies to target versions 10.2.1 or later.

Property	Description
Parameter Sets File	A ZIP file that contains all the parameter sets for mappings or workflows configured for the Data Integration Service applications in a .config file. Use a text editor to create the parameter sets file and name the file as `ApplicationParameterSet.config`, and then zip the .config file. In the `<Parameter Sets>` file, define the parameter sets that you want to configure for the selected Data Integration Service applications. Use a comma as a delimiter to separate the parameter sets in the file. **Note:** This property is applies to target versions 10.2.1 or later.
Parameter File	An XML file that lists user-defined parameters and their assigned values for mappings or workflows configured for the Data Integration Service applications. To create a parameter file, create a folder with the name of the Data Integration Service application and move the XML files that contain the mapping or workflow parameters to the folder, and then zip the folder. **Note:** This property applies to target versions 10.2.1 or later.

The following table describes the **Additional** and **Advanced** properties for source metadata settings on the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Extracts metadata from the data source.
Auto assign Connections	Indicates that the connections are automatically assigned.
Enable Reference Resources	Extracts metadata about assets that are not included in this resource, but referred to in the resource. Examples include source and target tables in streaming mapping, streaming sources such as Apache Kafka, and source tables and files from Tableau reports.
Retain Unresolved Reference Assets	Retains unresolved reference assets in the catalog after you assign connections. Retaining unresolved reference assets help you view the complete lineage. The unresolved assets include deleted files, temporary tables, and other assets that are not present in the primary resource.
Detailed Lineage	Enables detailed lineage for the resource. Select to extract and ingest metadata related to transformation logic for assets that include transformations. A transformation indicates generation, modification, or passage of data between source and target connections. A transformation logic displays the mappings or data flow relation types between source assets and target assets related to the asset you select in Enterprise Data Catalog.

Property	Description
Memory	The memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1. - `-DINFA_TRUSTSTORE=<trust store path>`. Path to the Informatica truststore file. - `-DINFA_TRUSTSTORE_PASSWORD=<trust store encrypted key>`. Encrypted truststore password.

Sample Parameter Set File

The following example shows a sample parameter set file:

```
Application_Map_rel_conn_param,Parameter_Set_Rel_Conn,Parameter_Set_Rel_Conn1
Application_Map_rel_schema_param,Parameter_Set_Rel_Schema
Application_Map_rel_srctable_param,Parameter_Set_Rel_ScrTblname
Application_Map_rel_conn_param
```

Parameter File Requirements

A parameter file is an XML file that lists user-defined parameters and their assigned values for mappings or workflows configured for the Data Integration Service applications. Use a text editor or the Informatica Developer Client to create a parameter file. To enable the Informatica Platform resource to read parameter values from a parameter file, the file must have a .xml extension. You can group parameters and variables in different sections of the parameter file.

Note: A parameter file must include only one workflow or mapping. Enterprise Data Catalog does not support parameter files that include multiple workflows or mappings.

Creating a Parameter File

To create a parameter file, perform the following steps:

1. Create a folder with the same name as the Data Integration Service application.
2. Create parameter files in the following format `<file name>.xml`.

   Note: Each parameter file should include a specific workflow or mapping.
3. Place all the parameter file definitions of the workflow or mapping within the folder.
4. Zip the folder. The zip file can contain multiple folders containing parameter files.

Sample Parameter File

The following example shows a sample parameter file that contains a workflow:

```
<?xml version="1.0" encoding="UTF-8" standalone="no"?>
```

```
<root>
    <project name="ABC">
        <workflow name="Name1">
            <parameter name="Param1">Parameter1</parameter>
        </workflow>
    </project>
</root>
```

Scanner Binaries File

The ScannerBinaries.zip file only includes the Informatica Platform resource binary files for the current release. The binary files for earlier Informatica versions are included in the ExtendedScannerBinaries.zip file.

PowerCenter

You can configure a PowerCenter resource type to extract metadata from PowerCenter repository objects. Use PowerCenter to extract data from multiple sources, transform the data according to business logic you build in the client application, and load the transformed data into file and relational targets.

Objects Extracted

Enterprise Data Catalog extracts mapping metadata from the PowerCenter source.

Permissions to Configure the Resource

- Make sure that you configure the `Access Repository Manager` privilege for the user who accesses the PowerCenter repository.
- Configure read permission on the PowerCenter data source for the user account that you use to access the PowerCenter data source.
- Ensure that you run the `pmrep ObjectExport` command to export the mappings in PowerCenter.

Prerequisites

If the domain is SSL-enabled and cluster is Kerberos-enabled, perform the following steps:

- Copy the SSL certificate in all the cluster machines. Ensure that the infa_truststore.jks file path is common across all the cluster machines.
- To get the encrypted password, navigate to `<INFAHOME>/server/bin` file path, and then type `pmpasswd <Truststore Password>` in the command line.
- Configure the following JVM parameters for the resource:

  - `-DINFA_TRUSTSTORE=<trust store path>`
  - `-DINFA_TRUSTSTORE_PASSWORD=<trust store encrypted key>`

Connecting to an SSL-enabled PowerCenter Resource

Perform the following steps before you create the PowerCenter resource if you want to connect to an SSL-enabled PowerCenter data source:

1. Disable the Catalog Service.
2. Run the following command to export the certificates from the domain truststore of PowerCenter:
   a. $INFAINSTALL/java/bin/keytool -export -keystore
   b. $INFAINSTALL/services/shared/security/infa_truststore.jks -alias <alias_name> -file

c. $INFAINSTALL/services/shared/security/certExportFromSSLDomainTruststore.cert

3. On the node where Enterprise Data Catalog runs, run the following command to import the certificate into the Informatica domain truststore:
   a. $LDMINSTALL/java/bin/keytool -import -file certExportFromSSLDomainTruststore.cert -alias <new_alias_name>-keystore
   b. $LDMINSTALL/services/shared/security/infa_truststore.jks -storepass pass2038@infaSSL

   Note: Make sure that you use a new alias name when you import the certificate.
4. Copy the truststore and the keystore files from the PowerCenter domain to the Informatica domain truststore.
5. Run the following commands to configure the environment variables:
   a. Edit the `infaservice.sh` script on the Informatica domain using the following command: $INFAINSTALL/tomcat/bin/infaservice.sh
   b. In the script, add the following line to set the environment variable: INFA_TRUSTSTORE=<the location where you copied the truststore and keystore files.> Ensure that you add the line after the following command in the script: unset INFA_TRUSTSTORE INFA_TRUSTSTORE_PASSWORD INFA_KEYSTORE INFA_KEYSTORE_PASSWORD
   c. For a Hortonworks cluster, open the YARN application and set the INFA_TRUSTSTORE=<the location where you copied the truststore and keystore files.> variable in the advanced YARN environment variables section, and restart YARN.
   d. For a Cloudera cluster, edit the yarn-env.sh script and set the INFA_TRUSTSTORE=<the location where you copied the truststore and keystore files.> variable, and restart YARN.
6. Recycle the Catalog Service.

Parameter Files

If a PowerCenter source repository uses parameter files in sessions and workflows, you can configure Catalog Administrator to read the parameter files when you create the PowerCenter resource. A parameter file is a .zip file that contains all the parameters and variables and their associated values configured for workflows, worklets, or sessions in PowerCenter repository. PowerCenter parameters can represent flat file sources, flat file lookups, flat file targets, relational connections, expressions at the transformation level, or objects in SQL overrides. The PowerCenter resource parses the parameter files and substitutes the parameter values to extract metadata for the flat file sources, flat file lookups, flat file targets, relational connections, and objects in SQL overrides.

Generating a parameter file archive

You can create a parameter file archive manually within a directory or use the parameter file utility to create the archive. The parameter file utility converts PowerCenter files into a format that the Enterprise Data Catalog PowerCenter resource can parse. You must run the parameter file utility before you run the PowerCenter resource.

Note: Before you run the parameter file utility, verify that you have set the JAVA_HOME variable.

Running the parameter file utility

1. Download the parameter file utility from the following location: `<Informatica installation directory>/services/CatalogService/ScannerBinaries/PCParamFileUtility`
2. Copy the parameter file utility to the machine where PowerCenter runs.

3. Encrypt the Informatica domain and PowerCenter passwords:
   - For Microsoft Windows, use the pmpasswd.exe file. See How to use pmpasswd utility to encrypt the password in Windows where Informatica server or client is installed?
   - For Linux, use the pmpasswd file. See Using the Pmpasswd Utility.
4. Navigate to the following location: `<Informatica installation directory>/services/CatalogService/ScannerBinaries/PCParamFileUtility/config`
5. In the config.properties file, enter values for the following parameters:

Parameter	Description
domain.jdbc.driver=	Informatica domain driver.
domain.jdbc.url=	Informatica domain URL.
domain.jdbc.user=	Informatica domain user name.
domain.jdbc.password=	Informatica domain encrypted password.
pcrs.jdbc.driver=	PowerCenter driver.
pcrs.jdbc.url=	PowerCenter URL.
pcrs.jdbc.user=	PowerCenter user name.
pcrs.jdbc.password=	PowerCenter encrypted password.
integration.service=	Integration service name. **Note:** That value can be empty If you do not have any integration service related to the corresponding PowerCenter service.

6. On the machine where PowerCenter runs, depending on your operation system, perform one of the following actions:
   - For Microsoft Windows, run the generateParam.bat file.
   - For Linux, run the following command: `generateParam.sh`

     Note: If you did not configure the passwords in the configuration file, a prompt appears asking you to provide passwords.

The utility generates a .zip file that contains the converted files. Upload the file to the PowerCenter resource before you run it.

Creating the parameter files manually within a directory

1. Create a directory with the same name as the directory configured for the PowerCenter repository in the PowerCenter repository.
2. Create parameter files named as `<workflow name>.prm`.
3. Place all the parameter definitions that are needed for the workflow and the corresponding sessions in the directory you created.
4. Zip the folder. The zip file can contain multiple folders containing parameter files.

Create multiple parameter files

Create parameter files named as `<PowerCenter folder name>.<workflow name>.prm`. Each parameter file applies to the workflow with which it is named.

Create a single parameter file

Create a single parameter file named as `parameters.prm`, and then zip the parameter file. The parameter file must contain the parameters that apply to all the workflows within the PowerCenter folders.

Parameter File Requirements

Use a text editor to create the parameter file. To enable the PowerCenter resource to read parameter values from a parameter file, the file must have a .prm extension. You must group parameters and variables in different sections of the parameter file. Each section is preceded by a heading that identifies the folder, workflow, worklet, and session to which you want to pass parameter or variable values. You define parameters and variables directly below the heading, entering each parameter or variable on a new line.

The following table describes the headings that define each section in the parameter file and the scope of the parameters and variables that you define in each section:

Heading	Scope
[Global]	All folders, workflows, worklets, and sessions.
[folder name.WF:workflow name]	The named workflow and all sessions within the workflow.
[folder name.WF:workflow name.WT:worklet name]	The named worklet and all sessions within the worklet.
[folder name.WF:workflow name.WT:worklet name.WT:worklet name...]	The nested worklet and all sessions within the nested worklet.
[folder name.WF:workflow name.ST:session name] -or- [folder name.WF:workflow name.WT:worklet name.ST:session name] -or- [folder name.WF:workflow name.WT:worklet name.WT:worklet name.ST:session name] -or- [folder name.session name] -or- [session name]	The named session.

Sample Parameter File

The following example shows a sample `parameters.prm` file:

```
[Map_Param.WF:WF_Src_Tgt_map_param_case.ST:s_src_tgt_tbl_override_default_map_param]
$$Src_OwnName=MM_PERF6
$$Src_TblName=TBL_SAME_COL
$$Tgt_Tbl_Prefix=MM_PERF6
$$Tgt_TblName=INVENTORY_Q4_2005
[Param_lookup.WF:wf_M_LKP_schema_tble
$$LKP_SCHEMA=TEST_DATA
$$LKP_TBL=LKP_TBL_PARAM
[Param_lookup.WF:wf_M_LKP_schema_tbl_sess_param]
$Param_Lkp_Schema=TEST_DATA
```

```
$Param_Lkp_Tbl=LKP_TBL_PARAM[
Param_session.WF:wf_session_param.ST:s_session_param]
$Param_Schema_Name=CROSS_RESOURCE_LINKING_DUP
$Param_SrcTbl_Name=SRC_TBL_NAME_OVERRIDE_PARAM
$Param_TgtTbl_Name=TGT_TBL_NAME_OVERRIDE_PARAM
[Param_Sql_override.WF:wf_M_schema_table_map_parm_sql.ST:s_M_schema_table_map_parm_sql]
$$Map_Schema_Name=CROSS_RESOURCE_LINKING
$$Map_Tbl_Name=SRC_TBL_NAME_OVERRIDE_DUP
```

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The following table describes the properties for the PowerCenter resource:

Property	Description
Gateway Host Name or Address	PowerCenter domain gateway host name or address.
Gateway Port Number	PowerCenter domain gateway port number.
Informatica Security Domain	LDAP security domain name if one exists. Otherwise, enter "Native."
Repository Name	Name of the PowerCenter repository.
Repository User Name	Username for the PowerCenter repository.
Repository User Password	Password for the PowerCenter repository.
PowerCenter Version	PowerCenter repository version. **Note:** Informatica does not provide support for PowerCenter versions earlier than 9.6.0.
PowerCenter Code Page	Code page for the PowerCenter repository.

The following table describes the **Additional** and **Advanced** properties for source metadata settings in the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Select to extract metadata from the data source. **Note:** You can run Data Flow Analytics on the resource even if you have not enabled the source metadata setting.
Parameter File	Specify the parameter file zip that you want to attach from a local system.
Auto assign Connections	Specifies whether Enterprise Data Catalog assigns the connection is automatically.
Enable Reference Resources	Option to extract metadata about assets that are not included in this resource, but referred to in the resource. Examples include source and target tables in PowerCenter mappings, and source tables and files from Tableau reports.
Repository subset	Enter the file path list separated by semicolons for the Informatica PowerCenter Repository object. **Note:** If you want to run Data Flow Analytics on the resource, you need to enter the file path for the Informatica PowerCenter Repository object. If you do not enter the file path to run Data Flow Analytics, by default, the catalog runs Data Flow Analytics on all the PowerCenter Repository objects.
Detailed Lineage	Select to extract and ingest metadata related to transformation logic for assets that include transformations. A transformation indicates generation, modification, or passage of data between source and target connections. A transformation logic displays the mappings or data flow relation types between source assets and target assets related to the asset you select in Enterprise Data Catalog. **Note:** To view the detailed lineage for assets that include web service transformations in the Catalog, you must select the **Enable Reference Resources** option.
Memory	Specify the memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** Informatica recommends that you use the **High** memory value if you enable the Data Flow Analytics. For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. The default value is 1.

The following table describes the **Repository** details and **Advanced** properties for data flow analytics settings in the **Metadata Load Settings** tab:

Property	Description
Enable Data Flow Analytics	Select to enable Data Flow Analytics and view analytical data about PowerCenter mappings in the catalog. **Note:** You must configure Data Asset Analytics for the Catalog Service to run Data Flow Analytics on the resource.
Select PowerCenter Repository Type	Type of database for the PowerCenter repository. Select from the following options: - Oracle - Microsoft SQL Server - IBM DB2 - Sybase - PostgreSQL
Username	Name for the PowerCenter repository database user account.
Password	Password for the PowerCenter repository database user account.
Database Connection String	JDBC connection string to connect to the secure database, including the host name and port number and the security parameters for the database.
Custom Options	Specify the custom options to run the scanner job.

Note: Effective in version 10.5, Data Flow Analytics is available for technical preview. Technical preview functionality is supported but is unwarranted and is not production-ready. Informatica recommends that you use these features in non-production environments only. Informatica intends to include the preview functionality in an upcoming GA release for production use, but might choose not to in accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.

SQL Server Integration Service

SQL Server Integration Service (SSIS) is a component of the Microsoft SQL Server database that can be used to perform data migration tasks.

Deprecation Notice: Effective in version 10.5.0.1, the SQL Server Integration Service resource is deprecated. Informatica intends to drop support for the SQL Server Integration Service resource in a future release.

Informatica suggests that you configure advanced scanners for these sources in the Advanced Scanners tool before we drop support for the resources. When you create an advanced scanner, you can configure additional functionality such as embedded procedural calls and additional lineage detail. For more information, see the advanced scanners documentation.

IBM InfoSphere DataStage

IBM® InfoSphere® DataStage® is an ETL platform that is used to integrate data across multiple systems.

Deprecation Notice: Effective in version 10.5.1, the IBM InfoSphere DataStage resource is deprecated. Informatica intends to drop support for the IBM InfoSphere DataStage resource in a future release.

Informatica suggests that you configure advanced scanners for these sources in the Advanced Scanners tool before we drop support for the resources. When you create an advanced scanner, you can configure additional functionality such as embedded procedural calls and additional lineage detail.

For more information, see "IBM InfoSphere DataStage" in the *Advanced Scanners Configuration Guide*.

CHAPTER 8

Configuring Data Modeling Resources

This chapter includes the following topics:

- Overview of Data Modeling Resources, 146
- Erwin, 146
- SAP PowerDesigner, 151

Overview of Data Modeling Resources

The Data Modeling resource group includes the Data Modeling resources. When you create and run a data modeling resource, the data modeling scanner extracts and migrates the metadata from the associated resource to the catalog. You can view the migrated results in Enterprise Data Catalog.

Erwin

Erwin is computer software for data modeling.

Objects Extracted

Enterprise Data Catalog extracts metadata from the following objects in an Erwin data source:

- IErwin
- DataElement
- LogicalModel
- PhysicalModel
- Column or Attribute
- Index Member
- Data Object
- Index
- Unnamed RelationshipEnd

- Candidate Key Group
- View Column
- Primary Key Column
- Table or Entity
- Foreign Key Group
- Relationship
- Column
- Identity
- Sequence
- Trigger
- View Relationship
- View
- Primary Key Attribute
- DataSource
- Entity
- Database Schema
- Domain
- Synonym
- ER Diagram
- Subject Area
- User-Defined
- Table
- Argument
- Default Value
- Stored Procedure
- Subtype Relationship Role
- Min/Max
- Model
- Subtype Relationship
- UDP
- UDPs
- Resource

Prerequisites

You must install the Enterprise Data Catalog Agent to extract metadata from Erwin Data Modeler running on Microsoft Windows. Also, you must enable the Enterprise Data Catalog Agent to interact with the desktop. For information on how to enable the Enterprise Data Catalog Agent to interact with the desktop, see the How to enable the EDC Agent to Interact with the Desktop KB article.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The following tables list the properties that you must configure to add an Erwin data modeling resource: The **General** tab includes the following properties:

Property	Description
Erwin Scanner Type	Select any of the following options to specify the erwin scanner type: - Erwin 8.x File. Click **Choose** to upload a .xml file from which you want to extract metadata. - Erwin 9.x File. Click **Choose** to upload a .xml file from which you want to extract metadata. - Erwin 8.x Data Modeler (Single model from mart). Specify the following parameters if you select this option: - Agent URL. URL to the Enterprise Data Catalog agent that runs on a Microsoft Windows Server. **Note:** Make sure that you specify the URL in the following format: `http://<hostname>:<connector_port>/MIMBWebServices` - Database type. Select the database server type configured for Mart Server, from the drop-down list. - Database server. Specify the Mart Server to which you want to connect. - Database Name. Specify the name of the database server to which you want to connect. - Authentication. Select one of the following options based on the authentication mode configured to connect to the Mart Server: - Server Authentication. Specify the user name and password to connect to the Mart Server. - Database Authentication. Specify the database user name and password to connect to the Mart Server. - Windows Authentication. Select this option to specify that the Windows authentication must be used for authentication of the database username and password to connect to the Mart Server. - Model (Optional). Specify the erwin model locator string. Use the following format for specifying the locator string: `mmart://<database name>/<path>/<model name>`. - Erwin 9.x Data Modeler (Single model from mart). Specify the following parameters if you select this option: - Agent URL. URL to the Enterprise Data Catalog agent that runs on a Microsoft Windows Server. **Note:** Make sure that you specify the URL in the following format: `http://<hostname>:<connector_port>/MIMBWebServices` - Server Name. Specify the Mart Server to which you want to connect. - Server Port. Specify the port number on the Mart Server that you can use to connect. - Use IIS (Optional). Select this option if you have configured connecting to the Mart Server with Microsoft IIS Web Server. The port number is dynamically assigned for connections using an IIS Server. If an IIS Server is not used to connect, you must specify the **Server Port** number. - Use SSL(Optional). Select this option if you have configured Secure Socket Layer (SSL) authentication to connect to the Mart Server. - Application Name. Specify the name of the application to which you want to connect to on the Mart Server. - Authentication. Select one of the following options to specify the authentication mode to use for the connection to the Mart Server: - Server Authentication. Specify the user name and password to connect to the Mart Server. - Database Authentication. Specify the database user name and password to connect to the Mart Server. - Windows Authentication. Select this option to specify that the Windows authentication must be used for authentication of the database username and password to connect to the Mart Server. - Model (Optional). Specify the erwin model locator string. Use the following format for specifying the locator string: `mmart://<database name>/<path>/<model name>`.

The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Source Metadata	Select to extract metadata from the data source.
Import UDPs	Specify how you want to import property definitions and values User Defined Properties (UDP). In ERwin, a user-defined property is a property definition object that has a default value. The object to which the UDP applies can have an explicit value or no assigned value. If an object has no assigned value, ERwin assigns the default value of the property definition to the UDP. You can select any of the following options: - As metadata. Imports explicit values as Property Value objects. This option retains the default value on the Property Type and does not import implicit values. - As metadata, migrate default values. Imports explicit and implicit values as Property Value objects. - In description, migrate default values. Appends the property name and value to the object description property for explicit and implicit values. - Both, migrate default values. Imports the UDP value as metadata and in the object description. Default is **As metadata**.
Import relationship name	Specify how you want import relationship names from erwin: - From relationship name. Use the relationship name property. - From derived name. Use the derived name property. Default is **From relationship name**.
Import column order from	Specify how you want to import the position of columns in tables: - Column order. Imports the position of the columns from the order of the columns displayed in the erwin physical view. - Physical order. Imports the position of the columns from the order in which the columns are stored in the database. Default is **Physical order**.
Import owner schemas	Select this option if you want to import owner schemas.
Move entities to subject areas	Select this option to move entities to user-defined subject areas as conventional UML design packages with their own namespace.
Auto Assign Connections	Select this option to specify that the connection must be assigned automatically.
Memory	Specifies the memory required to run the scanner job. Select one of the following values based on the data set size imported: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal

Property	Description
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. The default value is 1.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

SAP PowerDesigner

Create a resource based on the SAP PowerDesigner resource type to extract metadata from an SAP PowerDesigner data source. You need to specify the connection details when you configure the resource.

Objects Extracted

Enterprise Data Catalog extracts metadata from the following objects in an SAP PowerDesigner data source:

- *Table
- *View
- Procedure
- Sequence
- User
- Association
- *Synonym
- Business rule
- Dimension
- Fact
- Physical data model
- Physical diagram
- Multidimensional diagram

Note: * indicates the objects for which the resource extracts lineage information.

Prerequisites

You must install the Enterprise Data Catalog Agent to extract metadata from SAP PowerDesigner running on Microsoft Windows.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The following table describes the resource connection properties:

Property	Description
Agent URL	URL to the Enterprise Data Catalog agent that runs on a Microsoft Windows Server.
SAP PowerDesigner Model Type	The PowerDesigner version used to create the Physical Data Modeling (PDM) files.
Remove Binary Characters	Removes binary characters from the PDM file before the file is parsed.
Import UDPs	The resource does not support this option in the current version of Enterprise Data Catalog.
Append Volumetric Information	Imports the value for the number of occurrences of an object and append the information to the description property of an object.
Remove RTF	Removes Rich Text Formatting (RTF) from annotations.
Reverse Role Names	Reverses the role names for each relationship in the model.
Auto Assign Connections	Automatically assigns connections to the resource.
File	Select the PDM file that includes the data model from where you want to import metadata.

The following table describes the **Additional** and **Advanced** properties for source metadata settings on the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Extracts metadata from the data source.
Memory	The memory required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to specific values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate multiple key-pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1. - `-DINFA_TRUSTSTORE=<trust store path>`. Path to the Informatica truststore file. - `-DINFA_TRUSTSTORE_PASSWORD=<trust store encrypted key>`. Encrypted truststore password.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

CHAPTER 9

Configuring Database Management Resources

This chapter includes the following topics:

- Overview of Database Management Resources, 154
- IBM DB2, 155
- IBM DB2 for z/OS, 157
- IBM Netezza, 160
- JDBC, 163
- Microsoft SQL Server, 167
- Oracle, 171
- SAP HANA, 174
- Sybase, 177
- Teradata, 180
- Configure Case Sensitivity for Database Management Resources, 183

Overview of Database Management Resources

The Database Management resource group includes the database management resources. When you create and run a database management resource, the database management scanner extracts and migrates the metadata from the associated resource to the catalog. You can view the migrated results in Enterprise Data Catalog.

IBM DB2

You can configure an IBM DB2 resource type to extract metadata from IBM DB2 databases.

Objects Extracted

Enterprise Data Catalog extracts metadata of the following schema objects from an IBM DB2 resource:

- Tables
- Partitioned Table
- Range Clustered Table
- Multidimensional Clustering Table
- Constraints (NOT NULL , UNIQUE , CHECK , PRIMARY KEY , FOREIGN KEY , DEFAULT)
- Alias
- Sequences
- Views
- Triggers
- Function
- Procedure
- Index
- Materialized Query Table

Permissions to Configure the Resource

The following table lists the schema objects and system tables on which you must configure SELECT permissions for the IBM DB2 database user account:

Schema Objects	System Tables
- Tables - Views - Indexes - Packages - Procedures - Functions - Sequences - Triggers - Synonyms	- SYSCAT.SCHEMATA - SYSCAT.TRIGGERS - SYSCAT.INDEXES - SYSCAT.INDEXCOLUSE - SYSCAT.REFERENCES - SYSCAT.TABCONST - SYSCAT.KEYCOLUSE - SYSCAT.COLCHECKS - SYSCAT.DATATYPES - SYSCAT.COLUMNS - SYSCAT.TABLES - SYSCAT.FUNCTIONS - SYSCAT.PROCEDURES - SYSCAT.PACKAGES - SYSCAT.TRIGGERS - SYSCAT.SEQUENCES - SYSCAT.INDEXES - SYSCAT.CHECKS - SYSCAT.VIEWS - SYSCAT.PACKAGES - SYSCAT.FUNCDEP - SYSCAT.TRIGDEP - SYSCAT.VIEWDEP

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The following table describes the connection properties for the IBM DB2 resource type:

Property	Description
User	Name of the user account that connects to IBM DB2 database.
Password	Password for the user account that connects to IBM DB2 database.
Host	Fully qualified host name of the machine where IBM DB2 database is hosted.
Port	Port number for the IBM DB2 database.
Database	The DB2 connection URL used to access metadata from the database.

The following table describes the **Additional** and **Advanced** properties for source metadata settings on the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Select to extract metadata from the data source.
Import system objects	Specifies the system objects to import.
Schema	Specifies a list of database schema.
Source Metadata Filter	You can include or exclude tables and views from the resource run. Use semicolons (;) to separate the table names and view names. For more information about the filter field, see "Source Metadata and Data Profile Filter" on page 16.
Import stored procedures	Specifies the stored procedures to import.
Case Sensitive	Specifies that the resource is configured for case insensitivity. Select one of the following values: - True. Select this check box to specify that the resource is configured as case sensitive. - False. Clear this check box to specify that the resource is configured as case insensitive. The default value is False.

Property	Description
Agent Options	Specify the Enterprise Data Catalog Agent options to run the scanner job.
Memory	Specify the memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1. - `-DskipExternalDatabaseObjects=true`. Excludes all external objects such as schema, table, database and view.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

You can enable data discovery for an IBM DB2 resource. For more information about enabling data discovery, see the Enable Data Discovery topic.

You can enable composite data domain discovery for an IBM DB2 resource. For more information about enabling composite data domain discovery, see the Composite Data Domain Discovery topic.

IBM DB2 for z/OS

Configure an IBM DB2 for z/OS resource type to extract metadata from IBM DB2 for z/OS databases.

Objects Extracted

Enterprise Data Catalog extracts metadata of the following schema objects from an IBM DB2 for z/OS resource:

- Store Procedure
- Function
- Synonym
- Trigger
- Table
- Views
- Schema

Permissions to Configure the Resource

When you load an IBM DB2 for z/OS resource, Enterprise Data Catalog extracts metadata from the IBM DB2 for z/OS subsystem. Enterprise Data Catalog selects the metadata from the DB2 catalog tables.

Configure SELECT privileges on the following system tables in the IBM DB2 for z/OS user account that you use to connect to the IBM DB2 for z/OS data source:

System Tables	Metadata Extracted
- SYSIBM.SYSSYNONYMS - SYSIBM.SYSTABLES - SYSIBM.SYSROUTINES - SYSIBM.SYSSEQUENCES - SYSIBM.SYSDATATYPES	Schemas
- SYSIBM.SYSTABLES - SYSIBM.SYSTABLESPACE	- Tables - Views
- SYSIBM.SYSVIEWS	- Alias
- SYSIBM.SYSTABLES - SYSIBM.SYSCOLUMNS - SYSIBM.SYSTABLESPACE	Columns
- SYSIBM.SYSKEYS - SYSIBM.SYSINDEXES	Primary keys
- SYSIBM.SYSRELS - SYSIBM.SYSFOREIGNKEYS - SYSIBM.SYSINDEXES	Foreign keys
- SYSIBM.SYSCHECKS - SYSIBM.SYSCHECKDEP	Check constraints
- SYSIBM.SYSINDEXES - SYSIBM.SYSKEYS	Indexes
- SYSIBM.SYSSYNONYMS - SYSIBM.SYSTABLES	Synonyms
SYSIBM.SYSTRIGGERS	Triggers
SYSIBM.SYSTRIGGERS	Stored procedures
SYSIBM.SYSROUTINES	Functions
SYSIBM.SYSSEQUENCES	Sequences
- SYSIBM.SYSSEQUENCESDEP - SYSIBM.SYSSEQUENCES	Identity
SYSIBM.SYSDATATYPES	User-defined types

Prerequisites

To access metadata from z/OS subsystems using PowerExchange, perform the following steps:

- Verify that the PowerExchange client is installed in the domain machine.
- Include the PWX_HOME path in the `<INFA_HOME>/tomcat/temp/envars.txt` file.

For more information about installing and configuring PowerExchange, see the PowerExchange Installation and Upgrade guide.

Resource Connection Properties

The following table describes the properties for the IBM DB2 for z/OS resource:

Property	Description
Location	Node name in the dbmover.cfg file on the machine where the Catalog Service runs that points to the PowerExchange Listener on the z/OS system. **Note:** Enterprise Data Catalog uses PowerExchange for DB2 for z/OS to access metadata from z/OS subsystems.
User	Name of the user account that connects to IBM DB2 for z/OS database.
Password	Password for the user account that connects to IBM DB2 for z/OS database.
Encoding	Code page for the IBM DB2 for z/OS subsystem.
Sub System ID	Name of the DB2 subsystem.

The following table describes the **Additional** property for source metadata settings on the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Enables metadata extraction
Database	Click **Select** to specify a list of databases that you want to import. You can use one of the following options from the Select Database dialog box to import the databases: - Select from List. Select the required database from a list of available databases. - Select using regex. Provide SQL regular expressions such as * or ? to include databases that match the expression.
Schema	Click **Select** to specify the schemas that you want to import. You can use one of the following options from the Select Schema dialog box to import the schemas: - Select from List. Select the required schemas from a list of available schemas. - Select using regex. Provide SQL regular expressions such as * or ? to include schemas that match the expression. **Note:** If you use regular expression to select the databases that you want to import, you must not use regular expression to select the schemas.

Property	Description
Source Metadata Filter	You can include tables and views from the resource run. Use semicolon (;) to separate the table names and view names. When the list is empty, the resource run includes all the available tables and views. You can use regular expressions such as * and ? to specify the list of tables and views. For more information about the filter field, see "Source Metadata and Data Profile Filter" on page 16.
Case Sensitive	Specifies that the resource is configured for case sensitivity. - Select the check box to specify that the resource is configured as case sensitive. - Clear the check box to specify that the resource is configured as case insensitive. By default, the resource is configured as case sensitive.

To view the log information of the IBM DB2 for z/OS resource, see the `LDM.log` file.

You can enable data discovery for an IBM DB2 for z/OS resource. For more information about enabling data discovery, see the Enable Data Discovery topic.

You can enable composite data domain discovery for an IBM DB2 for z/OS resource. For more information about enabling composite data domain discovery, see the Composite Data Domain Discovery topic.

Note: You cannot run profiling using an IBM DB2 for z/OS resource that contains multiple databases with the same schema name. Make sure the schema names of the databases that you select need to be unique.

IBM Netezza

You need to set up multiple configuration properties when you create a resource to extract metadata from IBM Netezza databases.

Objects Extracted

Enterprise Data Catalog extracts metadata of the following objects from an IBM Netezza resource:

- Column
- Database
- Data type
- Distribution key
- Distribution key column
- Expression
- External table
- Foreign key
- Materialized view
- Primary key
- Schema
- Sequence
- Stored Procedure
- Synonym

- Table
- Unique key constraint
- View

Permissions to Configure the Resource

Configure the LIST privilege for the Netezza user to extract metadata from a Netezza database. The LIST privilege grants the Netezza user access to all database objects in the Netezza database through Enterprise Data Catalog. If required, you can assign database permissions to limit access to individual database objects.

Configure SELECT permissions for the Netezza user account on the following system views:

- _V_TABLE_DIST_MAP
- _V_TABLE
- _V_DATABASE
- _v_relation_keydata
- _v_relation_column
- _v_view
- _v_synonym
- _v_datatype
- _v_sequence

Prerequisites

Download the JDBC driver JAR file from the Internet and copy the files to the `NetezzaScanner` directory in the `<INFA_HOME>/services/CatalogService/ScannerJars/externalDependencies` directory. You need not recycle the Catalog Service.

Alternatively, you can perform the following steps to complete the prerequisites:

1. Download the JDBC driver file and copy the file to the `<INFA_HOME>/services/CatalogService/ScannerBinaries` directory.
2. Open the `<INFA_HOME>/services/CatalogService/ScannerBinaries/CustomDeployer/scannerDeployer.xml` file and add the following lines in the file:

```
<ExecutionContextProperty isLocationProperty="true"
dependencyToUnpack="netezza.zip"> <PropertyName>NetezzaScanner_DriverLocation</
PropertyName> <PropertyValue>scanner_miti/netezza/Drivers</PropertyValue> </
ExecutionContextProperty>
```

3. Save the `scannerDeployer.xml` file.
4. Recycle the Catalog Service.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.

Information	Description
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The following table describes the connection properties for the IBM Netezza resource type:

Property	Description
Host	Host name or IP address of the machine where the database management server runs.
Port	Port number for the Netezza database.
User	Name of the user account used to connect to the Netezza database.
Password	Password for the user account used to connect to the Netezza database.
Database	ODBC data source connect string for a Netezza database. Enter the data source name of the Netezza DSN if you created one.

The following table describes the **Additional** and **Advanced** properties for source metadata settings on the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Select to extract metadata from the data source.
Schema	Specifies a list of semicolon-separated database schema.
Source Metadata Filter	You can include or exclude tables and views from the resource run. Use semicolons (;) to separate the table names and view names. For more information about the filter field, see "Source Metadata and Data Profile Filter" on page 16.
Case Sensitive	Specifies that the resource is configured for case insensitivity. Select one of the following values: - True. Select this check box to specify that the resource is configured as case sensitive. - False. Clear this check box to specify that the resource is configured as case insensitive. The default value is False.

Property	Description
Memory	Specify the memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1. - `-DskipExternalDatabaseObjects=true`. Excludes all external objects such as schema, table, database and view.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

You can enable data discovery for an IBM Netezza resource. For more information about enabling data discovery, see the Enable Data Discovery topic.

You can enable composite data domain discovery for an IBM Netezza resource. For more information about enabling composite data domain discovery, see the Composite Data Domain Discovery topic.

JDBC

You can use a JDBC connection to access tables in a database.

Objects Extracted

JDBC with MySQL

Enterprise Data Catalog extracts metadata of the following schema objects from a MySQL and AS400 databases that uses JDBC for connectivity:

- Tables
- Constraints (NOT NULL , UNIQUE , CHECK , PRIMARY KEY , FOREIGN KEY , DEFAULT)
- Clone Table
- Indexes
- Views
- Triggers
- Function
- Procedure
- Materialized View
- MV Trigger

JDBC with AS400

- Tables
- Constraints (NOT NULL , UNIQUE , CHECK , PRIMARY KEY , FOREIGN KEY , DEFAULT)
- Alias
- Duplicate Table
- Sequences
- Views
- Triggers
- Function
- Procedure
- Index
- User defined data types

Permissions to Configure the Resource

You must configure the CONNECT and SELECT_CATALOG_ROLE permissions for the database user account that you use to access the MySQL or AS400 databases.

Prerequisites

Download the JDBC driver JAR file from the Internet and copy the files to the `JDBCScanner` directory in the `<INFA_HOME>/services/CatalogService/ScannerJars/externalDependencies` directory. You need not recycle the Catalog Service.

Alternatively, you can perform the following steps to complete the prerequisites:

1. Download the JDBC driver file and copy the file to the `<INFA_HOME>/services/CatalogService/ScannerBinaries` directory.
2. Open the `<INFA_HOME>/services/CatalogService/ScannerBinaries/CustomDeployer/scannerDeployer.xml` file and add the following lines in the file:

```
<ExecutionContextProperty isLocationProperty="true"
dependencyToUnpack="genericJDBCJars.zip"> <PropertyName>JDBCScanner_DriverLocation</
PropertyName> <PropertyValue>scanner_miti/genericJDBC/Drivers</PropertyValue> </
ExecutionContextProperty>
```

3. Save the `scannerDeployer.xml` file.

4. Recycle the Catalog Service.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The following table describes the connection properties for the JDBC resource:

Property	Description
Driver class	Name of the JDBC driver class. For more information on JDBC driver class examples, refer to "Examples of JDBC Driver Classes" on page 219.
URL	Connection string to connect to the database.
User	Database user name.
Password	Password for the database user name.
Agent URL	Optional. URL to the Enterprise Data Catalog agent that runs on a Microsoft Windows Server. **Note:** Make sure that you specify the URL in the following format: `http://<hostname>:<connector_port>/MIMBWebServices`

Note: To extract metadata from multiple schemas of a source using the JDBC resource type, you can specify semicolon-separated schema names when you create the resource. You can type in the multiple schema names in the **Schema** field on the **Metadata Load Settings** page.

The following table describes the **Additional** and **Advanced** properties for source metadata settings on the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Select to extract metadata from the data source.
Catalog	Catalog name. **Note:** You cannot use Catalog option for JDBC or ODBC sources.
Schema	Specifies a list of schemas to import.

Property	Description
Source Metadata Filter	You can include or exclude tables and views from the resource run. Use semicolons (;) to separate the table names and view names. For more information about the filter field, see "Source Metadata and Data Profile Filter" on page 16.
Case sensitivity	Select one of the following options to specify if the database is configured for case sensitivity: - Auto. If you select this option, the resource uses the JDBC API and tries to find if the database is configured for case sensitivity. If the JDBC API is not available for the resource, the resource uses the not case sensitive mode. - Case Sensitive. Select this option to specify that the database is configured for case sensitivity. - Case Insensitive. Select this option to specify that the database is configured as case insensitive.
View definition extracting SQL	Specifies the database specific SQL query to retrieve the view definition text.
Synonyms lineage SQL	Specifies the database specific SQL query to retrieve the synonym lineage. The following are the two columns that the query returns: - Full Synonym Name - Full Table Name
Optional Scope	Specifies the database object types to import, such as Tables and Views, Indexes, and Procedures. Specify a list of optional database object types that you want to import. The list can have zero or more database object types, which are separated by semicolons. For example, Keys and Indexes, and Stored Procedures.
Import stored procedures	Specifies the stored procedures to import. The default value is True or False whatever the case might be.
Agent Options	Specify the Enterprise Data Catalog Agent options to run the scanner job.
Memory	Specify the memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article.

Property	Description
Custom Options	Custom JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. Verify that the value is a number. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1. - `-DskipExternalDatabaseObjects=true`. Excludes all external objects such as schema, table, database and view. - `-DcustomSampling=<Sampling Option>`. Overrides the SQL query and limits or samples the number of rows read from the database. The **Sampling Option** of the resource is overridden with this custom option that you set for JDBC connection. See the following samples of various databases for reference: - `-DcustomSampling='fetch first 10000 ROWS ONLY'` fetches the first 10000 rows from the PostgreSQL database. - `-DcustomSampling='LIMIT 10000'` limits the number of rows read from the Vertica database to 10000. - `-DcustomSampling='TABLESAMPLE(10 PERCENT)'` samples the Databricks table to 10 percent. **Note:** This argument is applicable for column profiling and data domain discovery and there is no impact on similarity discovery.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

You can enable data discovery for a JDBC resource. For more information about enabling data discovery, see the Enable Data Discovery topic.

You can enable composite data domain discovery for a JDBC resource. For more information about enabling composite data domain discovery, see the Composite Data Domain Discovery topic.

Microsoft SQL Server

You can configure a Microsoft SQL Server resource type to extract metadata from Microsoft SQL Server databases. Verify that you configure the VIEW DEFINITION permission for the SQL database and configure the SELECT permission for the *sys.sql_expression_dependencies* of the database.

Objects Extracted

Enterprise Data Catalog extracts metadata of the following schema objects from a Microsoft SQL Server resource:

- Tables
- Views
- Extended stored procedure
- Table (user-defined)
- Synonym

- UNIQUE constraint
- Triggers
- Procedures
- Index
- SQL table-valued-function
- Users
- Partition Scheme
- Partition Function
- XML Schema Collections
- User defined Data type
- Rules
- Defaults
- Constraints
- SQL_INLINE_TABLE_VALUED_FUNCTION
- SQL_SCALAR_FUNCTION
- SEQUENCE

Permissions to Configure the Resource

Before you create a Microsoft SQL Server resource, configure the permissions for the Microsoft SQL Server database user account that you use to connect to Microsoft SQL Server. Enterprise Data Catalog uses SQL Server authentication to connect to the Microsoft SQL Server database. The user account that you use to connect to Microsoft SQL Server must be an SQL Server login account. Configure the VIEW DEFINITION permission for the user account. Configuring this permission lists all the schemas from where the user can load metadata. You must also configure the SELECT permission for the sys.sql_expression_dependencies for the database.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The following table describes the properties for the Microsoft SQL Server resource:

Property	Description
User	Name of the Microsoft SQL Server user account that connects to the Microsoft SQL Server database. The Catalog Service uses Microsoft SQL Server authentication to connect to the Microsoft SQL Server database.
Password	Password for the user account that connects to the Microsoft SQL Server database.
Host	Host name of the machine where Microsoft SQL Server runs.
Port	Port number for the Microsoft SQL Server database engine service.
Database	Name of the Microsoft SQL Server database. **Note:** The database name and the resource name must not be same.
Instance	Microsoft SQL Server instance name.

The following table describes the **Additional** and **Advanced** properties for source metadata settings on the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Select to extract metadata from the data source.
Import system objects	Specifies the system objects to import. The default value is True or False whatever the case might be.
Schema	Specifies a list of semicolon-separated database schema.
Source Metadata Filter	You can include or exclude tables and views from the resource run. Use semicolons (;) to separate the table names and view names. For more information about the filter field, see "Source Metadata and Data Profile Filter" on page 16.
Import stored procedures	Specifies the stored procedures to be imported. The default value is True or False whatever the case might be.
Case Sensitive	Specifies that the resource is configured for case insensitivity. Select one of the following values: - True. Select this check box to specify that the resource is configured as case sensitive. - False. Clear this check box to specify that the resource is configured as case insensitive. The default value is False.
Memory	Specify the memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal

Property	Description
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1. - `-DskipExternalDatabaseObjects=true`. Excludes all external objects such as schema, table, database and view.
Track Data Source Changes	Vew metadata source change notifications in Enterprise Data Catalog.

You can enable data discovery for a Microsoft SQL Server resource. For more information about enabling data discovery, see the Enable Data Discovery topic. You can enable composite data domain discovery for a Microsoft SQL Server resource. For more information about enabling composite data domain discovery, see the Composite Data Domain Discovery topic.

Configuring Enterprise Data Catalog Agent for the Resource on a Windows-authenticated Server

To configure the Enterprise Data Catalog Agent for Microsoft SQL Server, you can perform the following steps:

1. Verify that you installed the Enterprise Data Catalog Agent as a Windows service. For information on how to install the agent, see Appendix D: Enterprise Data Catalog Agent in the *Catalog Administrator Guide*. To configure the service, you can specify the Active Directory ID used to access the Microsoft SQL Server database as the login ID.
2. Stop the Apache Tomcat 9.0 MITI service.
3. Download the JDBC driver for Microsoft SQL Server (version 7.4.1) to a temporary location.
4. Run the .exe file to extract the files to the `Microsoft JDBC Driver 7.4.1 for SQL Server` folder in the temporary location.
5. From the extracted folder, copy the `\sqljdbc_7.4.1\enu\auth\x64\sqljdbc_auth.dll` file to the following directories on your machine:
   - `C:\Windows\System32`
   - `<EDC Agent Installation_Directory>\jre\bin`
6. Copy the `\sqljdbc_7.4.1\enu\jre8\sqljdbc42.jar` file to the `<EDC Agent Installation_Directory>\java\Jdbc\mssql` directory on your machine.
7. In the `<EDC Agent Installation_Directory>\java\Jdbc\mssql` directory, rename the `sqljdbc41.jar` file to `sqljdbc41.jar.old`.
8. In the Windows Services console, double-click on the Apache Tomcat 9.0 MITI service, select the **Log On** tab.
9. Enter the service account credentials to connect to the Microsoft SQL Server database on the domain, and click OK to start the service.
10. Add the service account to the Administrators group on your machine.

11. In Catalog Administrator, create a Microsoft SQL Server resource with the following Enterprise Data Catalog Agent URL: `http://<HostName>:<Port>/MIMBWebServices/`
12. Run the resource.

Viewing Lineage of External Schema and Database Objects

You can view detailed lineage of external schema and database objects in Enterprise Data Catalog. To view the lineage of the external schema and database objects, use the **Connection Assignment** option.

Note: For Connection Assignment, make sure that the destination or target data sources that represent the external schema and database objects are available in Enterprise Data Catalog.

Oracle

Configure an Oracle resource to extract metadata from Oracle databases.

Objects Extracted

Enterprise Data Catalog extracts metadata of the following schema objects from an Oracle resource:

- Table
- Synonym
- View
- Different datatypes
- Precision and scale information for all data types
- Comments
- Trigger
- Function
- Procedure
- Condition
- Integrity constraints
- Index-organized table
- Materialized View

Note: The following points are applicable from version 10.4.1.2 onwards:

- Enterprise Data Catalog extracts the pluggable database object if you configure the Oracle Multitenant Database.
- If you configure the Oracle Multitenant Database, before you rerun the Oracle resource, you must take a backup of the resource enrichments using the BackupContents command.

Permissions to Configure the Resource

You must configure the CONNECT and SELECT_CATALOG_ROLE permissions for the database user account that you use to access the Oracle database.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The following table describes the properties for the Oracle resource type:

Property	Description
User	Name of the user account that connects to the Oracle database.
Password	Password for the user account that connects to the Oracle database.
Host	Fully qualified host name of the machine where the Oracle database is hosted.
Port	Port number for the Oracle database engine service.
Service	Unique identifier for the Oracle database server. **Note:** You must not use the System ID (SID) as the unique identifier for the Oracle database server.

The following table describes the **Additional** and **Advanced** properties for source metadata settings on the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Select to extract metadata from the data source.
Import system objects	Specifies the system objects to import. The default value is True or False whatever the case might be.
Schema	Specifies a list of semicolon-separated database schema.
Source Metadata Filter	You can include or exclude tables and views from the resource run. Use semicolons (;) to separate the table names and view names. For more information about the filter field, see "Source Metadata and Data Profile Filter" on page 16.
Import stored procedures	Specifies the stored procedures to import. The default value is True or False whatever the case might be.

Property	Description
Import Private and Public Synonyms	Specifies private and public synonyms to import. Select one of the following values: - True. Select this check box to import private and public synonyms. - False. Clear this check box if you do not want to import the private and public synonyms. The default value is False.
Case Sensitive	Specifies that the resource is configured for case insensitivity. Select one of the following values: - True. Select this check box to specify that the resource is configured as case sensitive. - False. Clear this check box to specify that the resource is configured as case insensitive. The default value is False.
Import Database Name	Extracts the database name of the data source from the `v$database` variable. Enable the option when you want to extract metadata from an Oracle Real Application Clusters (Oracle RAC) data source.
Agent Options	Specify the Enterprise Data Catalog Agent options to run the scanner job.
Memory	Specify the memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1. - `-DskipExternalDatabaseObjects=true`. Excludes all external objects such as schema, table, database and view.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

You can enable data discovery for an Oracle resource. For more information about enabling data discovery, see the Enable Data Discovery topic.

You can enable composite data domain discovery for an Oracle resource. For more information about enabling composite data domain discovery, see the Composite Data Domain Discovery topic.

SAP HANA

Create a resource based on the SAP HANA resource type to extract metadata from an SAP HANA data source. You need to specify the connection details when you configure the resource.

Objects Extracted

Enterprise Data Catalog extracts metadata from the following objects in an SAP HANA data source:

- Table
- Synonym
- Column
- Attribute
- View
- Graphical Calculation View
- View Column
- Schema
- Stored procedures and functions object count

  Note: The resource does not extract lineage information for stored procedures and functions.
- SQL conditions
- Measures

Permissions to Configure the Resource

Verify that you configure the following permissions for the username that you use to connect to the database:

- `SELECT METADATA` on the schemas for which you want to import metadata.
- `SELECT` on the `_SYS_REPO.ACTIVE_OBJECT` table and the `_SYS_BI` and `_SYS_BIC` system schemas.

Prerequisites

Download the `ngdbc.jar` driver JAR files from the Internet and copy the files to the `SAPHANAScanner` directory in the `<INFA_HOME>/services/CatalogService/ScannerJars/externalDependencies` directory. You need not recycle the Catalog Service.

Alternatively, you can perform the following steps to complete the prerequisites:

1. Download the `ngdbc.jar` JAR files from the Internet and include the drivers in the `saphanaJars.zip` file.
2. Copy the `saphanaJars.zip` file to the `<INFA_HOME>/services/CatalogService/ScannerBinaries` directory.
3. Open the `<INFA_HOME>/services/CatalogService/ScannerBinaries/CustomDeployer/scannerDeployer.xml` file and add the following lines in the file:

   ```
   <ExecutionContextProperty isLocationProperty="true"
   dependencyToUnpack="saphanaJars.zip"> <PropertyName>SAPHANAScanner_DriverLocation</
   PropertyName> <PropertyValue>scanner_miti/SAPHANA/Drivers</PropertyValue> </
   ExecutionContextProperty>
   ```
4. Save the `scannerDeployer.xml` file.
5. Recycle the Catalog Service.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The following table describes the resource connection properties:

Property	Description
User	Username to connect to the database.
Password	Password to connect to the database.
Host	Host name or IP address of the machine that hosts the database.
Port	Port number of the machine that hosts the database. Default is 39013.
Database	Name of the database instance.

The following table describes the **Additional** and **Advanced** properties for source metadata settings on the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Extracts metadata from the data source.
Import system objects	Imports system objects.
Schema	Specify the schemas for which you want to import metadata. If you do not specify any schema, Enterprise Data Catalog imports all schemas in the database. To select the required schemas, perform the following steps: 1. Click **Select**. The **Select Schema** dialog box appears. 2. Select the list of schemas from the **Schema** list. 3. Click **Select using regex** if you want to specify an SQL query to identify the schema names that match the pattern in the query.

Property	Description
Source Metadata Filter	Specify a list of tables that you want to include or exclude when the resource collects metadata. Default is **All**, which includes all the tables in the database. You can specify a list of tables that you want to include or exclude by separating the table names with a semicolon (;). You can use expressions similar to SQL expressions along with the regular operators such as AND, OR, and NOT to include or exclude tables in the list of tables.
Import Stored Procedures	Select the option to import program objects from the database such as stored procedures and functions.
Case Sensitive	Select the option if you want to retain the case sensitivity for metadata in the lineage diagram.
Package	Specify a list of package names that you want to import. Package names are case sensitive. Ensure that you separate package names with a semicolon (;). You can leave the option blank to import all the packages. If you want to specify sub package names, ensure that you specify a sub package name in the following format: `<package name>.<sub package name>`.
Memory	Specify the memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to specific values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the YARN environment. Use a comma to separate multiple key-pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1. - `-DINFA_TRUSTSTORE=<trust store path>`. Path to the Informatica truststore file. - `-DINFA_TRUSTSTORE_PASSWORD=<trust store encrypted key>`. Encrypted truststore password.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

Sybase

You can configure a Sybase resource to extract metadata from Sybase databases.

Objects Extracted

Enterprise Data Catalog extracts metadata of the following schema objects from a Sybase resource:

- Table
- View
- Clustered Index
- Procedure
- Triggers
- Rule
- Unique Index
- Non Clustered Index
- Index
- Function
- User defined data type
- Schema
- Column
- ForeignKey
- Stored Procedure
- PrimaryKeyConstraint

Permissions to Configure the Resource

Before you create a Sybase resource, configure the permissions for the Sybase ASE database user account that you use to connect to the Sybase ASE database.

Configure SELECT permissions on the following schema objects:

- Tables
- Views
- Columns
- Procedures

Configure SELECT permissions on the following system tables:

- sysobjects
- syscomments
- sysdepends
- syscolumns
- sysindexes
- sysconstraints
- sysreferences
- sysusers

- systypes
- master.dbo.sysservers
- master.dbo.sysconfigures
- @@<servername

Prerequisites

Download JDBC driver file

Complete the prerequisites to create a Sybase resource, download the JDBC driver JAR file from the Internet and copy the files to the `SybaseScanner` directory in the `<INFA_HOME>/services/CatalogService/ScannerJars/externalDependencies` directory. You need not recycle the Catalog Service.

Alternatively, you can perform the following steps to complete the prerequisites:

1. Download the JDBC driver file and copy the file to the `<INFA_HOME>/services/CatalogService/ScannerBinaries` directory.
2. Open the `<INFA_HOME>/services/CatalogService/ScannerBinaries/CustomDeployer/scannerDeployer.xml` file and add the following lines in the file:

```
<ExecutionContextProperty isLocationProperty="true"
dependencyToUnpack="sybase_jars.zip">
<PropertyName>SybaseScanner_DriverLocation</PropertyName>
<PropertyValue>scanner_miti/sybase_jars/Drivers</PropertyValue> </
ExecutionContextProperty>
```

3. Save the `scannerDeployer.xml` file.
4. Recycle the Catalog Service.

Update the JDBC driver file

To replace the obsolete JDBC driver jar with latest driver jar, perform the following steps:

1. Open the `<INFA_HOME>/services/CatalogService/ScannerBinaries/` directory.
2. Replace the `sybase-jdbc3` driver jar with `sybase-jdbc4` in the sybase_jars.zip file.
3. Recycle the Catalog Service.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The following table describes the properties for the Sybase resource:

Property	Description
Host	Host name of the machine where Sybase database is hosted.
Port	Port number for the Sybase database engine service.
User	Database user name.
Password	The password for the database user name.
Database	Name of the database.

The following table describes the **Additional** and **Advanced** properties for source metadata settings on the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Select to extract metadata from the data source.
Schema	Specify a list of database or scheme to import.
Imported stored procedures	Specifies the stored procedures to import. The default value is True or False whatever the case might be.
Case Sensitive	Specifies that the resource is configured for case insensitivity. Select one of the following values: - True. Select this check box to specify that the resource is configured as case sensitive. - False. Clear this check box to specify that the resource is configured as case insensitive. The default value is False.
Memory	Specify the memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal

Property	Description
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1. - `-DskipExternalDatabaseObjects=true`. Excludes all external objects such as schema, table, database and view.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

You can enable data discovery for a Sybase resource. For more information about enabling data discovery, see the Enable Data Discovery topic.

You can enable composite data domain discovery for a Sybase resource. For more information about enabling composite data domain discovery, see the Composite Data Domain Discovery topic.

Teradata

Teradata is one of the ETL resource types in Enterprise Data Catalog. Configure a Teradata resource type to extract metadata from Teradata databases.

Objects Extracted

- User
- Tables
- Column
- Datatype
- Views
- Index
- IndexColumn
- PrimaryKeyConstraint
- ForeignKey
- UniqueConstraint
- CheckConstraint
- Procedure
- TableTrigger
- Database

- Functions
- Macros
- Schema
- Stored Procedures

Permissions to Configure the Resource

Configure the permissions for the Teradata database user account that you use to connect to the Teradata database.

The following table lists the schema objects and system tables on which you must configure SELECT permissions for the Teradata database user account:

Schema Objects	System Tables
- Tables - Views - Indexes - Packages - Procedures - Functions - Sequences - Triggers - Synonyms - Database	- DBC.DBCINFO - DBC.TABLETEXT - DBC.HOSTSINFO - DBC.DATABASES - DBC.TABLES - DBC.COLUMNS - DBC.ALL_RI_CHILDREN - DBC.INDICES - DBC.TRIGGERS - DBC.SHOWTBLCHECKS - DBC.TVFIELDS - DBC.TVM - DBC.UDTINFO - TDQCD.STATSRECS - DBC.DBASE - DBC.IDCOL - DBC.UDFINFO

Prerequisites

To complete the prerequisites to create a Teradata resource, download the JDBC driver JAR file from the Internet and copy the files to the `TeradataScanner` directory in the `<INFA_HOME>/services/CatalogService/ScannerJars/externalDependencies` directory. You need not recycle the Catalog Service.

Alternatively, you can perform the following steps to complete the prerequisites:

1. Download the JDBC driver files and copy the file to the `<INFA_HOME>/services/CatalogService/ScannerBinaries` directory.
2. Open the `<INFA_HOME>/services/CatalogService/ScannerBinaries/CustomDeployer/scannerDeployer.xml` file and add the following lines in the file:

```
<ExecutionContextProperty isLocationProperty="true"
dependencyToUnpack="teradataJars.zip"> <PropertyName>TeradataScanner_DriverLocation</
PropertyName> <PropertyValue>scanner_miti/teradata/Drivers</PropertyValue> </
ExecutionContextProperty>
```

3. Save the `scannerDeployer.xml` file.
4. Recycle the Catalog Service.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The following table describes the properties for the Teradata resource type:

Property	Description
User	Name of the user account that connects to the Teradata database.
Password	Password for the user account that connects to the Teradata database.
Host	Fully qualified host name of the machine where the Teradata database is hosted. **Note:** To connect to a Teradata resource through an LDAP server, you must specify / LOGMECH=LDAP to the host name.

The following table describes the **Additional** and **Advanced** properties for source metadata settings on the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Select to extract metadata from the data source.
Import system objects	Specifies the system objects to import. The default value is True or False whatever the case might be.
Schema	Specifies a list of semicolon-separated database schema.
Source Metadata Filter	You can include or exclude tables and views from the resource run. Use semicolons (;) to separate the table names and view names. For more information about the filter field, see "Source Metadata and Data Profile Filter" on page 16.
Import stored procedures	Specifies the stored procedures to import. The default value is True or False whatever the case might be.
Case Sensitive	Specifies that the resource is configured for case insensitivity. Select one of the following values: - True. Select this check box to specify that the resource is configured as case sensitive. - False. Clear this check box to specify that the resource is configured as case insensitive. The default value is False.
Agent Options	Specify the Enterprise Data Catalog Agent options to run the scanner job.

Property	Description
Fetch Views Data Types	Specifies to import the views data type.
Memory	Specify the memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1. - `-DskipExternalDatabaseObjects=true`. Excludes all external objects such as schema, table, database and view.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

You can enable data discovery for a Teradata resource. For more information about enabling data discovery, see the Enable Data Discovery topic.

You can enable composite data domain discovery for a Teradata resource. For more information about enabling composite data domain discovery, see the Composite Data Domain Discovery topic.

Configure Case Sensitivity for Database Management Resources

Before you change the case-sensitivity configured for any database management resource, you must first purge the resource.

After you purge the resource, you can run the resource and perform connection assignments.

If you want to retain the existing enrichments associated with the resource, you can perform a bulk export of the enrichments before you purge the resource.

Note: A bulk export does not export all collaboration categories, such as ratings or comments.

If you do not want to purge the resource, you can select or clear the Case-sensitivity option of the resource to revert the changes.

CHAPTER 10

Configuring Hadoop Multi-Part File Handler Resources

This chapter includes the following topics:

- Overview of Hadoop Multi-Part File Handler Resources , 184
- Hadoop Multi-Part File Handler, 184

Overview of Hadoop Multi-Part File Handler Resources

The Hadoop Multi-Part File Handler resource group includes the Hadoop Multi-Part File Handler resource. The Hadoop Multi-Part File Handler resource handles the link between Hadoop Multi-Part files to their respective targets. When you run the resource, the Hadoop Multi-Part File Handler scanner extracts and migrates the metadata results in the associated resource to the catalog. You can view the migrated results in Enterprise Data Catalog.

Hadoop Multi-Part File Handler

Use the Hadoop Multi-Part File Handler resource to fetch lineage from a combined multi-part file to a relational target.

Prerequisites

Perform the following steps to complete the prerequisites:

- In the Data Integration Service where the data engineering mappings are deployed, set the **HDFSRetainOriginialTargetFile** custom property value as **True**.
- Configure and run the relational and the HDFS metadata sources that are used to create the data engineering mappings.
- Configure and run the Infomatica Platform metadata resource.
- For the required relational and HDFS metadata sources, assign the connections using the **Connection Assignment** option in the Catalog.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
HDFS Resource Name	Name of the HDFS resource.
Source Relational Resource Name	Name of the relational source.
Target Relational Resource Name	Name of the target relational source.

The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Source Metadata	Extracts metadata from the data source.
Memory	The memory required to run the scanner job. Select one of the following values based on the data set size imported: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure the scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.

CHAPTER 11

Configuring File Management Resources

This chapter includes the following topics:

- Overview of File Management Resources, 186
- File System, 186
- OneDrive, 190
- Microsoft Sharepoint, 193

Overview of File Management Resources

The File Management resource group includes all the resources related to file management. When you create and run a file management resource, the file management scanner extracts and migrates the metadata results from the associated resource to the catalog. You can view the migrated results in Enterprise Data Catalog.

You can create the following file management resources in Enterprise Data Catalog:

- File System
- OneDrive
- Microsoft Sharepoint

File System

You can choose the **File System** resource to import metadata from files in Windows and Linux file systems into the catalog. You can choose the **Local File** option to import the metadata from the files located on a local machine. You can choose the **SFTP** or Secure File Transfer Protocol option to import the files located

on a remote Linux machine and choose the **SMB/CIFS** or Server Message Block/Common Internet File System option to import the files located on a remote Windows machine.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

In the **General** tab, you can enter a name and brief description for the resource. In the **Additional properties** section, choose **File System** as the resource.

Choose the following options in the **Connection Properties** section to import the metadata from the files located on the local machine:

Property	Description
File Protocol	Choose **Local File** to scan the files in the local Windows or Linux machine.
Path	Specify the absolute path of the file from where you want to import the metadata into the catalog. Make sure that the absolute path that you enter in this field is available on the machine where you have installed Enterprise Data Catalog and all the cluster nodes.
Test Connection	Click **Test Connection** to verify the connection to the specified location.

Choose the following options in the **Connection Properties** section to import the metadata from the files located on a Linux machine:

Property	Description
File Protocol	Choose **SFTP** to scan the files on a remote Linux machine.
User Name	Enter the user name to access the Linux machine.
Password	Enter the password to access the Linux machine.
Host	Specify the host name or IP address of the Linux machine.
Port	Specify the port number on which the SFTP file protocol is configured on the machine. The default value is 22.
Path	Specify the network or shared path of the file from where you want to import the metadata into the catalog.
Test Connection	Click **Test Connection** to verify the connection to the specified location.

Choose the following options in the **Connection Properties** section to import the metadata from the files located on a Windows machine:

Property	Description
File Protocol	Choose **SMB/CIFS** to scan the files located on a remote Windows machine.
User Name	Enter the user name to access the Windows machine.
Password	Enter the password to access the Windows machine.
Physical Location	Applies to flat files. Enter the absolute path of the shared directory on the Windows machine. For example, assume that the files are located in the `c:\user1\1021\SMB\test_files\test1` and `c:\user1\1021\SMB\test_files\test2` folders. You map the `c:\user1\1021\SMB` location which results in the `\\SMB\` shared directory. In this scenario, enter `\\SMB\` in the Physical Location field. For more information about how Enterprise Data Catalog uses the physical location field for lineage, see *Appendix F - Lineage between PowerCenter Resources and Flat Files*.
Host	Specify the host name or IP address of the machine.
Path	Specify the network or shared path from where you want to import the metadata into the catalog. For example, enter `\\SMB\test_files\` in the Path field.
Test Connection	Click **Test Connection** to verify the connection to the specified location.

The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Source Metadata	Extracts and ingests metadata from the data source.
File Types	Select any or all of the following file types from which you want to extract metadata: - All. Use this option to specify if you want to extract metadata from all file types. - Select. Use this option to specify that you want to extract metadata from specific file types. Perform the following steps to specify the file types: 1. Click **Select**. The **Select Specific File Types** dialog box appears. 2. Select the required files from the following options: - Extended unstructured formats. Use this option to extract metadata from file types such as audio files, video files, image files, and ebooks. - Structured file types. Use this option to extract metadata from file types such as JSON, XML, text, and delimited files. - Unstructured file types. Use this option to extract metadata from file types such as Microsoft Excel, Microsoft PowerPoint, Microsoft Word, web pages, compressed files, emails, and PDF. 3. Click Select. **Note:** You can select **Specific File Types** option in the dialog box to select files under all the categories.
Treat Files Without Extension As	Select one of the following options to identify files without an extension: - None - Avro - Parquet

Property	Description
Enter File Delimiter	You can specify one of the following delimiters: - Single-character delimiter. Specify the file delimiter if the file from which you extract metadata uses a delimiter other than the following delimiters: - Comma (,) - Horizontal tab (\t) - Semicolon (;) - Colon (:) - Pipe symbol (|) Verify that you enclose the delimiter in single quotes. For example, `'$'`. Use a comma to separate multiple delimiters. For example, `'$','%','&'` - Multi-character delimiter. Specify multiple characters as a delimiter within single quotes for CSV files. For example, `'|$#'`. If you specify a multi-character delimiter, do not specify any other delimiter for the scanner. When you run a file system resource with a multi-character delimiter, the scanner considers the files that contain the delimiter as valid CSV files and the remaining files as unstructured files while processing the files. The following resources do not support multi-character delimiter for CSV files in Enterprise Data Catalog: - File System resource that uses SFTP or SMB/CIFS protocol. - Amazon S3 - OneDrive - SharePoint - Microsoft Azure Blob Storage - Azure Data Lake Store
Other File Types	Extracts basic file metadata such as size of the file, path to the file, and time stamp information from other file types.
First Level Directory	Specify a directory or a list of directories under the source directory. If you leave this option blank, Enterprise Data Catalog imports all the files from the specified source directory. To specify a directory or a list of directories, you can perform the following steps: 1. Click **Select...**. The **Select First Level Directory** dialog box appears. 2. Select the required directories using one of the following options: - **Select from list**: select the required directories from a list of directories. - **Select using regex**: provide an SQL regular expression to select schemas that match the expression. **Note:** If you are selecting multiple directories, you must separate the directories using a semicolon (;).
Recursive scan	Recursively scans the subdirectories under the selected first-level directories. Recursive scan is required for partitioned file discovery.
Case Sensitive	Specifies that the resource is configured for case sensitivity. Select one of the following values: - True. Select this check box to specify that the resource is configured as case sensitive. - False. Clear this check box to specify that the resource is configured as case insensitive. The default value is True.
Memory	The memory required to run the scanner job. Select one of the following values based on the data set size imported: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal

Property	Description
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value should be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

You can enable data discovery for a **File System** resource. For more information about enabling data discovery, see the Enable Data Discovery topic.

You can enable composite data domain discovery for a File System resource. For more information about enabling composite data domain discovery, see the Composite Data Domain Discovery topic.

OneDrive

OneDrive is a file hosting and synchronization service operated by Microsoft as part of its suite of online office services.

Objects Extracted

Enterprise Data Catalog extracts only files from OneDrive source.

Permissions to Configure the Resource

Configure `read` permission on the Microsoft OneDrive data source for the user account that you use to access the data source.

Supported File Types

The OneDrive resource enables you to extract metadata from structured, unstructured, and extended unstructured files.

The structured files supported are:

- AVRO files
- Delimited files
- Text files
- JSON files
- Parquet files
- XML files

The unstructured files supported are:

- Apple files
- Compressed files
- Email

The extended unstructured files are:

- VB files
- ASP files
- TIF files
- LOG files
- CSS files
- ASPX files
- DLL files
- GIF files
- SQL files

Assign `read` permission to the files to extract metadata.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The following table describes the connection properties:

Property	Description
OneDrive URL	URL to access OneDrive.
User Name	User name to connect to OneDrive.
Password	Password to connect to OneDrive
Source Directory	The source directory from where all the files are scanned.

The following table describes the **Additional** and **Advanced** properties for source metadata settings on the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Extracts and ingests metadata from the data source.
File Types	Select any or all of the following file types from which you want to extract metadata: - All. Use this option to specify if you want to extract metadata from all file types. - Select. Use this option to specify that you want to extract metadata from specific file types. Perform the following steps to specify the file types: 1. Click **Select**. The **Select Specific File Types** dialog box appears. 2. Select the required files from the following options: - Extended unstructured formats. Use this option to extract metadata from file types such as audio files, video files, image files, and ebooks. - Structured file types. Use this option to extract metadata from file types, such as Avro, Parquet, JSON, XML, text, and delimited files. - Unstructured file types. Use this option to extract metadata from file types, such as Apple files, Microsoft Excel, Microsoft PowerPoint, Microsoft Word, web pages, compressed files, emails, and PDF. 3. Click Select. **Note:** You can select **Specific File Types** option in the dialog box to select files under all the categories.
Treat Files Without Extension As	Select one of the following options to identify files without an extension: - None - Avro - Parquet
Enter File Delimiter	Specify the file delimiter if the file from which you extract metadata uses a delimiter other than the following list of delimiters: - Comma (,) - Horizontal tab (\t) - Semicolon (;) - Colon (:) - Pipe symbol (|) Verify that you enclose the delimiter in single quotes. For example, `'$'`. Use a comma to separate multiple delimiters. For example, `'$','%','&'`
Other File Types	Extracts basic file metadata such as size of the file, path to the file, and time stamp information from other file types.
First Level Directory	Specify a directory or a list of directories under the source directory. If you leave this option blank, Enterprise Data Catalog imports all the files from the specified source directory. To specify a directory or a list of directories, you can perform the following steps: 1. Click **Select...**. The **Select First Level Directory** dialog box appears. 2. Select the required directories using one of the following options: - **Select from list**: select the required directories from a list of directories. - **Select using regex**: provide an SQL regular expression to select schemas that match the expression. **Note:** If you are selecting multiple directories, you must separate the directories using a semicolon (;).
Include Subdirectory	Select this option to import all the files in the subdirectories under the source directory.

Property	Description
Case Sensitive	Specifies that the resource is configured for case sensitivity. Select one of the following values: - True. Select this check box to specify that the resource is configured as case sensitive. - False. Clear this check box to specify that the resource is configured as case insensitive. The default value is True.
Memory	The memory required to run the scanner job. Select one of the following values based on the data set size imported: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

You can enable data discovery for an OneDrive resource. For more information about enabling data discovery, see the Enable Data Discovery topic.

You can enable composite data domain discovery for an OneDrive resource. For more information about enabling composite data domain discovery, see the Composite Data Domain Discovery topic.

Microsoft Sharepoint

SharePoint is a web-based collaborative platform that integrates with Microsoft Office. SharePoint is primarily a document management and storage system that is configurable as required.

Objects Extracted

Enterprise Data Catalog extracts only files from the Microsoft Sharepoint data source.

Permissions to Configure the Resource

Configure the `read` permission on the Microsoft SharePoint data source for the user account that you use to access the data source.

Supported File Types

Microsoft Sharepoint resource enables you to extract metadata from structured, unstructured, and extended unstructured files.

The structured files supported are:

- AVRO files
- Delimited files
- Text files
- JSON files
- Parquet files
- XML files

The unstructured files supported are:

- Apple files
- Compressed files
- Email

The extended unstructured files are:

- VB files
- ASP files
- TIF files
- LOG files
- CSS files
- ASPX files
- DLL files
- GIF files
- SQL files

Assign `read` permission to the files to extract metadata.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The following table describes the connection properties:

Property	Description
SharePoint URL	URL to access SharePoint.
User Name	User name to access SharePoint.

Property	Description
Password	Password to access SharePoint.
SharePoint Content Type	Specifies the type of SharePoint content. Select one of the following types of SharePoint content: - All. Scans contents from both list and library Sharepoint content type. - SharePoint List. Scans contents from the list SharePoint content type. - SharePoint Library. Scans contents from the library SharePoint content type.
Enable Subsite Scan	Select this option to scan subsites from the SharePoint site.
Include Nested Subsites	Specify this option to scan the nested subsites within the top-level subsite. This option applies when you select the Enable Subsite Scan option.

The following table describes the **Additional** and **Advanced** properties for source metadata settings on the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Extracts and ingests metadata from the data source.
File Types	Select any or all of the following file types from which you want to extract metadata: - All. Use this option to specify if you want to extract metadata from all file types. - Select. Use this option to specify that you want to extract metadata from specific file types. Perform the following steps to specify the file types: 1. Click **Select**. The **Select Specific File Types** dialog box appears. 2. Select the required files from the following options: - Extended unstructured formats. Use this option to extract metadata from file types such as audio files, video files, image files, and ebooks. - Structured file types. Use this option to extract metadata from file types, such as Avro, Parquet, JSON, XML, text, and delimited files. - Unstructured file types. Use this option to extract metadata from file types, such as Apple files, Microsoft Excel, Microsoft PowerPoint, Microsoft Word, web pages, compressed files, emails, and PDF. 3. Click Select. **Note:** You can select **Specific File Types** option in the dialog box to select files under all the categories.
Treat Files Without Extension As	Select one of the following options to identify files without an extension: - None - Avro - Parquet
Enter File Delimiter	Specify the file delimiter if the file from which you extract metadata uses a delimiter other than the following list of delimiters: - Comma (,) - Horizontal tab (\t) - Semicolon (;) - Colon (:) - Pipe symbol (|) Verify that you enclose the delimiter in single quotes. For example, `'$'`. Use a comma to separate multiple delimiters. For example, `'$','%','&'`
Other File Types	Extracts basic file metadata such as size of the file, path to the file, and time stamp information from other file types.

Property	Description
First Level Directory	Specify a directory or a list of directories under the source directory. If you leave this option blank, Enterprise Data Catalog imports all the files from the specified source directory. To specify a directory or a list of directories, you can perform the following steps: 1. Click **Select...**. The **Select First Level Directory** dialog box appears. 2. Select the required directories using one of the following options: - **Select from list**: select the required directories from a list of directories. - **Select using regex**: provide an SQL regular expression to select schemas that match the expression. **Note:** If you are selecting multiple directories, you must separate the directories using a semicolon (;).
Include Subdirectory	Select this option to import all the files in the subdirectories under the source directory. **Note:** This option is mandatory to scan files from SharePoint.
Case Sensitive	Specifies that the resource is configured for case sensitivity. Select one of the following values: - True. Select this check box to specify that the resource is configured as case sensitive. - False. Clear this check box to specify that the resource is configured as case insensitive. The default value is True.
Memory	The memory required to run the scanner job. Select one of the following values based on the data set size imported: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair value. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

You can enable data discovery for a SharePoint resource. For more information about enabling data discovery, see the Enable Data Discovery topic.

You can enable composite data domain discovery for a SharePoint resource. The Microsoft Sharepoint resource operates at one site per resource. For more information about enabling composite data domain discovery,see the Composite Data Domain Discovery topic.

CHAPTER 12

Configuring Streaming Source Resources

This chapter includes the following topics:

- Overview of Streaming Source Resources, 197
- Apache Kafka, 197

Overview of Streaming Source Resources

The Streaming Source group includes the Apache Kafka resource. When you create and run the resource, the Apache Kafka scanner extracts and migrates the results from the Apache Kafka data source to the catalog. You can view the migrated results in Enterprise Data Catalog.

Apache Kafka

Apache Kafka is an open-source distributed data streaming platform that provides a unified, high throughput, and low latency environment for handling real-time data.

Objects Extracted

The Apache Kafka resource extracts metadata from the schema details of the messages published to Kafka topics in an Apache Kafka data source.

Connect to an Apache Kafka Data Source Enabled for SSL

To connect to an Apache Kafka data source with a SSL enabled Schema Registry, perform the following steps:

1. Download the SSL certificates from the Schema Registry for Apache Kafka using a web browser.

   Note: Make sure that you import the SSL enabled certificate of the Schema Registry for Apache Kafka in the Certificates directory.
2. Copy the certificates to the `<INFA_HOME>/services/shared/security/` directory.

3. Go to the `<INFA_HOME>/source/java/jre/bin` directory and then run the following `keytool` command to import each copied certificate as a trusted certificate in to the Informatica domain keystore:

```
keytool -import -file <INFA_HOME>/services/shared/security/<certificate>.cer -alias
<alias name> -keystore <INFA_HOME>/services/shared/security/infa_truststore.jks -
storepass <Informatica domain keystore password>
```

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
Confluent Platform	Indicates if the selected platform is a stream data platform.
Schema Registry URL	A URL to access the schema registry. The URL syntax is: `http://host1:port1` **Note:** If the schema registry is enabled for SSL, the public certificate of the Confluent Schema Registry must be imported in to the Informatica truststore.
Broker List	Specify the list of brokers in the following format: `<host number>:<port number>`. To specify multiple brokers, use a comma as the delimiter.
Security Protocol	Security protocol for communication with brokers. Some of the valid values are PLAINTEXT, SSL, SASL_PLAINTEXT, and SASL_SSL.
SSL Truststore File Path	Location of the default `infa_truststore` file along with the file name, from the domain node. Default path: *`INFA_HOME`*`/services/shared/security/infa_truststore.jks`
SSL Truststore Password	Default password for the SSL truststore.
SSL KeyStore File Path	Location of the default `infa_keystore` file along with the file name, from the domain node. Default path: *`INFA_HOME`*`/services/shared/security/infa_keystore.jks`
SSL KeyStore Password	Default password for the SSL keystore.
SSL Key Password	SSL key password.
SASL Authentication Mechanism	SASL mechanism used for client connections. Specify GSSAPI for Kerberos and PLAIN for a simple user authentication.

Property	Description
SASL JAAS Configuration	Specify the JAAS login context parameters for SASL connections in the format supported by the JAAS configuration file. The format used is `loginModuleClass controlFlag (optionName=optionValue)*;` For brokers, the JAAS configuration must be prefixed with the listener prefix and the name of the SASL mechanism must be in lowercase.
Kerberos Configuration File Path	Fully qualified path of the Kerberos configuration file, if you use Kerberos authentication for the resource.
API Timeout in milliseconds	Timeout duration for the Apache Kafka API. Specify the timeout in milliseconds.

The following table describes the properties that you can configure in the **Source Metadata** section of the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Enables metadata extraction.
Topic Names or Patterns	List of topic names or wildcard patterns that must be included or excluded from the metadata scan. An empty list indicates that all available topics are included in the metadata scan. Use a comma to separate the topic names. For example, you can use the following wildcard patterns or regular expressions to include or exclude topics from the metadata scan: - ^(abc).* Selects the topics or patterns that start with abc. - <Topic name>_[^de]<Topic name>. Selects the topics that do not have the d and e characters in the example string. - <Topic name>[iJK]<Topic name>\d. Selects the topics that have the i,J, and K characters, which end with a digit in the example string. **Note:** - If the selected topics include incorrect XML or JSON content, the schema inferred by the resource might be incorrect. - If the selected topic includes all types of messages such as XML, JSON, and CSV, the schema inferred by the resource might be incorrect. - If there is no content in the selected topic, you cannot view the schema details in the resource.
Polling Strategy	Select one of the polling strategies: - Poll from Beginning - Poll from End
Memory	The memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal

Property	Description
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, INFO, or ERROR. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value must be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the multiple key value pairs. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1. - `-Djava.security.krb5.conf=<Path of the Kerberos configuration file>`. Extracts metadata from the data source enabled for Kerberos when the security protocol is SASL.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

Note:

- The Apache Kafka resource has a sampling size of 30 messages. The sampling size cannot be changed.
- The Apache Kafka resource supports external lineage with streaming mappings.

Running an Apache Kafka resource on Kerberos-enabled Cluster

If you want to run an Apache Kafka resource on a Kerberos-enabled cluster, perform the following steps:

1. Copy the krb.conf file to the following location: `<Install Directory>/data/ldmbcmev/Informatica/LDM20_309/source/services/shared/security/krb5.conf`
2. Copy the krb.conf file to `/etc` location on all the clusters where the Catalog Service is running.
3. Copy the keytab file to the `/opt` directory in the following locations:
   - Common location for all clusters where Catalog Service is running.
   - The domain machine.
   - The Kerberos cluster machine.
4. Add the machine details of the kdc host in the `etc/hosts` location of the domain machine and the cluster machine where the Catalog Service is running.

CHAPTER 13

Configuring No SQL Resources

This chapter includes the following topics:

- Overview of No SQL Resources, 201
- Apache Cassandra, 201
- MongoDB, 204

Overview of No SQL Resources

The No SQL resource group includes the No SQL resources. When you create and run a No SQL resource, the No SQL scanner extracts and migrates the metadata from the associated resource to the catalog. You can view the migrated results in Enterprise Data Catalog.

Apache Cassandra

Apache Cassandra is an open source, distributed, wide-column store, and a NoSQL database management system designed to handle large amounts of data across many commodity servers. It provides high data availability without failure.

Objects Extracted

The Apache Cassandra resource extracts metadata from the following assets in an Apache Cassandra data source:

- Keyspace
- Tables
- Views
- Function
- Fields

Permissions to Configure the Resource

Configure the `read` permission on the Apache Cassandra data source for the user account that you use to access the data source.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
Host	Host name or IP address of the Apache Cassandra server.
KeyStore Password	Password to access the Apache Cassandra KeyStore
KeyStore Path	Location of the Apache Cassandra KeyStore.
Local Datacenter	Name of the datacenter that contains the required node.
Password	Password of the user account to access the Apache Cassandra server.
Port	Port number of the Apache Cassandra server. Default port number is 9042
SSL Enabled	Option to enable SSL
Truststore Password	Password to access the Apache Cassandra Truststore.
Truststore Path	Location of the Apache Cassandra Truststore.
Username	User name to access the Apache Cassandra server.

The following table describes the properties that you can configure in the **Source Metadata** section of the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Enables metadata extraction.
Case Sensitive	Specifies that the resource is configured for case insensitivity. Select one of the following values: True. Select the check box to specify that the resource is configured as case sensitive. False. Clear the check box to specify that the resource is configured as case insensitive. The default value is True.

Property	Description
Keyspace	Option to import a particular database schema by specifying a list of keyspaces using the Browse API.
Memory	The memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, INFO, or ERROR. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value must be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the multiple key value pairs. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

You can enable domain discovery for Cassandra resource. For more information, see Enable Data Discovery and Composite Data Domain Discovery topics.

Prerequisite to Perform Domain Discovery on Cassandra Resource

Before you run profiles to perform domain discovery on Cassandra resource, make sure to configure the connection and Data Integration Service settings.

Connection setting:

1. In Informatica Administrator, click **Connection** tab in the **Manage** section.
2. In the **Domain Navigator** section, select Cassandra resource connection that you use to run profiles.
3. In the **Advanced Property** section, set **SQL identifier character** as `""(quotes)`
4. Click **OK**.
5. In the **Additional Connection Properties** field, add the following value: `EnableCaseSensitive=0`.
6. Click **OK**.

Data Integration Service setting:

1. In Informatica Administrator, click **Services and Nodes** tab in the **Manage** section.
2. Go to Data Integration Service.
3. In the **Advance Properties** field, set **Maximum Heap Size** as 4096M.
4. Click **OK**.
5. In the **Custom Properties** field, add the following value: `ExecutionContextOptions.JVMMaxMemory=1024M`
6. Click **OK**.

MongoDB

MongoDB is a cross-platform, document-based database that provides high performance, high availability, and scalability.

Objects Extracted

The MongoDB resource extracts metadata from the following assets in a MongoDB data source:

- Database
- Collections
- Fields
- Views

You can view external lineage of objects, nested objects, and all MongoDB data types except the array data type when you use the Informatica Intelligent Cloud Services, PowerCenter, and Data Engineering Integration mappings.

Note: The MongoDB resource does not display internal lineage, and relationship between views and collections.

Permissions to Configure the Resource

Configure the `read` permission on the MongoDB data source for the user account that you use to access the data source.

Connect to a MongoDB Data Source Enabled for SSL

To connect to a MongoDB data source enabled for SSL, perform the following steps:

1. Download the MongoDB SSL certificates using a web browser.

   Note: Make sure that you import the MongoDB Trust Services certificate in the Certificates directory.
2. Copy the certificates to the `<INFA_HOME>/services/shared/security/` directory.
3. Go to the `<INFA_HOME>/source/java/jre/bin` directory and then run the following `keytool` command to import each copied certificate as a trusted certificate in to the Informatica domain keystore:

   ```
   keytool -import -file <INFA_HOME>/services/shared/security/<certificate>.cer -alias
   <alias name> -keystore <INFA_HOME>/services/shared/security/infa_truststore.jks -
   storepass <Informatica domain keystore password>
   ```

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
Use Connection URL	URL to connect to the MongoDB server. Default value is Yes.
Connection URL	URL to connect to the MongoDB server. URL syntax for a standalone MongoDB server is `mongodb://<hostname>:<port>/<database>?retryWrites=true&w=majority` URL syntax for MongoDB as a service in the cluster is `mongodb+srv://<username>:<password>@<hostname>:<port>/test?retryWrites=true&w=majority`
Host	Host name or IP address of the MongoDB server.
Port	Port number of the MongoDB server.
SSL Enabled	Indicates whether the MongoDB server is enabled for SSL
Authentication	Type of authentication to connect to the MongoDB server. Select one of the following options: - None. - Username / Password. Enables role-based authentication.
Username	Username of the account to connect to the MongoDB server.
Password	Password of the account to connect to the MongoDB server.
Authentication Database	Authentication database where the user data is defined. By default, user data is stored in the admin database.

The following table describes the properties that you can configure in the **Source Metadata** section of the **Metadata Load Settings** tab:

Property	Description
Enable Source Metadata	Enables metadata extraction
Database	Imports the required database for the resource run. Use a semicolon to separate the database names. An empty string imports all databases. A database name imports all the collections that belong to the database.
Number of rows to sample	Specify the number of rows that the resource runs on. By default, you can choose a sample size of 20 rows. The maximum sample size in a resource run is 1000 rows. The MongoDB resource limits the sampling size to 30 documents, to identify the schema.
Sampling Option	Specify one of the following sampling options: - First N Rows - Random N Rows - Last N Rows
Source Metadata Filter	Specify a combination of regular expressions and wildcards to include or exclude specific assets in the resource run. Use a semicolon to separate the wildcard patterns.

Property	Description
Import system collections	Option to indicate whether the MongoDB collections should be imported. Select True if the user has the database owner privileges.
Memory	The memory value required to run a scanner job. Specify one of the following memory values: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	JVM parameters that you can set to configure scanner container. Use the following arguments to configure the parameters: - `-Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of scanner to values, such as DEBUG, INFO, or ERROR. Default value is INFO. - `-Dscanner.container.core=<No. of core>`. Increases the core for the scanner container. The value must be a number. - `-Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the multiple key value pairs. - `-Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Number of threads for execution	Number of threads required for the metadata extraction.
Track Data Source Changes	View metadata source change notifications in Enterprise Data Catalog.

Unsupported Features

- The MongoDB resource does not support indexed array, and virtual tables.
- For heterogenous arrays, the MongoDB resource fetches the key to infer the schema. The value of the heterogeneous arrays is not used to infer the schema.

CHAPTER 14

Configuring Informatica Platform Resources

This chapter includes the following topics:

- Informatica Platform Resources Overview, 207
- Informatica Data Quality, 207
- Informatica Similarity Discovery Resource, 212
- Troubleshooting, 216

Informatica Platform Resources Overview

Informatica Platform resource group includes the Informatica Data Quality and Informatica Similarity Discovery resource.

When you create and run the Informatica Data Quality resource, the scanner extracts and migrates the profile results from the profiling warehouse to the catalog. You can then view the migrated results in Enterprise Data Catalog. When you create and run the Informatica Similarity Discovery resource, the scanner discovers similar columns based on column name, column data pattern, and unique values in columns from the selected resources.

Informatica Data Quality

When you run a profile in Informatica Analyst or Informatica Developer, the Data Integration Service stores the profile results in the profiling warehouse. To view these results in Enterprise Data Catalog, you can create an Informatica Data Quality resource in Catalog Administrator and associate the profiling warehouse to the resource. When you run the resource, the Informatica Data Quality scanner extracts and migrates the profile results in the associated profiling warehouse to the catalog. The migrated profile results include column profile results, data domain discovery results, rule profiling results, scorecards and value frequencies. You can migrate the profile results from the current or earlier versions on any Informatica domain. Informatica Data Quality resource supports relational database systems.

In Catalog Administrator, you can create multiple profiling warehouse resources. When you create a profiling warehouse resource, you associate a profiling warehouse to the resource. Enterprise Data Catalog scans the

associated profiling warehouse and lists all the schemas with profile results. You can choose one or more schemas for which the Informatica Data Quality scanner migrates the results to the catalog.

When you run the Informatica Data Quality scanner, the catalog compares the timestamp of the results in the catalog with the migrated results. Enterprise Data Catalog displays the latest profile results and curation results based on the timestamp. Enterprise Data Catalog merges the inference results with the existing results in the catalog.

You can view the value frequency for a column or rule in the Developer tool or Analyst tool. The Informatica Data Quality scanner migrates the value frequencies for the columns along with the profile results to the catalog. The profiling warehouse stores a maximum of 16,000 values for a column. The scanner identifies the top 1,000 maximum values for a column and migrates these values to Enterprise Data Catalog.

Example

Assume that you have more than 1 million tables spread across 700 schemas and multiple databases in your enterprise. Over the years, you have run profiles on these schemas in the Developer tool or Analyst tool. All the profiling results reside in one or more profiling warehouse databases. Now, you want to implement Enterprise Data Catalog in your enterprise and want to access and view the existing profile results for the schemas in the catalog. Instead of running the profiles on the schemas and databases in Catalog Administrator, which is a time-consuming and resource-intensive effort, you decide to migrate the results to the catalog. Additionally, you want the developers and analysts to continue using the Developer tool and Analyst tool.

In this case, you can run the core scanners to extract the metadata from the schemas and databases to the catalog. Then, you can create a profiling warehouse resource, choose the schemas and connections for which you want to migrate the profile results, and run the profiling warehouse resource. The scanner migrates the profile results of the selected schemas to the catalog. This action saves time and effort. An added advantage is that the users of the Developer tool and Analyst tool can continue to use the tools and you can migrate the results as and when required.

Extracting Rule Profile Results

You can apply rules and run rule profiles in the Developer tool and the Analyst tool. You can use the following methods to create or apply rules in the column profiles:

- In the Developer tool, create a mapplet and validate it as a rule. The rule appears as a reusable rule in the Analyst tool. You can apply the rule to column profiles in the Analyst tool and Developer tool.
- In the Analyst tool, create a rule specification and generate a mapplet. You can apply the rule specification to column profiles in the Analyst tool. In the Developer tool, validate the mapplet as a rule. The rule appears as a reusable rule that you can use in the column profiles.
- You can use predefined rules in the column profiles. Informatica provides the predefined rules with the Developer tool and Analyst tool.

When you run Informatica Data Quality resource, the resource extracts the rule profile results to the catalog. You cannot apply expression rules to the profile. You cannot associate business terms with rule asset in the Enterprise Data Catalog. To learn how to create and apply rules to the profiles in the Developer tool and Analyst tool, see *Informatica Data Discovery Guide*.

Extracting Scorecard Results

You can create and edit a scorecard in the Developer tool and Analyst tool. A scorecard has multiple components, such as metrics, metric groups, scores, and thresholds. Scorecards help an enterprise to measure the value of data quality at the metric level. You can create metric groups to group related metrics. After you run profile, you can add columns from the profile results as metric to a scorecard. When you run a scorecard, the Analyst tool and Developer tool generates weighted average value for each metric group. When you run data quality resource, the resource extracts the scorecard results to the catalog.

You cannot associate business terms with scorecard asset in the Enterprise Data Catalog. To learn how to create, edit, and run scorecards in the Developer tool and Analyst tool, see *Informatica Data Discovery Guide*.

Objects Extracted

The Informatica Data Quality scanner extracts the following profiling metrics from the profiling warehouse to the catalog:

- Column profile results
  - Null, unique, and non-unique values.
  - Patterns
  - Data types
  - Maximum and minimum values
  - Standard deviation value
  - Average value
  - Sum value
- Data domain discovery results
  - Inference results
  - Curation results
- Rule profile results
  - Null, distinct, and non-distinct values
  - Patterns
  - Maximum and minimum values
  - Maximum and minimum length
  - Data types
- Scorecards
  - Metric groups
  - Metric scores
  - Quality scores
- Value frequencies

Prerequisites

Perform the following step to complete the prerequisites:

- Before you migrate the profiling results for a resource, you must run the respective resource in Catalog Administrator to extract the metadata into the catalog.
  For example, you want to migrate the profile results of the Oracle schema S1 to the catalog. Before you migrate the Oracle schema profile results, create and run an Oracle resource with S1 and use this resource to configure the profiling warehouse resource.
- Export data domains from the profiling warehouse domain to the Enterprise Data Catalog domain if the profiling warehouse and Enterprise Data Catalog are on two different domains.

Basic Information

The **General** tab includes the following basic information about the resource:

Information	Description
Name	The name of the resource.
Description	The description of the resource.
Resource type	The type of the resource.
Execute On	You can choose to execute on the default catalog server or offline.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
Extract	To extract the results from the profiling warehouse, choose one of the following options: - Rule - Scorecard - Profile - Profile and Value Frequency
Domain Name	Name of the Informatica domain. **Note:** This property applies to the versions 10.2.0 HotFix 2 and later.
Domain User Name	User name that the Data Integration Service uses to access the Model Repository Service.
Domain Password	Password for the Model repository user.
Security Domain	Name of the security domain to which the domain user belongs.
Domain Host	Informatica domain host name.
Domain Port	Informatica domain port number.
Node Name	Name of the node on which the Data Integration service runs.
Repository Service Name	Name of the Model Repository Service.
Profiling Warehouse Name	Name of the Profiling Warehouse.

The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Source Metadata	Select the option to extract metadata from the data sources.
Profiled Schema Connections	Select one or more schemas in the **Select Profiled Schema Connections** dialog box. The profiling warehouse scanner migrates the profiling results of the selected schemas to the catalog.

Property	Description
Cumulative	Select or clear the option as necessary. - Select the option if you want the scanner to scan all the profile results for a data source to extract the latest column results based on the timestamp. For more information about the option, see the examples in the "Troubleshooting" on page 216 topic. - Clear the option if you want to fetch the latest profile results.
Auto Accept Percentage	Enter a value from 0 to 100. Enterprise Data Catalog automatically accepts the data domains when the inference percentage exceeds the configured number.
Incremental	Select or clear the option as necessary. - Select the option if you want to migrate the delta of profile results in each run. In the first profiling warehouse resource run, the scanner migrates the profile results for all the tables.. In subsequent resource runs, only the delta of profile results are migrated. For more information about the option, see the examples in the "Troubleshooting" on page 216 topic. - Clear the option if you want to fetch the latest profile results.
Memory	Specify the memory required to run the scanner job. Select one of the following values based on the migrated data set size: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal
Custom Options	Enter the JVM parameters to configure the scanner container.
Track Data Source Changes	Select the option to view the metadata source change notification in Enterprise Data Catalog.

Informatica Data Quality Resource Tasks

When you run the Informatica Data Quality resource, the scanner performs the following tasks:

1. Verifies that the catalog has the source metadata for the resource.

   For example, to migrate Oracle profile results, you must first create and run the Oracle resource in Catalog Administrator.
2. Verifies that the matching resource has profile results.
3. Verifies that the matching resource has curation decisions.
4. If curation results do not exist, the scanner migrates the results to the catalog.
5. If curation results exist, the scanner compares the timestamp of the curation decisions in the catalog with the curation decision in the profiling warehouse. The scanner migrates and overwrites the curation results if the curation results in the profiling warehouse have the latest timestamp.

   For example, you reject the *SSN* data domain for a column in Employee table in the catalog on 05/01/19. You accept the *SSN* data domain for the same column in the Developer tool on 05/02/2019. You run the Informatica Data Quality scanner on 05/03/19. The curation decision for the data domain *SSN* in the profiling warehouse has the latest timestamp. The scanner migrates and overwrites the curation decision in the catalog. The data domain *SSN* displays accepted in the catalog.

Note: Enterprise Data Catalog displays **SYSTEM** in the **Assigned By** column for a data domain asset when you curate a data domain in the Developer tool or Analyst tool and the decision is migrated to the catalog.

6. Merges the data domain inferred results.

   For example, in the Developer tool, you accept the data domains *SSN* and *Age* in the Employee table. In Enterprise Data Catalog, you accept the data domains *Address* and *City* in the same Employee table. When you run the Informatica Data Quality scanner, the scanner merges the data domain accepted results and displays *SSN* , *Age*, *Address*, and *City* as inferred data domains for the Employee table in the catalog.
7. Automatically accepts a data domain if the inference percentage exceeds the **Auto Accept Percentage** value.
8. Identifies the top 1,000 maximum values for a column and migrates the values to the catalog.
9. In Catalog Administrator, when you click **Missing Report link** tab in the **Connection Assignment** section, the exported file does not display the details of source and target objects that are not linked.

Informatica Similarity Discovery Resource

As a data analyst or data architect, you can scan your enterprise data to find similar columns. You can run the Informatica Similarity Discovery resource to discover similar columns based on column name, column data pattern, and unique values factors.

In Catalog Administrator, you can create multiple Informatica Similarity Discovery resources. The Informatica Similarity Discovery resource runs on multiple resources to discover similar columns in the catalog. You can specify the resources on which you want to run similarity discovery.

When you create and run the Informatica Similarity Discovery resource, the resource scans the PostgreSQL database, compares the prepared data to infer similar columns, and persists the results into the PostgreSQL database. After you run the Informatica Similarity Discovery resource, you can view similar columns in Enterprise Data Catalog.

Prerequisite

Before you create and run the Informatica Similarity Discovery resource, ensure that you have selected the **Run Similarity Profile** option for resources on which you want to run the similarity discovery in Catalog Administrator.

Transformations supported in lineage

A transformation indicates generation, modification, or passage of data between source and target connections.

You can view the following transformations in the Lineage and Impact tab for an Informatica Similarity Discovery resource:

- Aggregator
- Datamasking
- Expr_filter
- Javatx
- Joiner

- Lookup
- Mappingwithsqltx
- Normalizer
- Rank
- Router
- Sorter
- Sqq_gen
- Union

Example

The catalog contains similar customer details spread across multiple resources. You might want to identify all the resources that contain similar customer details so that you can integrate the data into a single resource and remove the duplicate data. To achieve this goal, you can enable the **Run Similarity Profile** option for the identified resources, and then create and run the Informatica Similarity Discovery resource on those resources to identify and view similar columns in Enterprise Data Catalog.

For more examples of column similarity, see the Business Example topic in the *Informatica Catalog Administrator Guide*.

Resource Connection Properties

The **General** tab includes the following properties:

Property	Description
Name	Enter a name for the resource.
Description	Optionally, you can enter a brief description for the resource.
Resource type	Choose **Informatica Similarity Discovery** option as the resource type.
Resources to be grouped	Select the resources on which you want to run similarity discovery. By default, you can select four resources for similarity discovery. If you want to add more resources, you can configure the following custom property for the Catalog Service in Informatica Administrator: `LdmCustomOptions.scanner.simDiscoveryScanner.maxPermittedResourceCount` **Note:** After you run the Informatica Similarity Discovery resource with a set of selected resources for similarity discovery, you cannot select the same resources to run similarity discovery using another Informatica Similarity Discovery resource. For example, if you use resources R1, R2, and R3 to run similarity discovery, you cannot use resources R1, R2, and R3 again with another Similarity Discovery resource.

The **Metadata Load Settings** tab includes the following properties:

Property	Description
Enable Similarity Discovery	Select the option to run similarity discovery on the selected resources.
Features enabled	Select or clear the options as required. Select the features based on which you want to run similarity discovery. The following are the similarity discovery features: - Name. Determines similar columns across data sources based on the column names. The name match uses fuzzy string match to identify the columns that have similar names. - Patterns. Determines the column data pattern in the data sources and calculates the percentage of pattern overlap across column pairs for all data types except numeric data types. - Unique Values. Determines similar columns across data sources based on distinct values. The distinct values overlap is calculated as a percentage of distinct values that overlap in two columns. For more information about the similarity discovery features, see the Column Similarity topic.
Cumulative	Choose this option to retain the column similarity results from the previous scan in the next resource scan results. If you do not choose this option, the previous profile results are purged.
Memory	The memory required to run the scanner job. Select one of the following values based on the data set size imported: - Low - Medium - High **Note:** For more information about the memory values, see the *Tuning Enterprise Data Catalog Performance* article on **How To-Library Articles** tab in the Informatica Doc Portal .
Custom Options	Specify the custom options to run the scanner job.
Schema Filter	Specify the schema to scope for similarity discovery. Use the property to limit similarity analysis to the specified schemas. Use the full schema name in the filter. The name is case-insensitive. To specify multiple schemas, use a semicolon-separated list. For example, to include both the PROD_SCHEMA and IMP_SCHEMA, use PROD_SCHEMA; IMP_SCHEMA.
Table Filter	Specify include or exclude pattern tables in the selected schema filter. Use the Table Filter to control which specific tables are analyzed during Similarity Discovery. If a table matches both include and exclude patterns, the exclude pattern takes precedence. Table and object names are case-sensitive For example, you can use `customer*` to include objects whose names start with customer, and `*_tmp` to match temporary objects. To exclude temporary objects, use `NOT *_tmp`. To include more than one pattern, use a semicolon-separated list such as `orders*;invoice*`.
Track Data Source Changes	Select the option to view the metadata source change notification in Enterprise Data Catalog.

Resource tasks

When you run the Informatica Similarity Discovery resource, the scanner performs the following tasks:

1. Verifies whether the selected resources are enabled with the **Run Similarity Profile** option.

2. Scans the prepared data in the PostgreSQL database, compares the prepared data to infer similar columns, and persists the results into the PostgreSQL database.

The following matrix describes the type of similarity computation performed on the columns of different inferred data types:

Column type	Name	Pattern	Unique values
Numeric	Yes	No	No
String	Yes	Yes	Yes
Date	Yes	No	No

For more information about the column similarity process, see the Column Similarity Process topic.

Use Schema Filter and Table Filter in Similarity Discovery

After you configure the Schema Filter and Table Filter for an Informatica Similarity Discovery resource, Catalog Administrator applies the filters to control the scope of similarity analysis.

Filter Application Order

Similarity Discovery applies filters in the following sequence:

1. Schema Filter - Defines the schema scope
2. Table Filter - Applies within the defined schema scope

Example 1: Using `Schema Filter`
Use the following configuration:

- `Schema Filter`: `IMP_SCHEMA`
- `Table Filter`: Not configured

Result: Excludes all schemas that match `IMP_SCHEMA` and excludes all tables within the selected schema.

Example 2: Both `Schema Filter` **and** `Table Filter`
Use the following configuration:

- `Schema Filter`: `IMP_SCHEMA`
- `Table Filter`: `EMP%;NOT EMPLOYEES%`

Result: The scope is first limited to schemas matching `IMP_SCHEMA`. Within that scope, tables with names starting with EMP are included, but tables starting with EMPLOYEES are excluded. It also excludes all tables in schemas outside `IMP_SCHEMA` scope.

Example 3: `Table Filter`
Use the following configuration:

- `Schema Filter`: Not configured
- `Table Filter`: `CUSTOMERS;NOT %_TEMP`

Result: The analysis runs on all schemas by default. It includes tables named CUSTOMERS but excludes any table whose name ends with _TEMP.

Example 4: No Filters Configured

Use the following configuration:

- `Schema Filter`: Not configured
- `Table Filter`: Not configured

Result: Includes all schemas and tables without restriction.

Note: If a table matches both include and exclude patterns, the exclude pattern takes precedence.

Troubleshooting

In the Developer tool, I run a profile on the first three columns of a data source in the first run. Then, I run the profile on the next three columns in the data source. What will Enterprise Data Catalog display after I run the profiling warehouse resource?

By default, the catalog displays the last profile run results. If you want to view all the six columns in the catalog, then choose the **Cumulative** option in the profiling warehouse resource and run the resource. Enterprise Data Catalog displays the latest results for each column in the data source based on the timestamp.

How does the Cumulative option work?

Assume that you have an Employee table with columns C1, C2, C3, and C4. In the Developer tool, you create and run a profile on the table. The profile runs include R1, R2, and R3.

The following list shows the columns that you choose when you run the profile:

1. R1. You chose C1 and C2.
2. R2. You chose C3 and C4.
3. R3. You chose C1 and C4.

In Catalog Administrator, you create and run the profiling warehouse resource with the **Cumulative** option. The Informatica Data Quality scanner extracts the following latest column results based on the timestamp:

1. C1. The scanner extracts the latest results from R3.
2. C2. The scanner extracts the latest results from R1.
3. C3. The scanner extracts the latest results from R2.
4. C4. The scanner extracts the latest results from R3.

How does the Incremental option work?

1. In the Developer tool, you run a profile P1 on the Employee and Department tables at 10.00 AM on 05/02/19.
2. In the Catalog Administrator, you create a profiling warehouse resource PWH with the **Incremental** option, and run the resource at 10.15 AM on 05/02/19.
   The scanner migrates the profile results of P1 to the catalog.
3. You run P1 again at 10.00 AM on 05/03/19.
4. You run PWH at 10.15 AM on 05/03/19.
   The scanner does not migrate any results as there are no new results or change in timestamp.
5. You create and run profile P2 on the Employee and Department tables at 10.00 AM on 05/04/19.
6. You run PWH at 10.15 AM on 05/04/19.
   The scanner migrates P2 profile results to the catalog.

7. You run the profile on the Payroll table at 10.00 AM on 05/06/19.
8. You run PWH at 10.15 AM on 05/06/19.
   The scanner migrates only the profile results of the Payroll table because you chose the **Incremental** option. In this case, the profile results of the Payroll table is also called delta.

What results appear in the catalog if I run the profiling resource and profiling warehouse resource on the same table?

The catalog compares the timestamp of the profile results from the profiling resource run and the migrated profile results. It displays the results with the latest timestamp.

What job types are available for the profiling warehouse resource in the Monitoring tab?

The **Metadata Load** job type is available for the profiling warehouse resource.

What profiling statistics and progress operations appear In the Monitoring tab for the profiling warehouse resource?

In the **Statistics** tab, the following profiling statistic names and their values appear:

- Number of Profile Result Identifiers Fetched
- Number of Profile Results Fetched
- Value Frequency results for Tables extracted

In the **Progress** tab, the following progress operation names with their timestamp and outcomes appear:

- Profile Results Extraction
- Profile Results Identifiers Extraction
- Profile Results Ingestion
- VF Extraction

If I have multiple resource results in the catalog, which resource results are overwritten by the profiling warehouse scanner?

Assume that you have an Oracle schema S1. You create Oracle connections C1 and C2 in the Administrator tool.

In the Developer tool, you choose C1 to import the Oracle tables from S1, create a profile for the tables, and run the profile.

In Catalog Administrator, you choose C1 and S1 to create the Oracle resource R1. You choose C2 and S1 to create the Oracle resource R2. You run the resources. The catalog displays the resource results.

When you create a profiling warehouse resource, you choose the profiling warehouse connection C1. When you run the profiling warehouse resource, the catalog compares and overwrites the resource results of R1.

I see a mismatch in data domain names in the catalog. How do I resolve this issue?

This issue appears when the profiling warehouse and Enterprise Data Catalog are in different domains. To synchronize the data domains, export the data domains from the profiling warehouse domain and import them into the Enterprise Data Catalog domain.

What are the supported database types and their fully qualified JDBC connection strings that I can use for the profiling warehouse scanner?

The following table lists the sample JDBC connection strings for all the supported database types:

Database Type	Class Name Value	Connection String Value
Oracle	com.informatica.jdbc.oracle.OracleDriver	jdbc:informatica:oracle://<hostname>:<port>;SID=<sid>
DB2	com.informatica.jdbc.db2.DB2Driver	jdbc:informatica:db2://<hostname>:<port>;DatabaseName=<dbname>
SQL Server	com.informatica.jdbc.sqlserver.SQLServerDriver	jdbc:informatica:sqlserver://<host>:<port>;databaseName=<dbname>
Sybase	com.informatica.jdbc.sybase.SybaseDriver	jdbc:informatica:sybase://<host>:<port>;databaseName=<dbname>
MYSQL	com.informatica.jdbc.mysql.MySQLDriver	jdbc:informatica:mysql://<host>:<port>;databaseName=<dbname>

I run a profile on a source column that has a special character in its name. Why does the profile run fail?

If a source table or column has a special character in its name, or the name starts with a number, the profiling warehouse replaces each special character or the number with an underscore (_) character. Therefore, the table is not stored in the profiling warehouse by its original name and you can see two tables instead of the source table in Enterprise Data Catalog. When you run a profile, the profiling warehouse cannot update the table with the profiling results, and the profile run fails. You can observe the similar behavior when a reference resource is created on the Data Quality scanner.

Before you select columns on which you want to run the profile, remove special characters and numbers from the table and column names.

In a Developer tool, I run a profile to perform data domain discovery and migrate profile results to Enterprise Data Catalog. Then I perform curation on the profile results and again run Informatica Data Quality resource to migrate results to the Enterprise Data Catalog. Will I see the updated profile results in Enterprise Data Catalog?

You will not see the updated results in the Enterprise Data Catalog. To view the updated profile results make sure you run a profile after curation and then migrate the profile results to the Enterprise Data Catalog.

What are the resources that support Parquet file formats for profiling?

The following table lists resources and their supported Parquet file formats:

Resources	Parquet file formats supported
Amazon S3	Single and partitioned Parquet files
Azure Data Lake Store Gen2	Single and partitioned Parquet files
HDFS	Single Parquet file
Local File Systems	Single Parquet file

APPENDIX A

JDBC Driver Classes

This appendix includes the following topic:

- Examples of JDBC Driver Classes, 219

Examples of JDBC Driver Classes

You can use a JDBC connection and a driver class to access tables in a database.

The following table provides a few examples of JDBC driver classes that belong to different data sources:

Driver Vendor	Data Source	Driver Class Example
Simba	Databricks	com.simba.spark.jdbc4.Driver
CData	MongoDB	cdata.jdbc.mongodb.MongoDBDriver
Vertica	Vertica	com.vertica.jdbc.Driver
Amazon	Athena	com.simba.athena.jdbc.Driver
Oracle	MySQL	com.informatica.jdbc.mysql.MySQLDriver
Postgres	PostgreSQL	org.postgresql.Driver
Oracle	Oracle	com.informatica.jdbc.oracle.OracleDriver
IBM	DB2	com.informatica.jdbc.db2.DB2Driver
Microsoft	SQL Server	com.informatica.jdbc.sqlserver.SQLServerDriver
SAP	Sybase	com.informatica.jdbc.sybase.SybaseDriver

INDEX

R

resource
- attributes 15
- creating 15
- overview 15
- scanner 14
- scanner schedule 15

resource type property
- Atlas 56
- Axon 78
- business glossary 81
- Cassandra 201
- DatabaseScripts 128
- DataStage 145
- DIH 127
- Erwin 146
- GCS 53
- Glue 50
- Hadoop 184
- Hive 65
- IBM DB2 155
- IBM DB2 for z/OS 157
- IBM Netezza 160
- IICS 129
- Informatica MDM 131
- Informatica platform 133

resource type property *(continued)*
- JDBC 163
- Kafka 197
- MicroStrategy 88
- MongoDB 204
- Oracle 171
- PowerBI 119
- PowerCenter 138
- Qlik Sense 119
- Salesforce 45
- SAP BusinessObjects 99
- SAP BW 108
- SAP BW/4HANA 114
- SAP PowerDesigner 151
- SAP S/4HANA 72
- Snowflake 53
- SQL Server 167
- SSIS 144
- Sybase 177
- Teradata 180

S

Source Metadata Filter
- overview 16